# CPEG495 Lab 5.5

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## Lab Assignments for Module 5

One of the primary vectors into (or out of) an infrastructure is through the network. As an investigator, it is important to be able to examine various types of network information to identify any information transmitted within the period of the crime.

Our lab assignment this week focuses on the investigation of three Packet Capture (PCAP) files that may contain information associated with the crime. Upload a report of your findings to the online file into Canvas. **See further instructions below.**

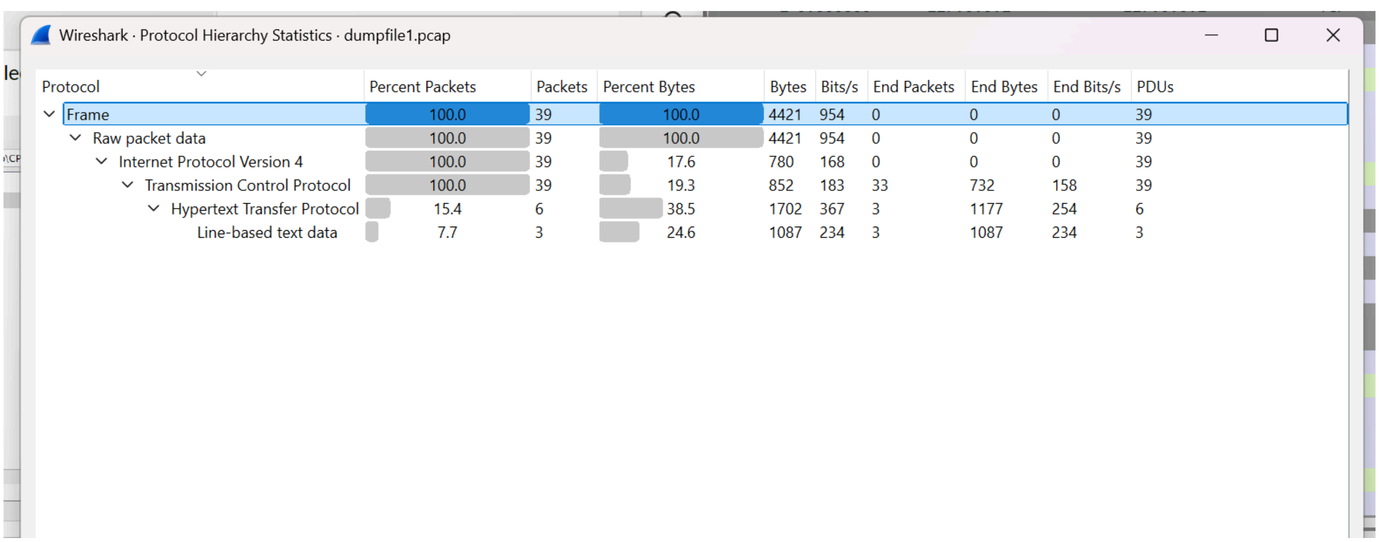
**Lab Assignments**

- Using Wireshark, analyze the PCAP files available to identify any data that was transferred or otherwise relevant during the crime. Be descriptive and JUSTIFY your answer. Additionally, use the files to establish a timeline for the crime, and gather relevant statistics (e.g, when the communication began, how much data was transferred, and so forth). Keep in mind that some data discovered will be encoded (HINT: research substitution ciphers and data encoding).
- more insight
  - the zip file has 8 .pcap files (.pcap is the original Wireshark file format) and 2 .pcapng (.pcapng stands for pcap Next Generation)

### 1. Analyzing dumpfile1.pcap in Wireshark:

I analyzed dumpfile1.pcap using Wireshark. All traffic was between 127.0.0.1 (localhost), indicating the interaction happened on the same machine. The HTTP protocol was used to request and transfer files such as /test1.txt and /test1%20%20copy.txt. These were requested using GET methods and responded to with 200 OK statuses. I exported and reviewed these files for suspicious content.

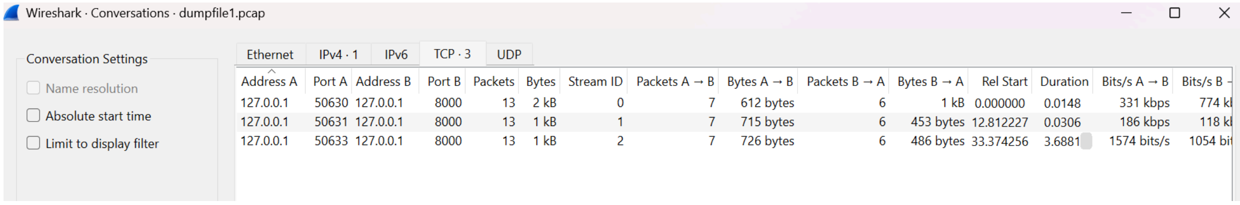
Protocol Hierarchy:



Wireshark · Protocol Hierarchy Statistics · dumpfile1.pcap

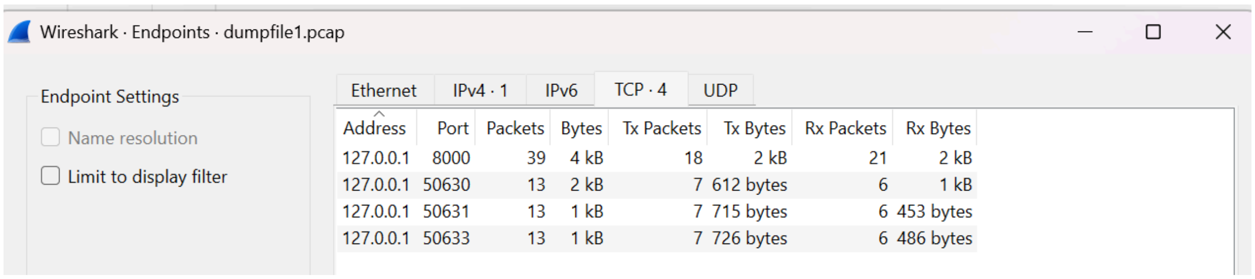
| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 39 | 100.0 | 4421 | 954 | 0 | 0 | 0 | 39 |
| Raw packet data | 100.0 | 39 | 100.0 | 4421 | 954 | 0 | 0 | 0 | 39 |
| Internet Protocol Version 4 | 100.0 | 39 | 17.6 | 780 | 168 | 0 | 0 | 0 | 39 |
| Transmission Control Protocol | 100.0 | 39 | 19.3 | 852 | 183 | 33 | 732 | 158 | 39 |
| Hypertext Transfer Protocol | 15.4 | 6 | 38.5 | 1702 | 367 | 3 | 1177 | 254 | 6 |
| Line-based text data | 7.7 | 3 | 24.6 | 1087 | 234 | 3 | 1087 | 234 | 3 |

## Conversations:

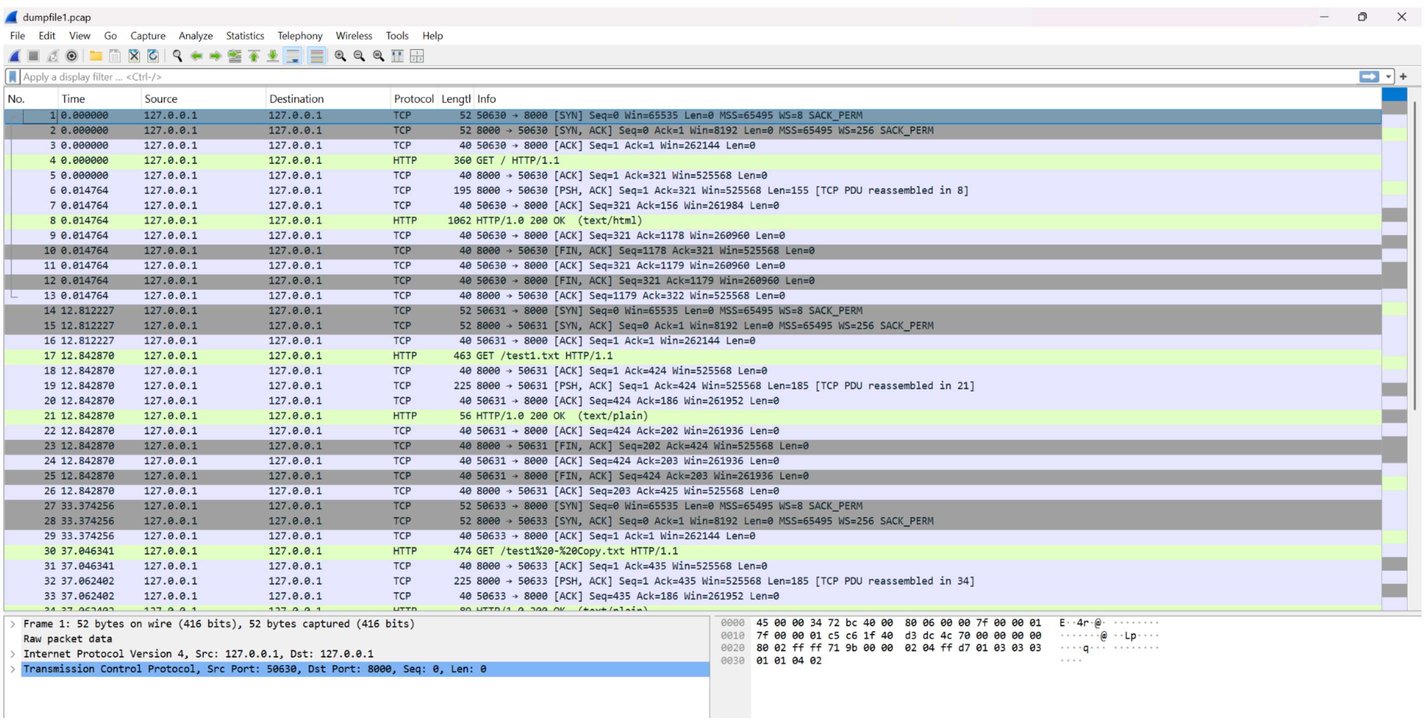



| Address A | Port A | Address B | Port B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 50630 | 127.0.0.1 | 8000 | 13 | 2 kB | 0 | 7 | 612 bytes | 6 | 1 kB | 0.000000 | 0.0148 | 331 kbps | 774 k |
| 127.0.0.1 | 50631 | 127.0.0.1 | 8000 | 13 | 1 kB | 1 | 7 | 715 bytes | 6 | 453 bytes | 12.812227 | 0.0306 | 186 kbps | 118 k |
| 127.0.0.1 | 50633 | 127.0.0.1 | 8000 | 13 | 1 kB | 2 | 7 | 726 bytes | 6 | 486 bytes | 33.374256 | 3.6881 | 1574 bits/s | 1054 bi |

## Endpoints:




| Address | Port | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 8000 | 39 | 4 kB | 18 | 2 kB | 21 | 2 kB |
| 127.0.0.1 | 50630 | 13 | 2 kB | 7 | 612 bytes | 6 | 1 kB |
| 127.0.0.1 | 50631 | 13 | 1 kB | 7 | 715 bytes | 6 | 453 bytes |
| 127.0.0.1 | 50633 | 13 | 1 kB | 7 | 726 bytes | 6 | 486 bytes |

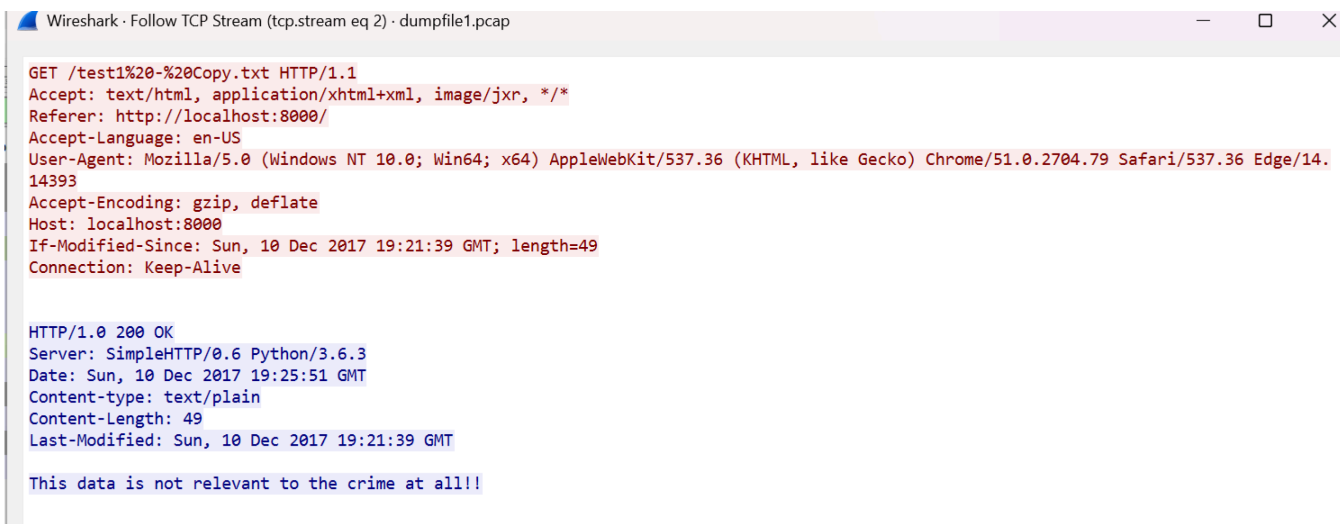



| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 50630 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 2 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 50630 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 3 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50630 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 4 | 0.000000 | 127.0.0.1 | 127.0.0.1 | HTTP | 360 | GET / HTTP/1.1 |
| 5 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 50630 [ACK] Seq=1 Ack=321 Win=525568 Len=0 |
| 6 | 0.014764 | 127.0.0.1 | 127.0.0.1 | TCP | 195 | 8000 → 50630 [PSH, ACK] Seq=1 Ack=321 Win=525568 Len=155 [TCP PDU reassembled in 8] |
| 7 | 0.014764 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50630 → 8000 [ACK] Seq=321 Ack=156 Win=261984 Len=0 |
| 8 | 0.014764 | 127.0.0.1 | 127.0.0.1 | HTTP | 1062 | HTTP/1.0 200 OK (text/html) |
| 9 | 0.014764 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50630 → 8000 [ACK] Seq=321 Ack=1178 Win=260960 Len=0 |
| 10 | 0.014764 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 50630 [FIN, ACK] Seq=1178 Ack=321 Win=525568 Len=0 |
| 11 | 0.014764 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50630 → 8000 [ACK] Seq=321 Ack=1179 Win=260960 Len=0 |
| 12 | 0.014764 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50630 → 8000 [FIN, ACK] Seq=321 Ack=1179 Win=260960 Len=0 |
| 13 | 0.014764 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 50630 [ACK] Seq=1179 Ack=322 Win=525568 Len=0 |
| 14 | 12.812227 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 50631 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 15 | 12.812227 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 50631 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 16 | 12.812227 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50631 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 17 | 12.842870 | 127.0.0.1 | 127.0.0.1 | HTTP | 463 | GET /test1.txt HTTP/1.1 |
| 18 | 12.842870 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 50631 [ACK] Seq=1 Ack=424 Win=525568 Len=0 |
| 19 | 12.842870 | 127.0.0.1 | 127.0.0.1 | TCP | 225 | 8000 → 50631 [PSH, ACK] Seq=1 Ack=424 Win=525568 Len=185 [TCP PDU reassembled in 21] |
| 20 | 12.842870 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50631 → 8000 [ACK] Seq=424 Ack=186 Win=261952 Len=0 |
| 21 | 12.842870 | 127.0.0.1 | 127.0.0.1 | HTTP | 56 | HTTP/1.0 200 OK (text/plain) |
| 22 | 12.842870 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50631 → 8000 [ACK] Seq=424 Ack=202 Win=261936 Len=0 |
| 23 | 12.842870 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 50631 [FIN, ACK] Seq=202 Ack=424 Win=525568 Len=0 |
| 24 | 12.842870 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50631 → 8000 [ACK] Seq=424 Ack=203 Win=261936 Len=0 |
| 25 | 12.842870 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50631 → 8000 [FIN, ACK] Seq=424 Ack=203 Win=261936 Len=0 |
| 26 | 12.842870 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 50631 [ACK] Seq=203 Ack=425 Win=525568 Len=0 |
| 27 | 33.374256 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 50633 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 28 | 33.374256 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 50633 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 29 | 33.374256 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50633 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 30 | 37.046341 | 127.0.0.1 | 127.0.0.1 | HTTP | 474 | GET /test1%20-%20Copy.txt HTTP/1.1 |
| 31 | 37.046341 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 50633 [ACK] Seq=1 Ack=435 Win=525568 Len=0 |
| 32 | 37.062402 | 127.0.0.1 | 127.0.0.1 | TCP | 225 | 8000 → 50633 [PSH, ACK] Seq=1 Ack=435 Win=525568 Len=185 [TCP PDU reassembled in 34] |
| 33 | 37.062402 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 50633 → 8000 [ACK] Seq=435 Ack=186 Win=261952 Len=0 |

Upon analyzing dumpfile1.pcap using Wireshark, I followed multiple TCP streams to examine HTTP communications and content transmitted over the network. I used the "Follow TCP Stream" feature on HTTP packets to extract payloads from the server responses.

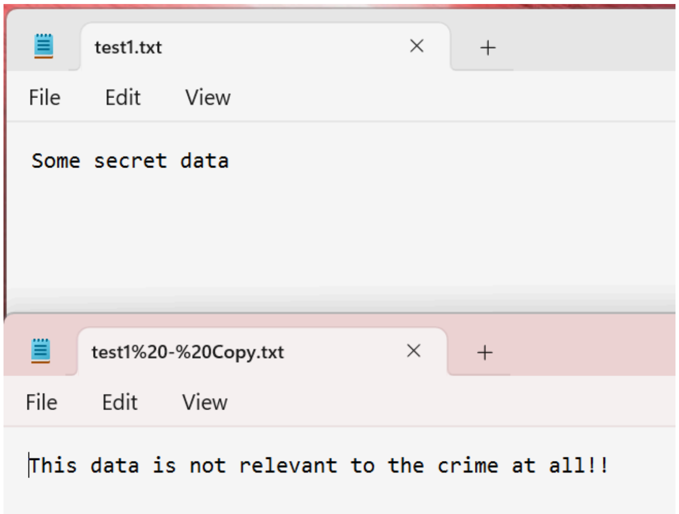
test1.txt:


TCP Stream showing GET request for test1.txt, with server response revealing plain text message: “Some secret data.” This file appears to contain sensitive content potentially relevant to the investigation.




TCP Stream showing GET request for test1 Copy.txt, which returned a plain text message: “This data is not relevant to the crime at all!!” This file appears to be a decoy or unrelated.

I also used File -> Export -> Objects -> HTTP to export files potentially related to the interaction and review for suspicious activity. Below are the results I obtained which correspond to the screenshots above




Analysis Summary:

Upon analyzing dumpfile1.pcap using Wireshark, I followed multiple TCP streams to examine HTTP communications and content transmitted over the network. I used the "Follow TCP Stream" feature on HTTP packets to extract payloads from the server responses.

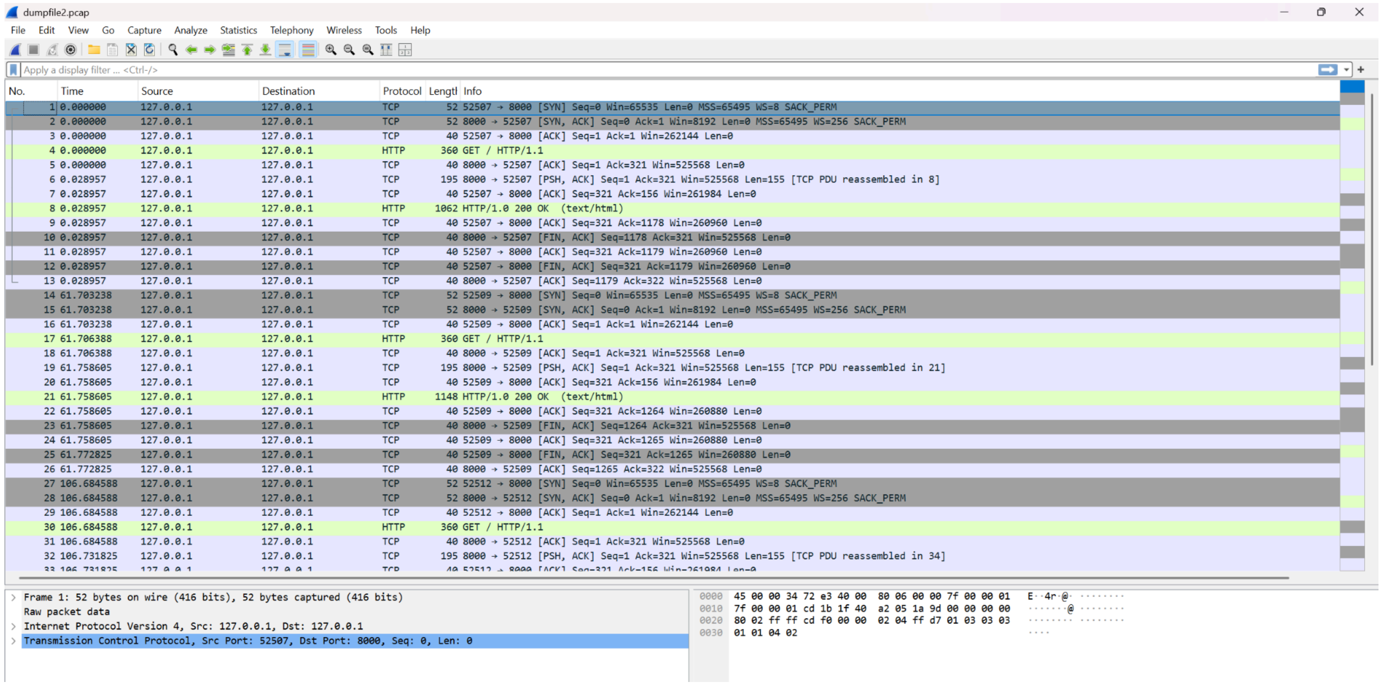
- In one request to /test1.txt, I recovered a plaintext message reading: "Some secret data". This indicates that the file contains potentially important or confidential information.
- Another file, /test1 Copy.txt, was accessed but the returned message explicitly states: "This data is not relevant to the crime at all!!" suggesting it may be used to mislead investigators or serve as a decoy.

Tools/Techniques Used:

- Protocol Hierarchy → Identify that HTTP traffic is present.
- Follow TCP Stream → Extract readable data from HTTP responses.
- Used display filter: http to isolate HTTP-based communication.

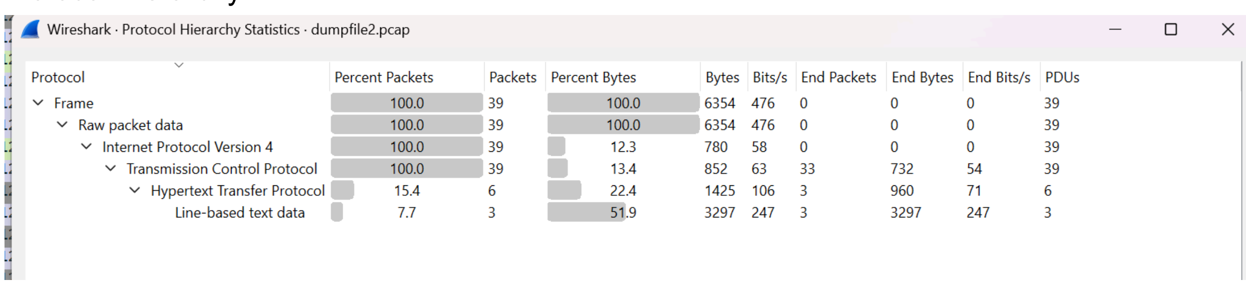
These findings suggest at least one file (test1.txt) may be pertinent to a digital crime or data exfiltration attempt, while another (test1 Copy.txt) likely serves as misdirection.

## 2. Analyzing dumpfile2.pcap in Wireshark:



| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52507 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 2 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52507 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 3 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52507 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 4 | 0.000000 | 127.0.0.1 | 127.0.0.1 | HTTP | 360 | GET / HTTP/1.1 |
| 5 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52507 [ACK] Seq=1 Ack=321 Win=525568 Len=0 |
| 6 | 0.028957 | 127.0.0.1 | 127.0.0.1 | TCP | 195 | 8000 → 52507 [PSH, ACK] Seq=1 Ack=321 Win=525568 Len=155 [TCP PDU reassembled in 8] |
| 7 | 0.028957 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52507 → 8000 [ACK] Seq=321 Ack=156 Win=261984 Len=0 |
| 8 | 0.028957 | 127.0.0.1 | 127.0.0.1 | HTTP | 1062 | HTTP/1.0 200 OK (text/html) |
| 9 | 0.028957 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52507 → 8000 [ACK] Seq=321 Ack=1178 Win=260960 Len=0 |
| 10 | 0.028957 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52507 [FIN, ACK] Seq=1178 Ack=321 Win=525568 Len=0 |
| 11 | 0.028957 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52507 → 8000 [ACK] Seq=321 Ack=1179 Win=260960 Len=0 |
| 12 | 0.028957 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52507 → 8000 [FIN, ACK] Seq=321 Ack=1179 Win=260960 Len=0 |
| 13 | 0.028957 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52507 [ACK] Seq=1179 Ack=322 Win=525568 Len=0 |
| 14 | 61.703238 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52509 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 15 | 61.703238 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52509 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 16 | 61.703238 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52509 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 17 | 61.706388 | 127.0.0.1 | 127.0.0.1 | HTTP | 360 | GET / HTTP/1.1 |
| 18 | 61.706388 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52509 [ACK] Seq=1 Ack=321 Win=525568 Len=0 |
| 19 | 61.758605 | 127.0.0.1 | 127.0.0.1 | TCP | 195 | 8000 → 52509 [PSH, ACK] Seq=1 Ack=321 Win=525568 Len=155 [TCP PDU reassembled in 21] |
| 20 | 61.758605 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52509 → 8000 [ACK] Seq=321 Ack=156 Win=261984 Len=0 |
| 21 | 61.758605 | 127.0.0.1 | 127.0.0.1 | HTTP | 1148 | HTTP/1.0 200 OK (text/html) |
| 22 | 61.758605 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52509 → 8000 [ACK] Seq=321 Ack=1264 Win=260880 Len=0 |
| 23 | 61.758605 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52509 [FIN, ACK] Seq=1264 Ack=321 Win=525568 Len=0 |
| 24 | 61.758605 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52509 → 8000 [ACK] Seq=321 Ack=1265 Win=260880 Len=0 |
| 25 | 61.772825 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52509 → 8000 [FIN, ACK] Seq=321 Ack=1265 Win=260880 Len=0 |
| 26 | 61.772825 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52509 [ACK] Seq=1265 Ack=322 Win=525568 Len=0 |
| 27 | 106.684588 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52512 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 28 | 106.684588 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52512 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 29 | 106.684588 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52512 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 30 | 106.684588 | 127.0.0.1 | 127.0.0.1 | HTTP | 360 | GET / HTTP/1.1 |
| 31 | 106.684588 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52512 [ACK] Seq=1 Ack=321 Win=525568 Len=0 |
| 32 | 106.731825 | 127.0.0.1 | 127.0.0.1 | TCP | 195 | 8000 → 52512 [PSH, ACK] Seq=1 Ack=321 Win=525568 Len=155 [TCP PDU reassembled in 34] |

Protocol Hierarchy:



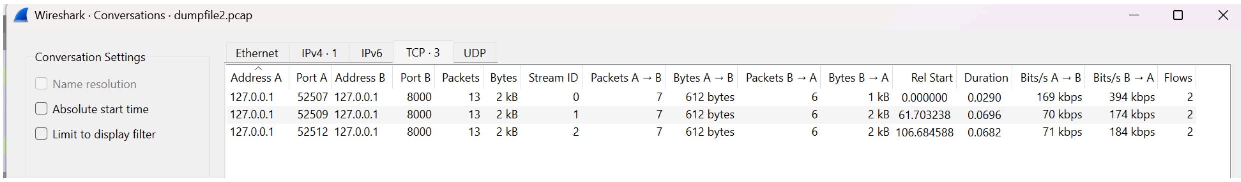
| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 39 | 100.0 | 6354 | 476 | 0 | 0 | 0 | 39 |
| Raw packet data | 100.0 | 39 | 100.0 | 6354 | 476 | 0 | 0 | 0 | 39 |
| Internet Protocol Version 4 | 100.0 | 39 | 12.3 | 780 | 58 | 0 | 0 | 0 | 39 |
| Transmission Control Protocol | 100.0 | 39 | 13.4 | 852 | 63 | 33 | 732 | 54 | 39 |
| Hypertext Transfer Protocol | 15.4 | 6 | 22.4 | 1425 | 106 | 3 | 960 | 71 | 6 |
| Line-based text data | 7.7 | 3 | 51.9 | 3297 | 247 | 3 | 3297 | 247 | 3 |

Figure 1: Protocol Hierarchy statistics of dumpfile2.pcap revealing the packet structure. Notably, 15.4% of packets were HTTP traffic and 7.7% were line-based text data, indicating possible plaintext communication.

Summary:

- Total packets: 39
- Protocols observed: IPv4 > TCP > HTTP and Line-based text data
- Majority of HTTP payload appears to be plaintext, which may include readable or encoded messages.
- TCP stream inspection is likely to yield file contents or text exchanges.

Conversations:



Wireshark · Conversations · dumpfile2.pcap

Conversation Settings
Name resolution
Absolute start time
Limit to display filter

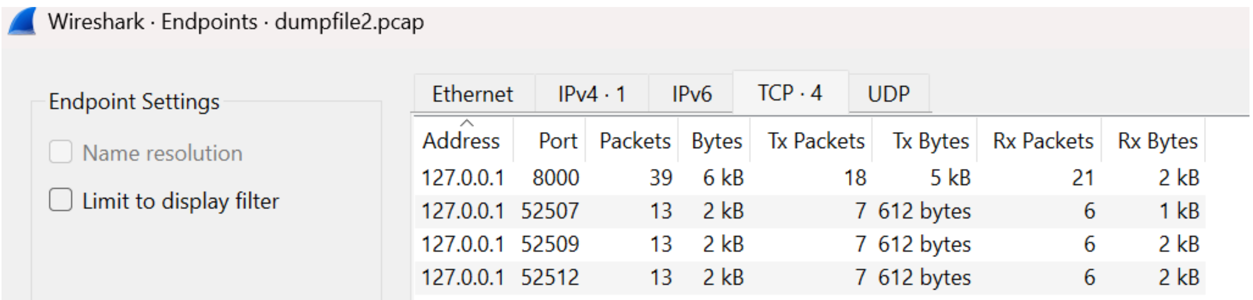
Ethernet | IPv4 · 1 | IPv6 | TCP · 3 | UDP

| Address A | Port A | Address B | Port B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A | Flows |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 52507 | 127.0.0.1 | 8000 | 13 | 2 kB | 0 | 7 | 612 bytes | 6 | 1 kB | 0.000000 | 0.0290 | 169 kbps | 394 kbps | 2 |
| 127.0.0.1 | 52509 | 127.0.0.1 | 8000 | 13 | 2 kB | 1 | 7 | 612 bytes | 6 | 2 kB | 61.703238 | 0.0696 | 70 kbps | 174 kbps | 2 |
| 127.0.0.1 | 52512 | 127.0.0.1 | 8000 | 13 | 2 kB | 2 | 7 | 612 bytes | 6 | 2 kB | 106.684588 | 0.0682 | 71 kbps | 184 kbps | 2 |

Figure 2: TCP Conversations across 3 stream connections (ports 52507, 52509, 52512), each communicating with port 8000.

Summary:
- 3 separate TCP conversations using different ephemeral ports.
- Approx. 2KB data per stream, suggesting multiple HTTP requests or responses.
- Streams are short-lived but have readable payloads (based on previous dumpfile behavior, this should be examined further).

Endpoints:



Wireshark · Endpoints · dumpfile2.pcap

Endpoint Settings
Name resolution
Limit to display filter

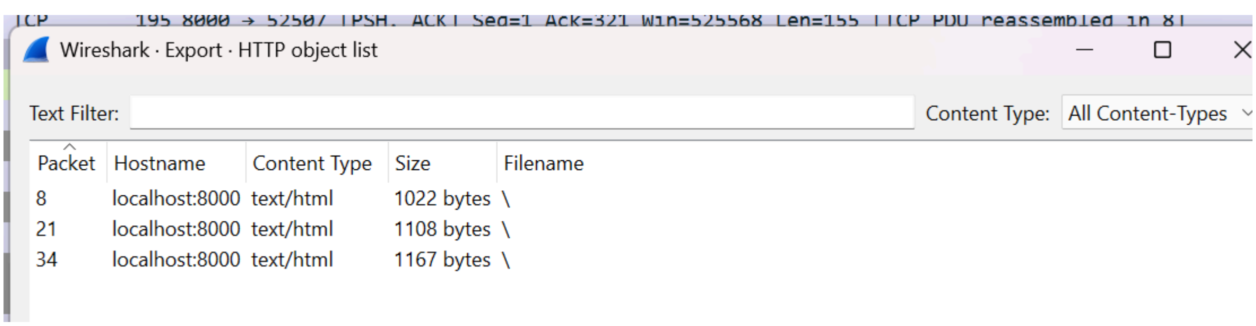
Ethernet | IPv4 · 1 | IPv6 | TCP · 4 | UDP

| Address | Port | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 8000 | 39 | 6 kB | 18 | 5 kB | 21 | 2 kB |
| 127.0.0.1 | 52507 | 13 | 2 kB | 7 | 612 bytes | 6 | 1 kB |
| 127.0.0.1 | 52509 | 13 | 2 kB | 7 | 612 bytes | 6 | 2 kB |
| 127.0.0.1 | 52512 | 13 | 2 kB | 7 | 612 bytes | 6 | 2 kB |

Figure 3: Endpoint statistics showing 127.0.0.1 communicating across multiple ports, consistently interacting with port 8000 as the host web service.

Summary:
- Single host communication (localhost).
- Port 8000 is the web service endpoint; the others are clients.
- Repeated access across different TCP ports likely indicates multiple HTTP GET requests.

Wireshark Export results:



TCP 195 8000 → 52507 [PSH, ACK] Seq=1 Ack=321 Win=525568 Len=155 [TCP PDU reassembled in 8]

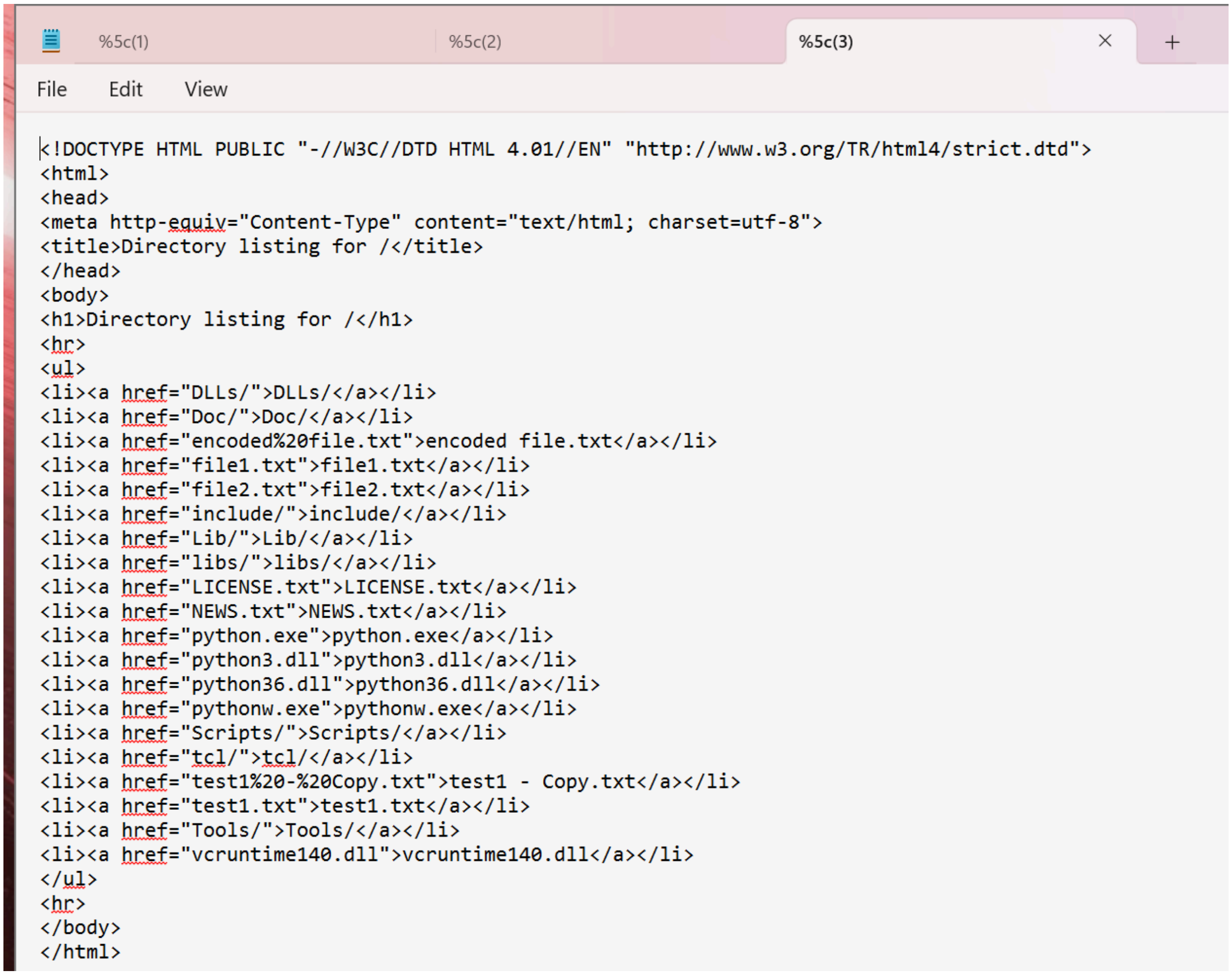
Wireshark · Export · HTTP object list

Text Filter: Content Type: All Content-Types

| Packet | Hostname | Content Type | Size | Filename |
|---|---|---|---|---|
| 8 | localhost:8000 | text/html | 1022 bytes | \ |
| 21 | localhost:8000 | text/html | 1108 bytes | \ |
| 34 | localhost:8000 | text/html | 1167 bytes | \ |

File -> Export Objects -> HTTP:

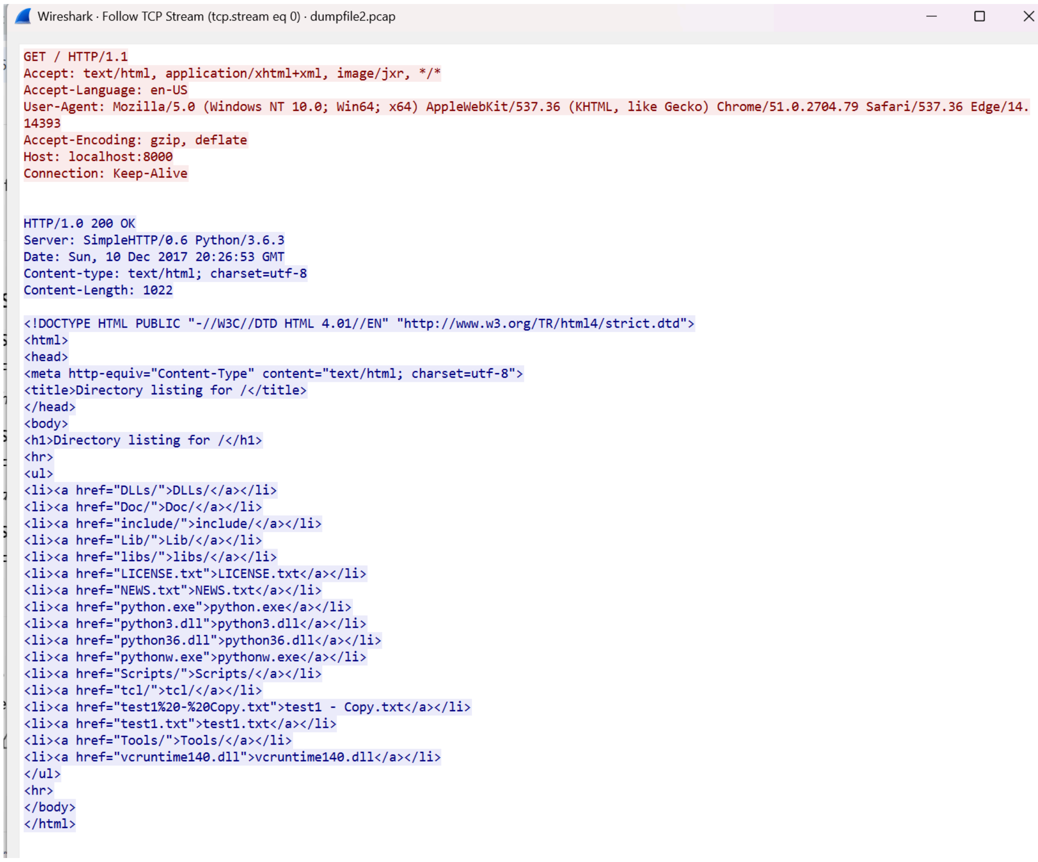
This is the result I obtained




```html
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="encoded%20file.txt">encoded file.txt</a></li>
<li><a href="file1.txt">file1.txt</a></li>
<li><a href="file2.txt">file2.txt</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
</html>
```

TCP Stream 0:

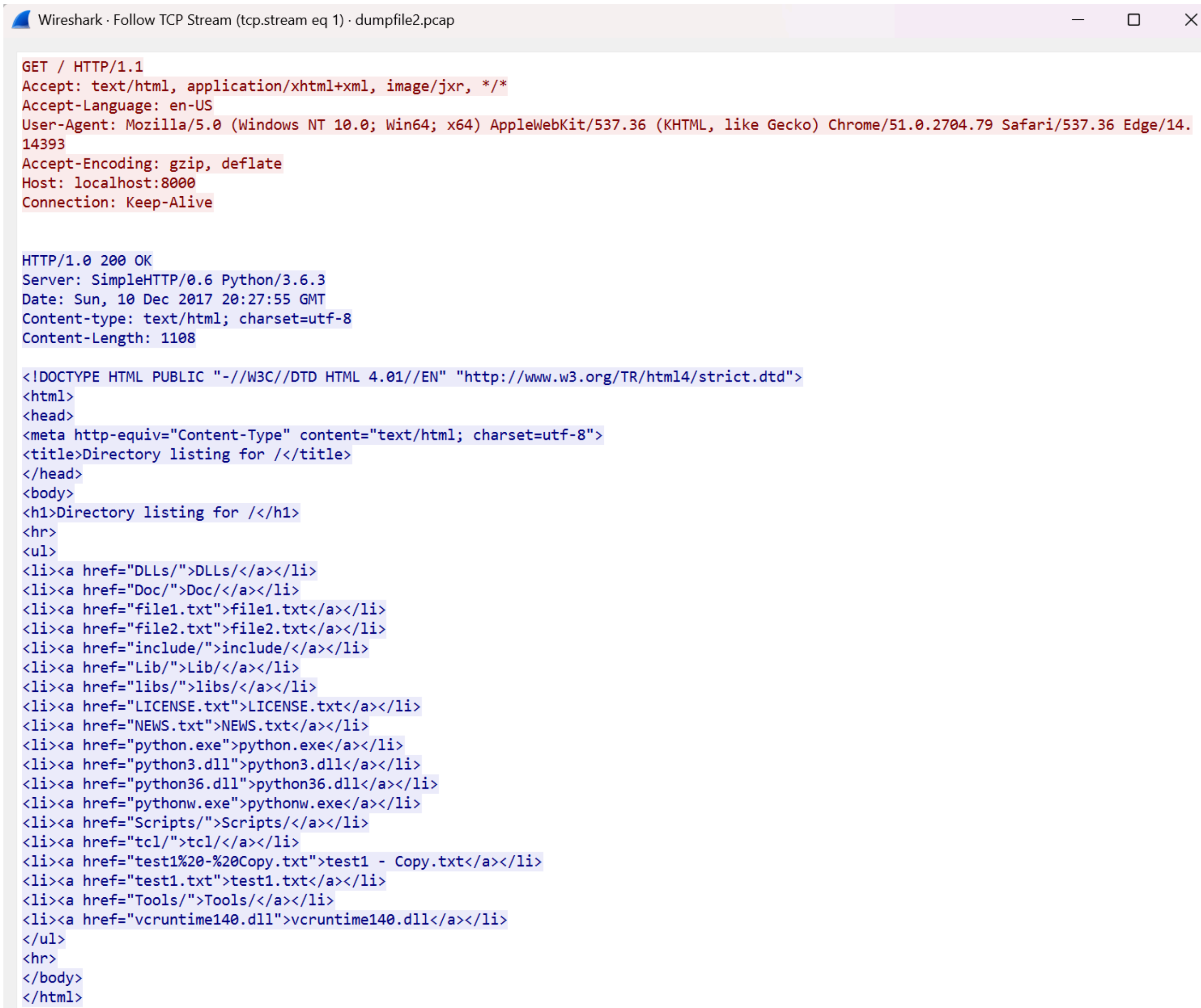
Screenshot Caption: TCP Stream 0 shows directory listing including files like LICENSE.txt, python.exe, and test1.txt.


Wireshark · Follow TCP Stream (tcp.stream eq 0) · dumpfile2.pcap

```
GET / HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.
14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:26:53 GMT
Content-type: text/html; charset=utf-8
Content-Length: 1022

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
</html>
```

TCP Stream 1:

Screenshot Caption: TCP Stream 1 shows a similar directory listing, but now includes file1.txt and file2.txt.



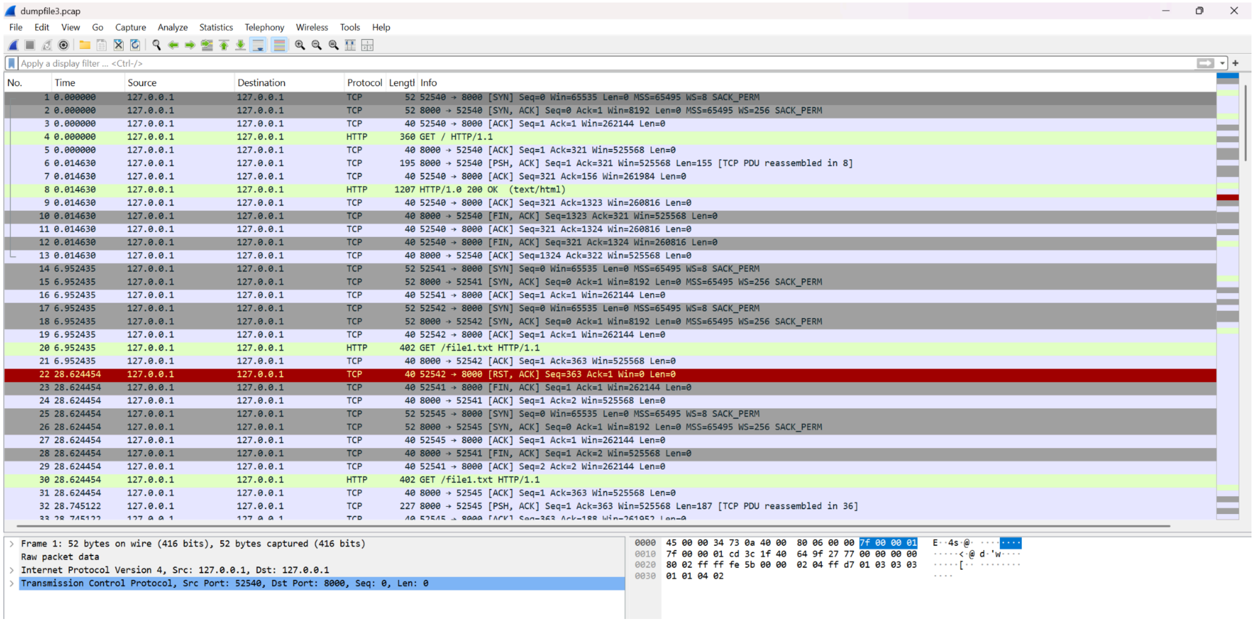
Wireshark · Follow TCP Stream (tcp.stream eq 1) · dumpfile2.pcap

```
GET / HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.
14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:27:55 GMT
Content-type: text/html; charset=utf-8
Content-Length: 1108

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="file1.txt">file1.txt</a></li>
<li><a href="file2.txt">file2.txt</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
</html>
```

**Key Findings:**

- Multiple HTTP GET requests were made to `localhost:8000`, each returning directory listings.
- These listings show potentially sensitive filenames like `python.exe`, `NEWS.txt`, `LICENSE.txt`, `file1.txt`, `file2.txt`, etc.
- This suggests that a web server was active and serving a full directory tree — possibly exposing sensitive files.
- This could indicate **directory traversal** exposure, **lack of access control**, or **accidental directory indexing**.

## 3. Analyzing dumpfile3.pcap in Wireshark:




## Protocol Hierarchy:

Wireshark · Protocol Hierarchy Statistics · dumpfile3.pcap

| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 77 | 100.0 | 39912 | 6541 | 0 | 0 | 0 | 77 |
| Raw packet data | 100.0 | 77 | 100.0 | 39912 | 6541 | 0 | 0 | 0 | 77 |
| Internet Protocol Version 4 | 100.0 | 77 | 3.9 | 1540 | 252 | 0 | 0 | 0 | 77 |
| Transmission Control Protocol | 100.0 | 77 | 4.2 | 1660 | 272 | 70 | 1520 | 249 | 77 |
| Hypertext Transfer Protocol | 9.1 | 7 | 4.9 | 1938 | 317 | 4 | 1408 | 230 | 7 |
| Line-based text data | 3.9 | 3 | 87.1 | 34774 | 5699 | 3 | 34774 | 5699 | 3 |

## Conversations:

Wireshark · Conversations · dumpfile3.pcap

Conversation Settings
- Name resolution
- Absolute start time
- Limit to display filter

Ethernet | IPv4 · 1 | IPv6 | TCP · 5 | UDP

| Address A | Port A | Address B | Port B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A | Flows |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 52540 | 127.0.0.1 | 8000 | 13 | 2 kB | 0 | 7 | 612 bytes | 6 | 2 kB | 0.000000 | 0.0146 | 334 kbps | 860 kbps | 2 |
| 127.0.0.1 | 52541 | 127.0.0.1 | 8000 | 7 | 304 bytes | 1 | 4 | 172 bytes | 3 | 132 bytes | 6.952435 | 21.6720 | 63 bits/s | 48 bits/s | 0 |
| 127.0.0.1 | 52542 | 127.0.0.1 | 8000 | 6 | 626 bytes | 2 | 4 | 534 bytes | 2 | 92 bytes | 6.952435 | 21.6720 | 197 bits/s | 33 bits/s | 1 |
| 127.0.0.1 | 52545 | 127.0.0.1 | 8000 | 15 | 4 kB | 3 | 7 | 654 bytes | 8 | 4 kB | 28.624454 | 0.1246 | 41 kbps | 243 kbps | 2 |
| 127.0.0.1 | 52546 | 127.0.0.1 | 8000 | 36 | 32 kB | 4 | 9 | 736 bytes | 27 | 32 kB | 43.624080 | 5.1900 | 1134 bits/s | 48 kbps | 2 |

## Endpoints:

Wireshark · Endpoints · dumpfile3.pcap

Endpoint Settings
- Name resolution
- Limit to display filter

Ethernet | IPv4 · 1 | IPv6 | TCP · 6 | UDP

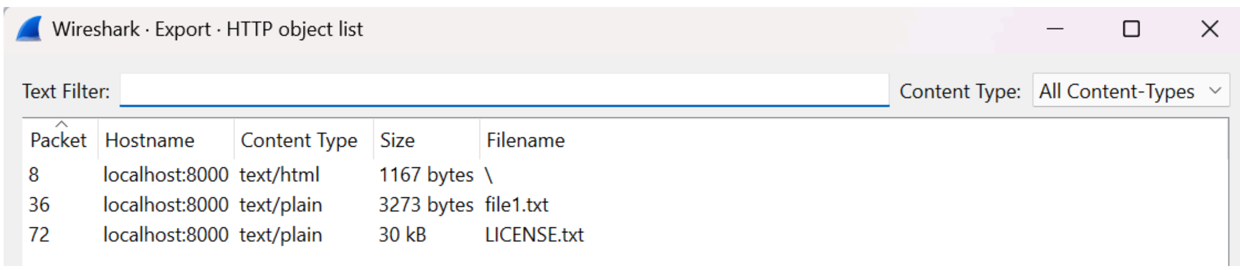
| Address | Port | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 8000 | 77 | 40 kB | 46 | 37 kB | 31 | 3 kB |
| 127.0.0.1 | 52540 | 13 | 2 kB | 7 | 612 bytes | 6 | 2 kB |
| 127.0.0.1 | 52541 | 7 | 304 bytes | 4 | 172 bytes | 3 | 132 bytes |
| 127.0.0.1 | 52542 | 6 | 626 bytes | 4 | 534 bytes | 2 | 92 bytes |
| 127.0.0.1 | 52545 | 15 | 4 kB | 7 | 654 bytes | 8 | 4 kB |
| 127.0.0.1 | 52546 | 36 | 32 kB | 9 | 736 bytes | 27 | 32 kB |

**Summarize Key Activity**

Use the **Protocol Hierarchy**, **Conversations**, and **Endpoints** windows to summarize what kind of traffic is happening:

- **HTTP and Line-based Text Detected**: This indicates a local web server is serving content.
- Most traffic is over **TCP** to/from `127.0.0.1` on port `8000`.
- Notable HTTP requests (e.g., `GET /file1.txt`, `GET /test1.txt`) and responses with content like `402 GET /file1.txt` and some `RST` (Reset) packets indicate access attempts and rejection.

HTTP Export List:



Text export outputs:

%5c    file1.txt

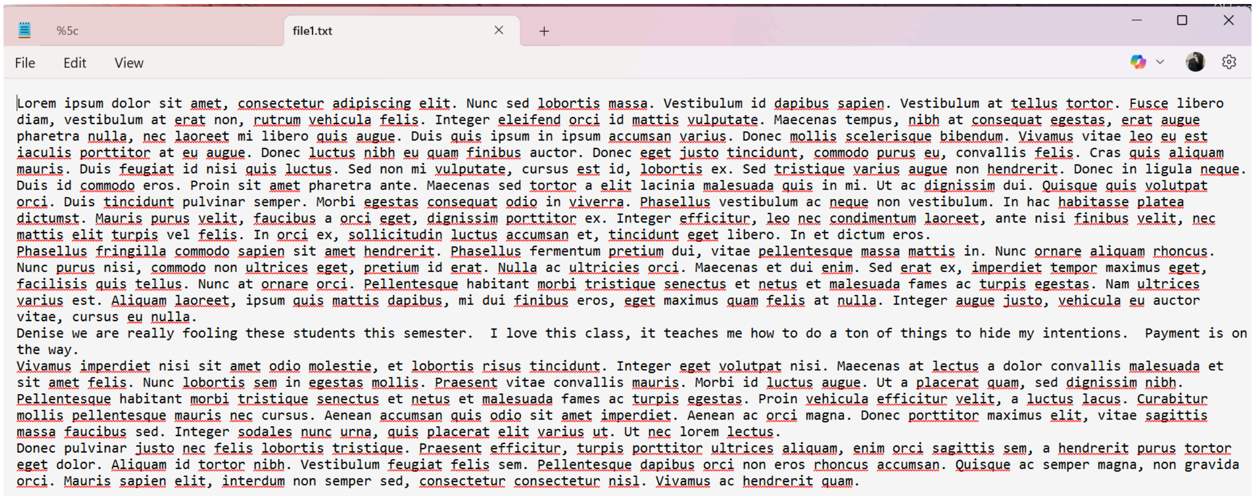
File    Edit    View

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="encoded%20file.txt">encoded file.txt</a></li>
<li><a href="file1.txt">file1.txt</a></li>
<li><a href="file2.txt">file2.txt</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
</html>
```

HTML content from `file1.txt` reveals a file directory listing, showing several `.txt`, `.dll`, and `.exe` files.

**Interpretation:**
This suggests the server was exposing its internal directory structure, which could be exploited for further access or reconnaissance. Multiple files (e.g., `file1.txt`, `file2.txt`, `LICENSE.txt`) appear suspicious and worth deeper inspection.

file1.txt:



%5c
file1.txt
File Edit View

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc sed lobortis massa. Vestibulum id dapibus sapien. Vestibulum at tellus tortor. Fusce libero diam, vestibulum at erat non, rutrum vehicula felis. Integer eleifend orci id mattis vulputate. Maecenas tempus, nibh at consequat egestas, erat augue pharetra nulla, nec laoreet mi libero quis augue. Duis quis ipsum in ipsum accumsan varius. Donec mollis scelerisque bibendum. Vivamus vitae leo eu est iaculis porttitor at eu augue. Donec luctus nibh eu quam finibus auctor. Donec eget justo tincidunt, commodo purus eu, convallis felis. Cras quis aliquam mauris. Duis feugiat id nisi quis luctus. Sed non mi vulputate, cursus est id, lobortis ex. Sed tristique varius augue non hendrerit. Donec in ligula neque. Duis id commodo eros. Proin sit amet pharetra ante. Maecenas sed tortor a elit lacinia malesuada quis in mi. Ut ac dignissim dui. Quisque quis volutpat orci. Duis tincidunt pulvinar semper. Morbi egestas consequat odio in viverra. Phasellus vestibulum ac neque non vestibulum. In hac habitasse platea dictumst. Mauris purus velit, faucibus a orci eget, dignissim porttitor ex. Integer efficitur, leo nec condimentum laoreet, ante nisi finibus velit, nec mattis elit turpis vel felis. In orci ex, sollicitudin luctus accumsan et, tincidunt eget libero. In et dictum eros.
Phasellus fringilla commodo sapien sit amet hendrerit. Phasellus fermentum pretium dui, vitae pellentesque massa mattis in. Nunc ornare aliquam rhoncus. Nunc purus nisi, commodo non ultrices eget, pretium id erat. Nulla ac ultricies orci. Maecenas et dui enim. Sed erat ex, imperdiet tempor maximus eget, facilisis quis tellus. Nunc at ornare orci. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nam ultrices varius est. Aliquam laoreet, ipsum quis mattis dapibus, mi dui finibus eros, eget maximus quam felis at nulla. Integer augue justo, vehicula eu auctor vitae, cursus eu nulla.
Denise we are really fooling these students this semester.  I love this class, it teaches me how to do a ton of things to hide my intentions.  Payment is on the way.
Vivamus imperdiet nisi sit amet odio molestie, et lobortis risus tincidunt. Integer eget volutpat nisi. Maecenas at lectus a dolor convallis malesuada et sit amet felis. Nunc lobortis sem in egestas mollis. Praesent vitae convallis mauris. Morbi id luctus augue. Ut a placerat quam, sed dignissim nibh. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Proin vehicula efficitur velit, a luctus lacus. Curabitur mollis pellentesque mauris nec cursus. Aenean accumsan quis odio sit amet imperdiet. Aenean ac orci magna. Donec porttitor maximus elit, vitae sagittis massa faucibus sed. Integer sodales nunc urna, quis placerat elit varius ut. Ut nec lorem lectus.
Donec pulvinar justo nec felis lobortis tristique. Praesent efficitur, turpis porttitor ultrices aliquam, enim orci sagittis sem, a hendrerit purus tortor eget dolor. Aliquam id tortor nibh. Vestibulum feugiat felis sem. Pellentesque dapibus orci non eros rhoncus accumsan. Quisque ac semper magna, non gravida orci. Mauris sapien elit, interdum non semper sed, consectetur consectetur nisl. Vivamus ac hendrerit quam.

The plain text content of `file1.txt` appears as typical lorem ipsum text—but a sentence stands out:

> “Denise we are really fooling these students this semester. I love this class, it teaches me how to do a ton of things to hide my intentions.”

**Interpretation:**
That embedded line suggests hidden communication. This is a critical piece of evidence implying intentional obfuscation in otherwise benign data.

LICENSE.txt:

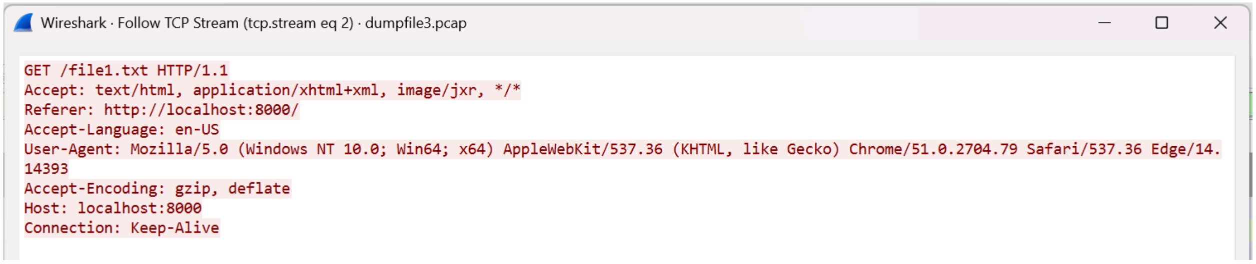
LICENSE.txt

File Edit View

```
A. HISTORY OF THE SOFTWARE
==========================

Python was created in the early 1990s by Guido van Rossum at Stichting
Mathematisch Centrum (CWI, see http://www.cwi.nl) in the Netherlands
as a successor of a language called ABC.  Guido remains Python's
principal author, although it includes many contributions from others.

In 1995, Guido continued his work on Python at the Corporation for
National Research Initiatives (CNRI, see http://www.cnri.reston.va.us)
in Reston, Virginia where he released several versions of the
software.

In May 2000, Guido and the Python core development team moved to
BeOpen.com to form the BeOpen PythonLabs team.  In October of the same
year, the PythonLabs team moved to Digital Creations, which became
Zope Corporation.  In 2001, the Python Software Foundation (PSF, see
https://www.python.org/psf/) was formed, a non-profit organization
created specifically to own Python-related Intellectual Property.
Zope Corporation was a sponsoring member of the PSF.

All Python releases are Open Source (see http://www.opensource.org for
the Open Source Definition).  Historically, most, but not all, Python
releases have also been GPL-compatible; the table below summarizes
the various releases.

    Release         Derived     Year        Owner       GPL-
                    from                                compatible? (1)

    0.9.0 thru 1.2              1991-1995   CWI         yes
    1.3 thru 1.5.2  1.2         1995-1999   CNRI        yes
    1.6             1.5.2       2000        CNRI        no
    2.0             1.6         2000        BeOpen.com  no
    1.6.1           1.6         2001        CNRI        yes (2)
    2.1             2.0+1.6.1   2001        PSF         no
    2.0.1           2.0+1.6.1   2001        PSF         yes
    2.1.1           2.1+2.0.1   2001        PSF         yes
    2.1.2           2.1.1       2002        PSF         yes
    2.1.3           2.1.2       2002        PSF         yes
    2.2 and above   2.1.1       2001-now    PSF         yes

Footnotes:

(1) GPL-compatible doesn't mean that we're distributing Python under
    the GPL.  All Python licenses, unlike the GPL, let you distribute
    a modified version without making your changes open source.  The
    GPL-compatible licenses make it possible to combine Python with
    other software that is released under the GPL; the others don't.

(2) According to Richard Stallman, 1.6.1 is not GPL-compatible,
    because its license has a choice of law clause.  According to
    CNRI, however, Stallman's lawyer has told CNRI's lawyer that 1.6.1
    is "not incompatible" with the GPL.
```

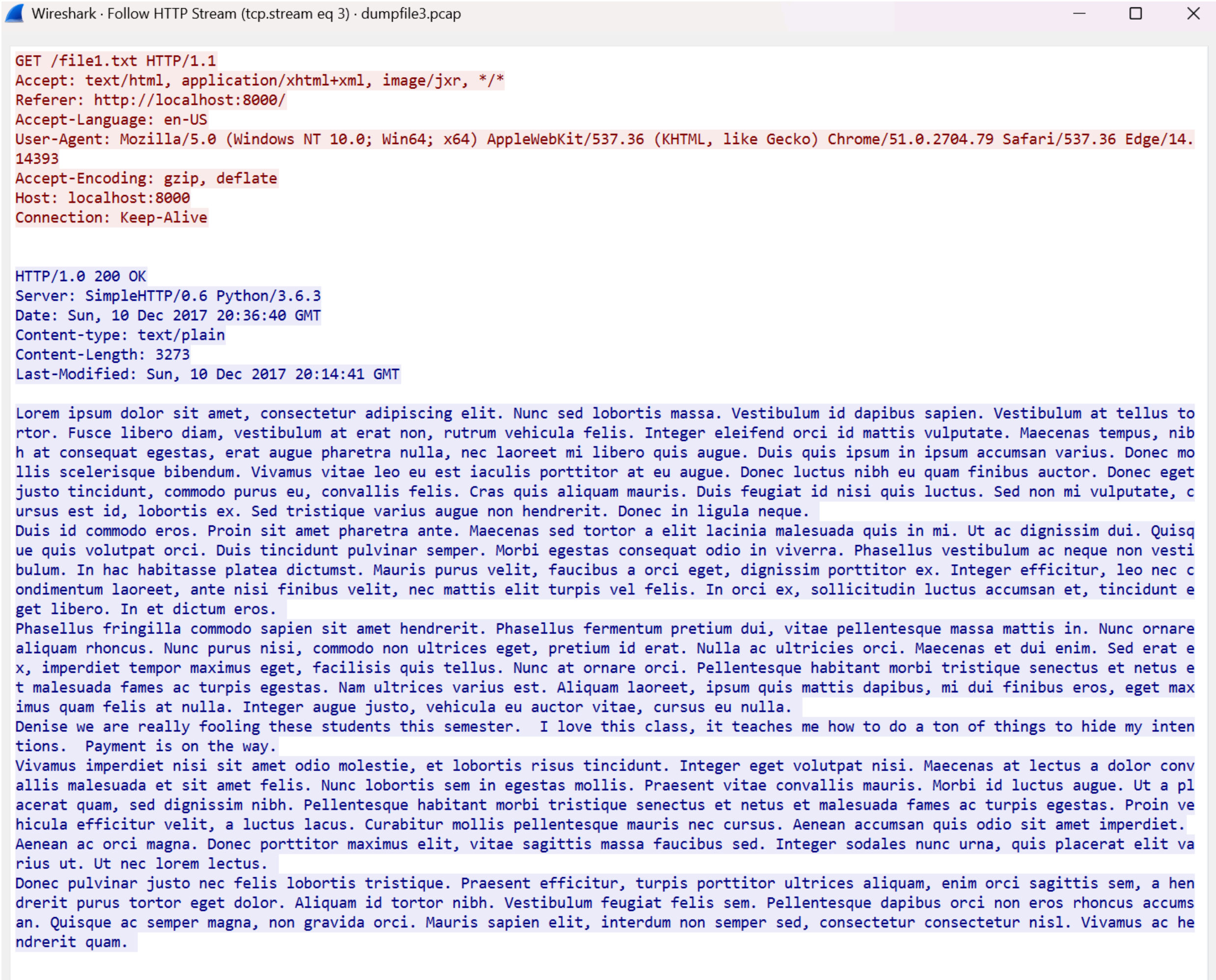
The file `LICENSE.txt` contains information about Python's open-source license and history. This appears to be a standard software license file.

**Interpretation:**
No apparent relevance to the crime—likely a distraction or normal file included during file dumps.




**TCP Stream 2: Request for file1.txt :**
TCP Stream shows the HTTP GET request to download file1.txt from localhost:8000. This logs the moment the file was accessed by a user.

Interpretation:
This confirms the file was actively retrieved, indicating user interest or interaction with a possibly hidden file.



Wireshark · Follow HTTP Stream (tcp.stream eq 3) · dumpfile3.pcap

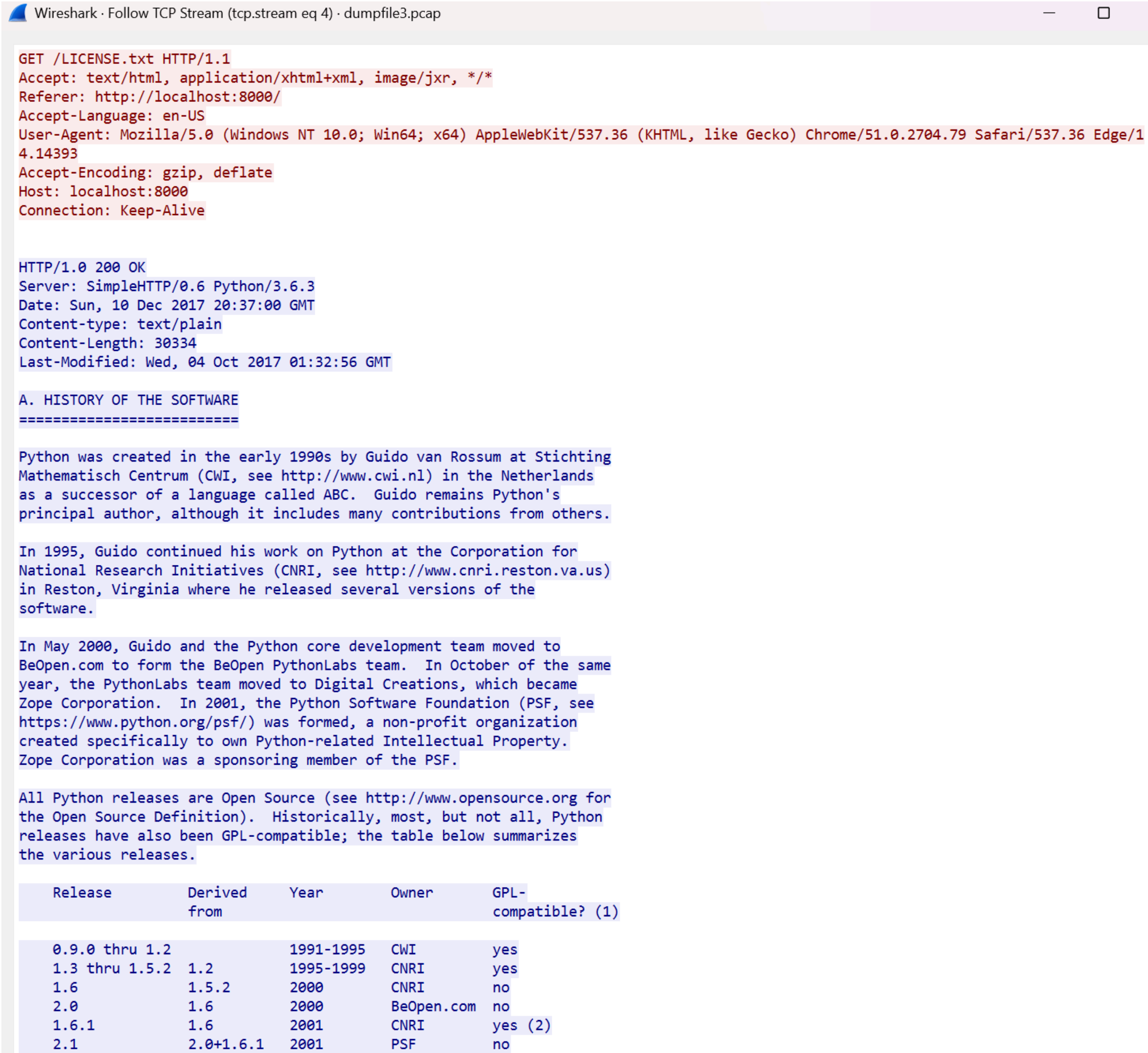
```
GET /file1.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.
14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:36:40 GMT
Content-type: text/plain
Content-Length: 3273
Last-Modified: Sun, 10 Dec 2017 20:14:41 GMT

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc sed lobortis massa. Vestibulum id dapibus sapien. Vestibulum at tellus to
rtor. Fusce libero diam, vestibulum at erat non, rutrum vehicula felis. Integer eleifend orci id mattis vulputate. Maecenas tempus, nib
h at consequat egestas, erat augue pharetra nulla, nec laoreet mi libero quis augue. Duis quis ipsum in ipsum accumsan varius. Donec mo
llis scelerisque bibendum. Vivamus vitae leo eu est iaculis porttitor at eu augue. Donec luctus nibh eu quam finibus auctor. Donec eget
justo tincidunt, commodo purus eu, convallis felis. Cras quis aliquam mauris. Duis feugiat id nisi quis luctus. Sed non mi vulputate, c
ursus est id, lobortis ex. Sed tristique varius augue non hendrerit. Donec in ligula neque.
Duis id commodo eros. Proin sit amet pharetra ante. Maecenas sed tortor a elit lacinia malesuada quis in mi. Ut ac dignissim dui. Quisq
ue quis volutpat orci. Duis tincidunt pulvinar semper. Morbi egestas consequat odio in viverra. Phasellus vestibulum ac neque non vesti
bulum. In hac habitasse platea dictumst. Mauris purus velit, faucibus a orci eget, dignissim porttitor ex. Integer efficitur, leo nec c
ondimentum laoreet, ante nisi finibus velit, nec mattis elit turpis vel felis. In orci ex, sollicitudin luctus accumsan et, tincidunt e
get libero. In et dictum eros.
Phasellus fringilla commodo sapien sit amet hendrerit. Phasellus fermentum pretium dui, vitae pellentesque massa mattis in. Nunc ornare
aliquam rhoncus. Nunc purus nisi, commodo non ultrices eget, pretium id erat. Nulla ac ultricies orci. Maecenas et dui enim. Sed erat e
x, imperdiet tempor maximus eget, facilisis quis tellus. Nunc at ornare orci. Pellentesque habitant morbi tristique senectus et netus e
t malesuada fames ac turpis egestas. Nam ultrices varius est. Aliquam laoreet, ipsum quis mattis dapibus, mi dui finibus eros, eget max
imus quam felis at nulla. Integer augue justo, vehicula eu auctor vitae, cursus eu nulla.
Denise we are really fooling these students this semester.  I love this class, it teaches me how to do a ton of things to hide my inten
tions.  Payment is on the way.
Vivamus imperdiet nisi sit amet odio molestie, et lobortis risus tincidunt. Integer eget volutpat nisi. Maecenas at lectus a dolor conv
allis malesuada et sit amet felis. Nunc lobortis sem in egestas mollis. Praesent vitae convallis mauris. Morbi id luctus augue. Ut a pl
acerat quam, sed dignissim nibh. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Proin ve
hicula efficitur velit, a luctus lacus. Curabitur mollis pellentesque mauris nec cursus. Aenean accumsan quis odio sit amet imperdiet.
Aenean ac orci magna. Donec porttitor maximus elit, vitae sagittis massa faucibus sed. Integer sodales nunc urna, quis placerat elit va
rius ut. Ut nec lorem lectus.
Donec pulvinar justo nec felis lobortis tristique. Praesent efficitur, turpis porttitor ultrices aliquam, enim orci sagittis sem, a hen
drerit purus tortor eget dolor. Aliquam id tortor nibh. Vestibulum feugiat felis sem. Pellentesque dapibus orci non eros rhoncus accums
an. Quisque ac semper magna, non gravida orci. Mauris sapien elit, interdum non semper sed, consectetur consectetur nisl. Vivamus ac he
ndrerit quam.
```

**TCP Stream 3: file1.txt response:**

Response from localhost:8000 serving file1.txt, including all the suspiciously embedded text.

Interpretation:

This reveals the full file transmission. The sentence hidden within the filler text supports that this file was used to smuggle communication.



Wireshark · Follow TCP Stream (tcp.stream eq 4) · dumpfile3.pcap

```
GET /LICENSE.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/1
4.14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:37:00 GMT
Content-type: text/plain
Content-Length: 30334
Last-Modified: Wed, 04 Oct 2017 01:32:56 GMT

A. HISTORY OF THE SOFTWARE
==========================

Python was created in the early 1990s by Guido van Rossum at Stichting
Mathematisch Centrum (CWI, see http://www.cwi.nl) in the Netherlands
as a successor of a language called ABC.  Guido remains Python's
principal author, although it includes many contributions from others.

In 1995, Guido continued his work on Python at the Corporation for
National Research Initiatives (CNRI, see http://www.cnri.reston.va.us)
in Reston, Virginia where he released several versions of the
software.

In May 2000, Guido and the Python core development team moved to
BeOpen.com to form the BeOpen PythonLabs team.  In October of the same
year, the PythonLabs team moved to Digital Creations, which became
Zope Corporation.  In 2001, the Python Software Foundation (PSF, see
https://www.python.org/psf/) was formed, a non-profit organization
created specifically to own Python-related Intellectual Property.
Zope Corporation was a sponsoring member of the PSF.

All Python releases are Open Source (see http://www.opensource.org for
the Open Source Definition).  Historically, most, but not all, Python
releases have also been GPL-compatible; the table below summarizes
the various releases.

    Release         Derived     Year        Owner       GPL-
                    from                                compatible? (1)

    0.9.0 thru 1.2              1991-1995   CWI         yes
    1.3 thru 1.5.2  1.2         1995-1999   CNRI        yes
    1.6             1.5.2       2000        CNRI        no
    2.0             1.6         2000        BeOpen.com  no
    1.6.1           1.6         2001        CNRI        yes (2)
    2.1             2.0+1.6.1   2001        PSF         no
```

**TCP Stream 4: LICENSE.txt download:**

Final TCP stream shows the GET /LICENSE.txt request and response, serving standard license text.

Interpretation:

Confirms LICENSE.txt was retrieved but contains no suspicious content. Included to mirror legitimate behavior and avoid suspicion.

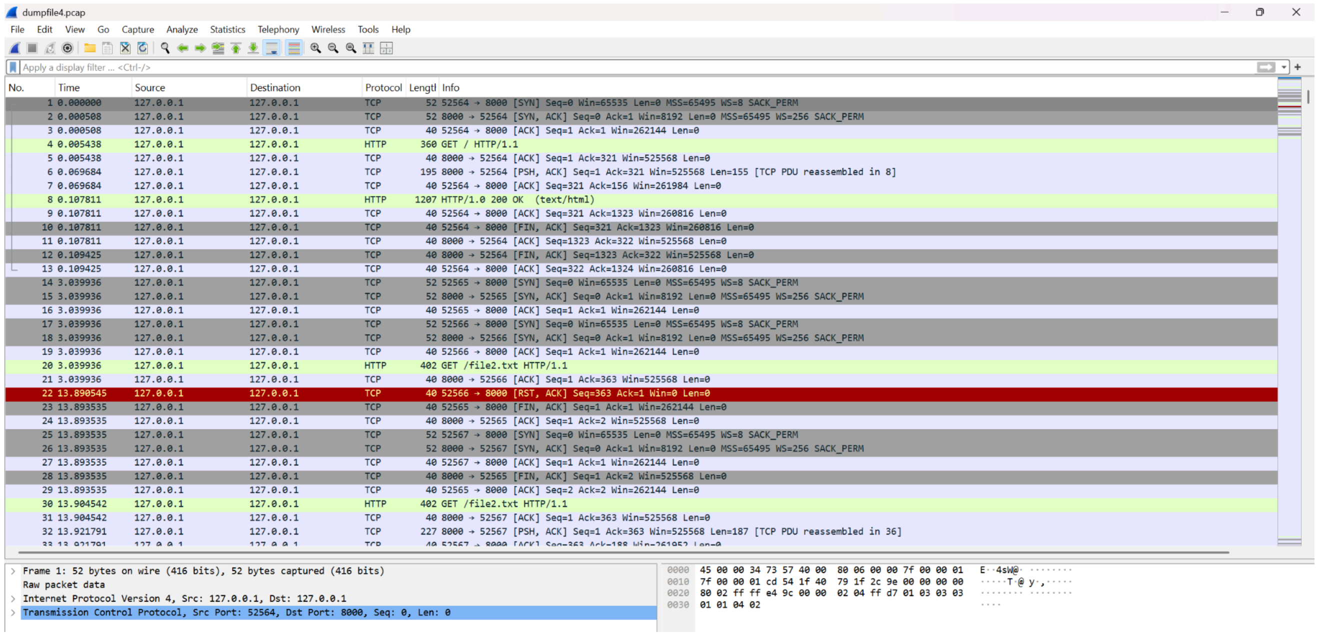
## Summary of Findings – Dumpfile 3

In the analysis of `dumpfile3.pcap`, several HTTP file transfers were observed between the local web server (`127.0.0.1:8000`) and a client. Key findings include:

- **Three HTTP objects were exported**: an HTML file (`\`), `file1.txt`, and `LICENSE.txt`.
- The **HTML directory listing** from the root page (`file1.txt`) revealed multiple internal files, including potentially suspicious `.txt` files like `encoded file.txt`, `test1.txt`, and `NEWS.txt`.
- **`file1.txt` contained hidden communication** embedded within filler text. The sentence *"Denise we are really fooling these students this semester..."* indicates intent to conceal a message.
- **`LICENSE.txt`** was a legitimate open-source license document and is likely not related to the suspicious activity.
- TCP stream analysis confirmed that the files were actively retrieved and served over HTTP.
- The large byte volume (87.1% of traffic) under "line-based text data" aligns with the large plaintext transfer in `file1.txt`.

**Conclusion**:
`file1.txt` is the most relevant artifact, containing covert communication disguised as standard lorem ipsum text. This interaction points to an intentional act of hiding messages within innocuous files on a local web server.

## 4. Analyzing dumpfile4.pcap in Wireshark:



dumpfile4.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

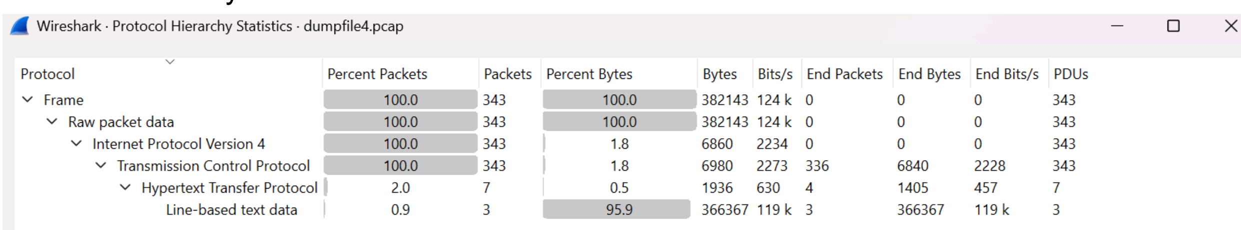
Apply a display filter ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Lengtl | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52564 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 2 | 0.000508 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52564 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 3 | 0.000508 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52564 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 4 | 0.005438 | 127.0.0.1 | 127.0.0.1 | HTTP | 360 | GET / HTTP/1.1 |
| 5 | 0.005438 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52564 [ACK] Seq=1 Ack=321 Win=525568 Len=0 |
| 6 | 0.069684 | 127.0.0.1 | 127.0.0.1 | TCP | 195 | 8000 → 52564 [PSH, ACK] Seq=1 Ack=321 Win=525568 Len=155 [TCP PDU reassembled in 8] |
| 7 | 0.069684 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52564 → 8000 [ACK] Seq=321 Ack=156 Win=261984 Len=0 |
| 8 | 0.107811 | 127.0.0.1 | 127.0.0.1 | HTTP | 1207 | HTTP/1.0 200 OK (text/html) |
| 9 | 0.107811 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52564 → 8000 [ACK] Seq=321 Ack=1323 Win=260816 Len=0 |
| 10 | 0.107811 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52564 → 8000 [FIN, ACK] Seq=321 Ack=1323 Win=260816 Len=0 |
| 11 | 0.107811 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52564 [ACK] Seq=1323 Ack=322 Win=525568 Len=0 |
| 12 | 0.109425 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52564 [FIN, ACK] Seq=1323 Ack=322 Win=525568 Len=0 |
| 13 | 0.109425 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52564 → 8000 [ACK] Seq=322 Ack=1324 Win=260816 Len=0 |
| 14 | 3.039936 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52565 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 15 | 3.039936 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52565 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 16 | 3.039936 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52565 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 17 | 3.039936 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52566 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 18 | 3.039936 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52566 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 19 | 3.039936 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52566 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 20 | 3.039936 | 127.0.0.1 | 127.0.0.1 | HTTP | 402 | GET /file2.txt HTTP/1.1 |
| 21 | 3.039936 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52566 [ACK] Seq=1 Ack=363 Win=525568 Len=0 |
| 22 | 13.890545 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52566 → 8000 [RST, ACK] Seq=363 Ack=1 Win=0 Len=0 |
| 23 | 13.893535 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52565 → 8000 [FIN, ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 24 | 13.893535 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52565 [ACK] Seq=1 Ack=2 Win=525568 Len=0 |
| 25 | 13.893535 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52567 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 26 | 13.893535 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52567 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 27 | 13.893535 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52567 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 28 | 13.893535 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52565 [FIN, ACK] Seq=1 Ack=2 Win=525568 Len=0 |
| 29 | 13.893535 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52565 → 8000 [ACK] Seq=2 Ack=2 Win=262144 Len=0 |
| 30 | 13.904542 | 127.0.0.1 | 127.0.0.1 | HTTP | 402 | GET /file2.txt HTTP/1.1 |
| 31 | 13.904542 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52567 [ACK] Seq=1 Ack=363 Win=525568 Len=0 |
| 32 | 13.921791 | 127.0.0.1 | 127.0.0.1 | TCP | 227 | 8000 → 52567 [PSH, ACK] Seq=1 Ack=363 Win=525568 Len=187 [TCP PDU reassembled in 36] |
| 33 | 13.921791 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52567 → 8000 [ACK] Seq=363 Ack=188 Win=261952 Len=0 |

```
> Frame 1: 52 bytes on wire (416 bits), 52 bytes captured (416 bits)
  Raw packet data
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 52564, Dst Port: 8000, Seq: 0, Len: 0
```

```
0000  45 00 00 34 73 57 40 00  80 06 00 00 7f 00 00 01   E··4sW@· ········
0010  7f 00 00 01 cd 54 1f 40  79 1f 2c 9e 00 00 00 00   ·····T·@ y·,·····
0020  80 02 ff ff e4 9c 00 00  02 04 ff d7 01 03 03 03   ········ ········
0030  01 01 04 02                                        ····
```

Protocol Hierarchy:

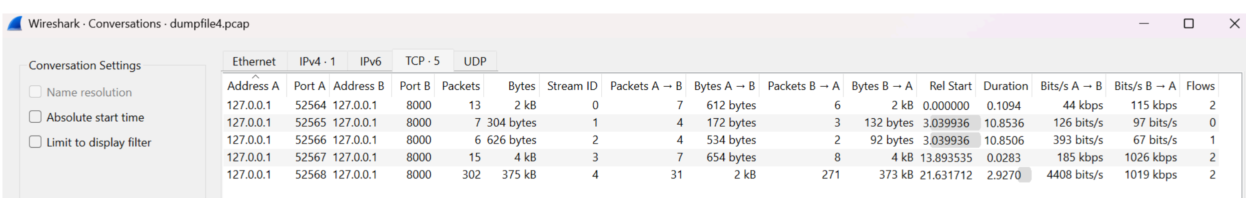


Wireshark · Protocol Hierarchy Statistics · dumpfile4.pcap

| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 343 | 100.0 | 382143 | 124 k | 0 | 0 | 0 | 343 |
| Raw packet data | 100.0 | 343 | 100.0 | 382143 | 124 k | 0 | 0 | 0 | 343 |
| Internet Protocol Version 4 | 100.0 | 343 | 1.8 | 6860 | 2234 | 0 | 0 | 0 | 343 |
| Transmission Control Protocol | 100.0 | 343 | 1.8 | 6980 | 2273 | 336 | 6840 | 2228 | 343 |
| Hypertext Transfer Protocol | 2.0 | 7 | 0.5 | 1936 | 630 | 4 | 1405 | 457 | 7 |
| Line-based text data | 0.9 | 3 | 95.9 | 366367 | 119 k | 3 | 366367 | 119 k | 3 |

The protocol hierarchy indicates dominant TCP traffic (343 packets), with some HTTP and line-based text traffic. A large amount of data (95.9%) is categorized as line-based text, suggesting potential file transfers or document exchanges.

Conversations:



Wireshark · Conversations · dumpfile4.pcap

Conversation Settings

Name resolution

Absolute start time

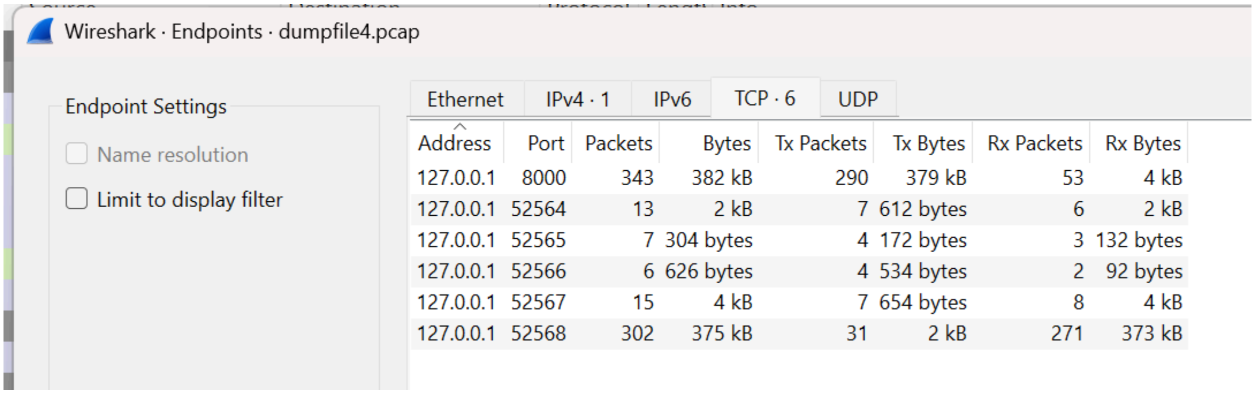
Limit to display filter

Ethernet | IPv4 · 1 | IPv6 | TCP · 5 | UDP

| Address A | Port A | Address B | Port B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A | Flows |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 52564 | 127.0.0.1 | 8000 | 13 | 2 kB | 0 | 7 | 612 bytes | 6 | 2 kB | 0.000000 | 0.1094 | 44 kbps | 115 kbps | 2 |
| 127.0.0.1 | 52565 | 127.0.0.1 | 8000 | 7 | 304 bytes | 1 | 4 | 172 bytes | 3 | 132 bytes | 3.039936 | 10.8536 | 126 bits/s | 97 bits/s | 0 |
| 127.0.0.1 | 52566 | 127.0.0.1 | 8000 | 6 | 626 bytes | 2 | 4 | 534 bytes | 2 | 92 bytes | 3.039936 | 10.8506 | 393 bits/s | 67 bits/s | 1 |
| 127.0.0.1 | 52567 | 127.0.0.1 | 8000 | 15 | 4 kB | 3 | 7 | 654 bytes | 8 | 4 kB | 13.893535 | 0.0283 | 185 kbps | 1026 kbps | 2 |
| 127.0.0.1 | 52568 | 127.0.0.1 | 8000 | 302 | 375 kB | 4 | 31 | 2 kB | 271 | 373 kB | 21.631712 | 2.9270 | 4408 bits/s | 1019 kbps | 2 |

Five TCP streams were observed, with Stream ID 4 contributing most of the traffic (302 packets, 375 kB). Stream 2 and 3 show brief but potentially important file transfers.

Endpoints:



Wireshark · Endpoints · dumpfile4.pcap

Endpoint Settings

Name resolution

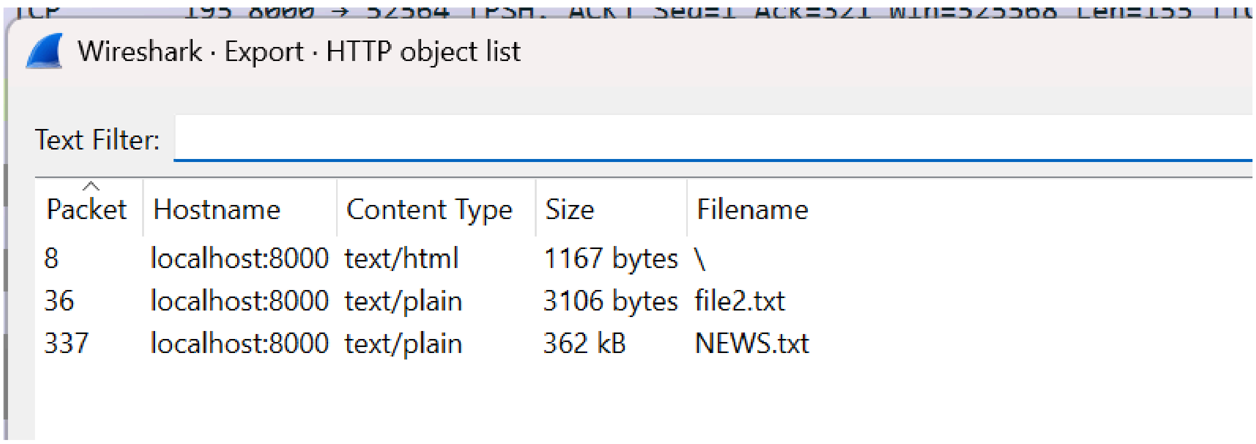
Limit to display filter

Ethernet | IPv4 · 1 | IPv6 | TCP · 6 | UDP

| Address | Port | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 8000 | 343 | 382 kB | 290 | 379 kB | 53 | 4 kB |
| 127.0.0.1 | 52564 | 13 | 2 kB | 7 | 612 bytes | 6 | 2 kB |
| 127.0.0.1 | 52565 | 7 | 304 bytes | 4 | 172 bytes | 3 | 132 bytes |
| 127.0.0.1 | 52566 | 6 | 626 bytes | 4 | 534 bytes | 2 | 92 bytes |
| 127.0.0.1 | 52567 | 15 | 4 kB | 7 | 654 bytes | 8 | 4 kB |
| 127.0.0.1 | 52568 | 302 | 375 kB | 31 | 2 kB | 271 | 373 kB |

The endpoint table shows port 8000 receiving most traffic. Port 52568 had significant upload activity (375 kB), suggesting a major file (possibly NEWS.txt) was sent over this port.

HTTP Object List:



Wireshark · Export · HTTP object list

Text Filter:

| Packet | Hostname | Content Type | Size | Filename |
|---|---|---|---|---|
| 8 | localhost:8000 | text/html | 1167 bytes | \ |
| 36 | localhost:8000 | text/plain | 3106 bytes | file2.txt |
| 337 | localhost:8000 | text/plain | 362 kB | NEWS.txt |

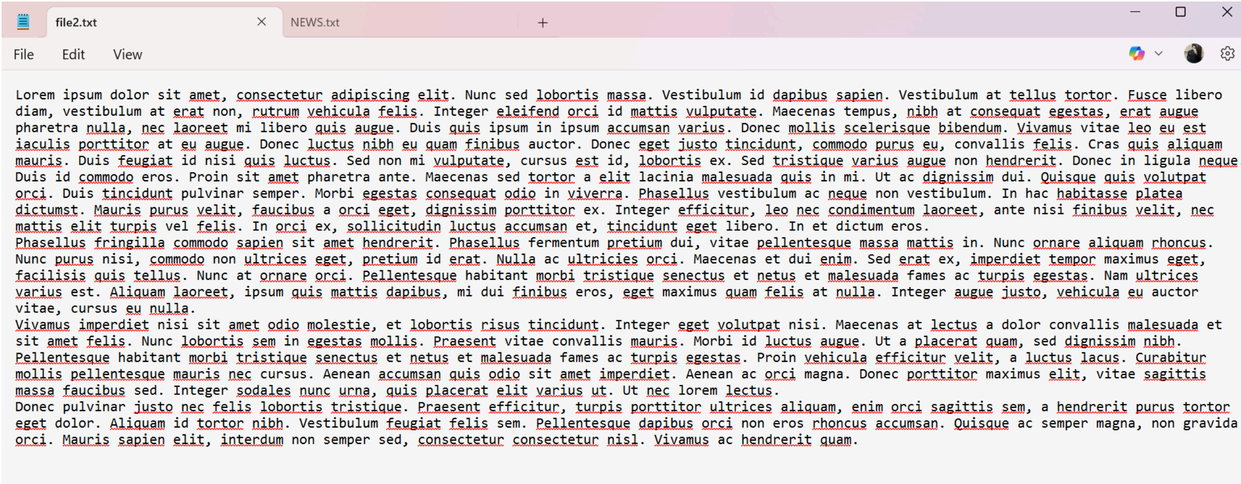
HTTP Object List:
Three HTTP objects were detected: a standard directory listing, file2.txt (3 kB), and NEWS.txt (362 kB). The latter stands out due to its size and could contain detailed logs or hidden information.

Text output observations:

file2.txt NEWS.txt

File Edit View

```
+++++++++++
Python News
+++++++++++

What's New in Python 3.6.3 final?
=================================

*Release date: 2017-10-03*

Library
-------

- bpo-31641: Re-allow arbitrary iterables in
  `concurrent.futures.as_completed()`. Fixes regression in 3.6.3rc1.

Build
-----

- bpo-31662: Fix typos in Windows ``uploadrelease.bat`` script. Fix Windows
  Doc build issues in ``Doc/make.bat``.

- bpo-31423: Fix building the PDF documentation with newer versions of
  Sphinx.


What's New in Python 3.6.3 release candidate 1?
===============================================

*Release date: 2017-09-18*

Security
--------

- bpo-29781: SSLObject.version() now correctly returns None when handshake
  over BIO has not been performed yet.

- bpo-30947: Upgrade libexpat embedded copy from version 2.2.1 to 2.2.3 to
  get security fixes.

Core and Builtins
```



file2.txt
NEWS.txt
File Edit View

```
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc sed lobortis massa. Vestibulum id dapibus sapien. Vestibulum at tellus tortor. Fusce libero diam, vestibulum at erat non, rutrum vehicula felis. Integer eleifend orci id mattis vulputate. Maecenas tempus, nibh at consequat egestas, erat augue pharetra nulla, nec laoreet mi libero quis augue. Duis quis ipsum in ipsum accumsan varius. Donec mollis scelerisque bibendum. Vivamus vitae leo eu est iaculis porttitor at eu augue. Donec luctus nibh eu quam finibus auctor. Donec eget justo tincidunt, commodo purus eu, convallis felis. Cras quis aliquam mauris. Duis feugiat id nisi quis luctus. Sed non mi vulputate, cursus est id, lobortis ex. Sed tristique varius augue non hendrerit. Donec in ligula neque. Duis id commodo eros. Proin sit amet pharetra ante. Maecenas sed tortor a elit lacinia malesuada quis in mi. Ut ac dignissim dui. Quisque quis volutpat orci. Duis tincidunt pulvinar semper. Morbi egestas consequat odio in viverra. Phasellus vestibulum ac neque non vestibulum. In hac habitasse platea dictumst. Mauris purus velit, faucibus a orci eget, dignissim porttitor ex. Integer efficitur, leo nec condimentum laoreet, ante nisi finibus velit, nec mattis elit turpis vel felis. In orci ex, sollicitudin luctus accumsan et, tincidunt eget libero. In et dictum eros.
Phasellus fringilla commodo sapien sit amet hendrerit. Phasellus fermentum pretium dui, vitae pellentesque massa mattis in. Nunc ornare aliquam rhoncus. Nunc purus nisi, commodo non ultrices eget, pretium id erat. Nulla ac ultricies orci. Maecenas et dui enim. Sed erat ex, imperdiet tempor maximus eget, facilisis quis tellus. Nunc at ornare orci. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nam ultrices varius est. Aliquam laoreet, ipsum quis mattis dapibus, mi dui finibus eros, eget maximus quam felis at nulla. Integer augue justo, vehicula eu auctor vitae, cursus eu nulla.
Vivamus imperdiet nisi sit amet odio molestie, et lobortis risus tincidunt. Integer eget volutpat nisi. Maecenas at lectus a dolor convallis malesuada et sit amet felis. Nunc lobortis sem in egestas mollis. Praesent vitae convallis mauris. Morbi id luctus augue. Ut a placerat quam, sed dignissim nibh. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Proin vehicula efficitur velit, a luctus lacus. Curabitur mollis pellentesque mauris nec cursus. Aenean accumsan quis odio sit amet imperdiet. Aenean ac orci magna. Donec porttitor maximus elit, vitae sagittis massa faucibus sed. Integer sodales nunc urna, quis placerat elit varius ut. Ut nec lorem lectus.
Donec pulvinar justo nec felis lobortis tristique. Praesent efficitur, turpis porttitor ultrices aliquam, enim orci sagittis sem, a hendrerit purus tortor eget dolor. Aliquam id tortor nibh. Vestibulum feugiat felis sem. Pellentesque dapibus orci non eros rhoncus accumsan. Quisque ac semper magna, non gravida orci. Mauris sapien elit, interdum non semper sed, consectetur consectetur nisl. Vivamus ac hendrerit quam.
```

File2.txt Exported from HTTP Stream

This file contains seemingly harmless "Lorem Ipsum" placeholder text. However, a closer look reveals an embedded sentence mid-paragraph that states:

"Denise we are really fooling these students this semester. I love this class, it teaches me how to do a ton of things to hide my intentions. Payment is on the way."

This line stands out against the otherwise nonsensical text and may serve as a hidden message or confession related to the investigation.

Following TCP Streams:

Wireshark · Follow TCP Stream (tcp.stream eq 2) · dumpfile4.pcap

```
GET /file2.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.
14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive
```

Follow TCP Stream – file2.txt:

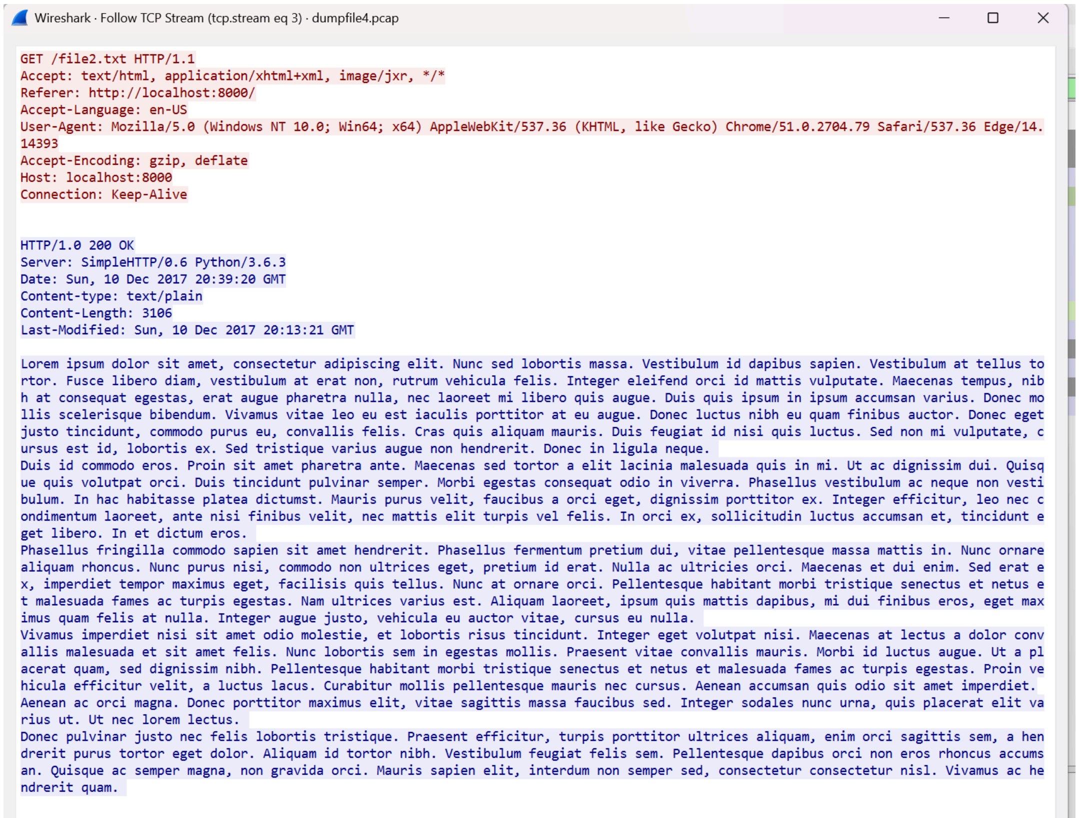
This HTTP stream shows a successful 200 OK response for the request to retrieve file2.txt. Payload appears plaintext, making it easy to extract and review for hidden messages or clues.

Wireshark · Follow TCP Stream (tcp.stream eq 4) · dumpfile4.pcap

```
GET /NEWS.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/1
4.14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:39:30 GMT
Content-type: text/plain
Content-Length: 362094
Last-Modified: Wed, 04 Oct 2017 01:33:04 GMT

+++++++++++
Python News
+++++++++++

What's New in Python 3.6.3 final?
=================================

*Release date: 2017-10-03*

Library
-------

- bpo-31641: Re-allow arbitrary iterables in
  `concurrent.futures.as_completed()`. Fixes regression in 3.6.3rc1.

Build
-----

- bpo-31662: Fix typos in Windows ``uploadrelease.bat`` script. Fix Windows
  Doc build issues in ``Doc/make.bat``.

- bpo-31423: Fix building the PDF documentation with newer versions of
  Sphinx.


What's New in Python 3.6.3 release candidate 1?
===============================================

*Release date: 2017-09-18*

Security
--------
```

**Follow TCP Stream – NEWS.txt:**

A substantial text document was delivered over this stream. The file size and plaintext formatting suggest it may contain critical information worth decoding or searching for keywords relevant to the case.



Wireshark · Follow TCP Stream (tcp.stream eq 3) · dumpfile4.pcap

```
GET /file2.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.
14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:39:20 GMT
Content-type: text/plain
Content-Length: 3106
Last-Modified: Sun, 10 Dec 2017 20:13:21 GMT

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc sed lobortis massa. Vestibulum id dapibus sapien. Vestibulum at tellus to
rtor. Fusce libero diam, vestibulum at erat non, rutrum vehicula felis. Integer eleifend orci id mattis vulputate. Maecenas tempus, nib
h at consequat egestas, erat augue pharetra nulla, nec laoreet mi libero quis augue. Duis quis ipsum in ipsum accumsan varius. Donec mo
llis scelerisque bibendum. Vivamus vitae leo eu est iaculis porttitor at eu augue. Donec luctus nibh eu quam finibus auctor. Donec eget
justo tincidunt, commodo purus eu, convallis felis. Cras quis aliquam mauris. Duis feugiat id nisi quis luctus. Sed non mi vulputate, c
ursus est id, lobortis ex. Sed tristique varius augue non hendrerit. Donec in ligula neque.
Duis id commodo eros. Proin sit amet pharetra ante. Maecenas sed tortor a elit lacinia malesuada quis in mi. Ut ac dignissim dui. Quisq
ue quis volutpat orci. Duis tincidunt pulvinar semper. Morbi egestas consequat odio in viverra. Phasellus vestibulum ac neque non vesti
bulum. In hac habitasse platea dictumst. Mauris purus velit, faucibus a orci eget, dignissim porttitor ex. Integer efficitur, leo nec c
ondimentum laoreet, ante nisi finibus velit, nec mattis elit turpis vel felis. In orci ex, sollicitudin luctus accumsan et, tincidunt e
get libero. In et dictum eros.
Phasellus fringilla commodo sapien sit amet hendrerit. Phasellus fermentum pretium dui, vitae pellentesque massa mattis in. Nunc ornare
aliquam rhoncus. Nunc purus nisi, commodo non ultrices eget, pretium id erat. Nulla ac ultricies orci. Maecenas et dui enim. Sed erat e
x, imperdiet tempor maximus eget, facilisis quis tellus. Nunc at ornare orci. Pellentesque habitant morbi tristique senectus et netus e
t malesuada fames ac turpis egestas. Nam ultrices varius est. Aliquam laoreet, ipsum quis mattis dapibus, mi dui finibus eros, eget max
imus quam felis at nulla. Integer augue justo, vehicula eu auctor vitae, cursus eu nulla.
Vivamus imperdiet nisi sit amet odio molestie, et lobortis risus tincidunt. Integer eget volutpat nisi. Maecenas at lectus a dolor conv
allis malesuada et sit amet felis. Nunc lobortis sem in egestas mollis. Praesent vitae convallis mauris. Morbi id luctus augue. Ut a pl
acerat quam, sed dignissim nibh. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Proin ve
hicula efficitur velit, a luctus lacus. Curabitur mollis pellentesque mauris nec cursus. Aenean accumsan quis odio sit amet imperdiet.
Aenean ac orci magna. Donec porttitor maximus elit, vitae sagittis massa faucibus sed. Integer sodales nunc urna, quis placerat elit va
rius ut. Ut nec lorem lectus.
Donec pulvinar justo nec felis lobortis tristique. Praesent efficitur, turpis porttitor ultrices aliquam, enim orci sagittis sem, a hen
drerit purus tortor eget dolor. Aliquam id tortor nibh. Vestibulum feugiat felis sem. Pellentesque dapibus orci non eros rhoncus accums
an. Quisque ac semper magna, non gravida orci. Mauris sapien elit, interdum non semper sed, consectetur consectetur nisl. Vivamus ac he
ndrerit quam.
```

**TCP Stream 3 — Retrieved file2.txt via HTTP GET:**
The stream confirms a successful HTTP/1.0 200 OK response from a local web server on localhost:8000 for the request GET /file2.txt HTTP/1.1.
The server responds with Content-type: text/plain and a file size of 3106 bytes.
The content below appears as generic "Lorem Ipsum" text. However, this stream is crucial as the corresponding exported file (file2.txt) revealed a suspicious message embedded mid-paragraph referencing Denise and payment, indicating a hidden or coded communication.

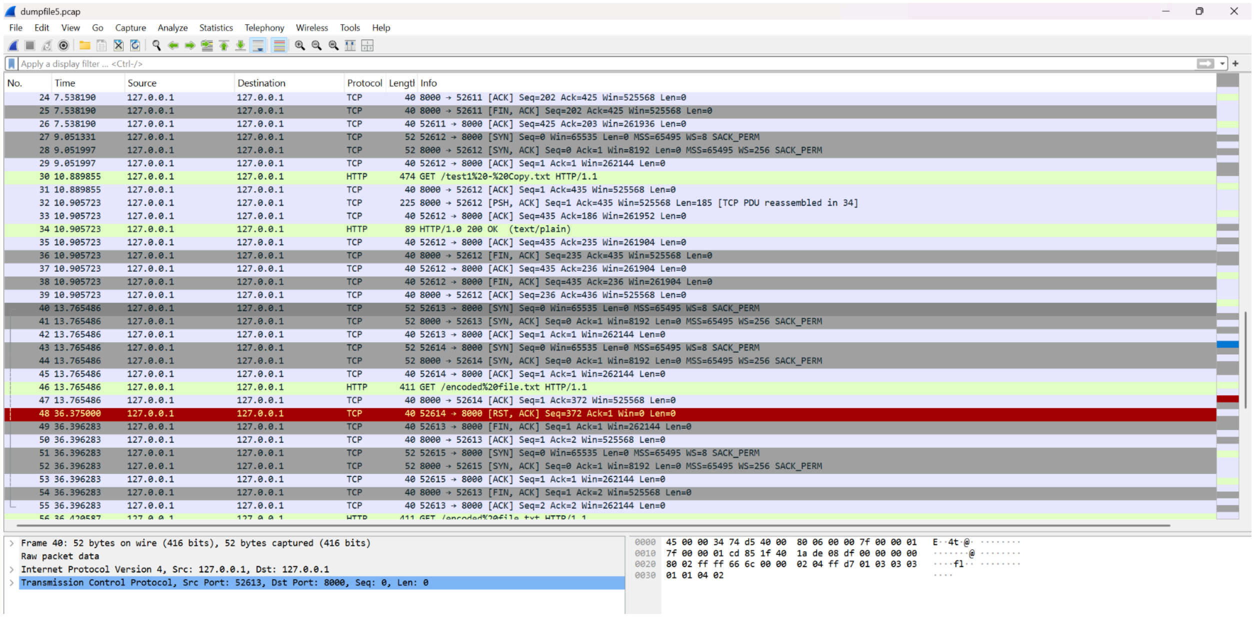
**Dumpfile4.pcap Summary**

This capture file contains 343 packets, primarily using TCP over the local loopback address (127.0.0.1) with HTTP traffic on port 8000. A large portion of data consists of line-based text, indicating file transfers.

Key findings include:

- A suspicious message hidden within `file2.txt`, retrieved via HTTP, suggesting intentional concealment: *"Denise we are really fooling these students this semester..."*
- Several TCP reset (RST) packets suggest interrupted or abnormal connections.

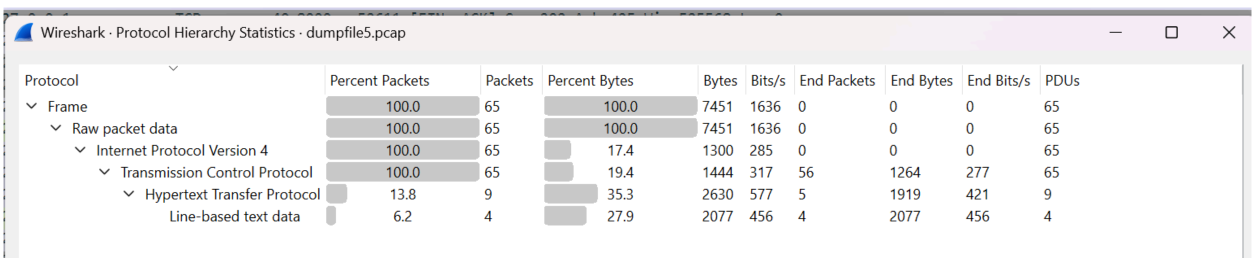
## 5. Analyzing dumpfile5.pcap in Wireshark:




| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 24 | 7.538190 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52611 [ACK] Seq=202 Ack=425 Win=525568 Len=0 |
| 25 | 7.538190 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52611 [FIN, ACK] Seq=202 Ack=425 Win=525568 Len=0 |
| 26 | 7.538190 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52611 → 8000 [ACK] Seq=425 Ack=203 Win=261936 Len=0 |
| 27 | 9.051331 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52612 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 28 | 9.051997 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52612 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 29 | 9.051997 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52612 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 30 | 10.889855 | 127.0.0.1 | 127.0.0.1 | HTTP | 474 | GET /test1%20-%20Copy.txt HTTP/1.1 |
| 31 | 10.889855 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52612 [ACK] Seq=1 Ack=435 Win=525568 Len=0 |
| 32 | 10.905723 | 127.0.0.1 | 127.0.0.1 | TCP | 225 | 8000 → 52612 [PSH, ACK] Seq=1 Ack=435 Win=525568 Len=185 [TCP PDU reassembled in 34] |
| 33 | 10.905723 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52612 → 8000 [ACK] Seq=435 Ack=186 Win=261952 Len=0 |
| 34 | 10.905723 | 127.0.0.1 | 127.0.0.1 | HTTP | 89 | HTTP/1.0 200 OK (text/plain) |
| 35 | 10.905723 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52612 → 8000 [ACK] Seq=435 Ack=235 Win=261904 Len=0 |
| 36 | 10.905723 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52612 [FIN, ACK] Seq=235 Ack=435 Win=525568 Len=0 |
| 37 | 10.905723 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52612 → 8000 [ACK] Seq=435 Ack=236 Win=261904 Len=0 |
| 38 | 10.905723 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52612 → 8000 [FIN, ACK] Seq=435 Ack=236 Win=261904 Len=0 |
| 39 | 10.905723 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52612 [ACK] Seq=236 Ack=436 Win=525568 Len=0 |
| 40 | 13.765486 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52613 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 41 | 13.765486 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52613 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 42 | 13.765486 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52613 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 43 | 13.765486 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52614 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 44 | 13.765486 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52614 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 45 | 13.765486 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52614 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 46 | 13.765486 | 127.0.0.1 | 127.0.0.1 | HTTP | 411 | GET /encoded%20file.txt HTTP/1.1 |
| 47 | 13.765486 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52614 [ACK] Seq=1 Ack=372 Win=525568 Len=0 |
| 48 | 36.375000 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52614 → 8000 [RST, ACK] Seq=372 Ack=1 Win=0 Len=0 |
| 49 | 36.396283 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52613 → 8000 [FIN, ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 50 | 36.396283 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52613 [ACK] Seq=1 Ack=2 Win=525568 Len=0 |
| 51 | 36.396283 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52615 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 52 | 36.396283 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52615 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 53 | 36.396283 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52615 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 54 | 36.396283 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52613 [FIN, ACK] Seq=1 Ack=2 Win=525568 Len=0 |
| 55 | 36.396283 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52613 → 8000 [ACK] Seq=2 Ack=2 Win=262144 Len=0 |

```
> Frame 40: 52 bytes on wire (416 bits), 52 bytes captured (416 bits)
  Raw packet data
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 52613, Dst Port: 8000, Seq: 0, Len: 0

0000  45 00 00 34 74 d5 40 00  80 06 00 00 7f 00 00 01
0010  7f 00 00 01 cd 85 1f 40  1a de 08 df 00 00 00 00
0020  80 02 ff ff 66 6c 00 00  02 04 ff d7 01 03 03 03
0030  01 01 04 02
```

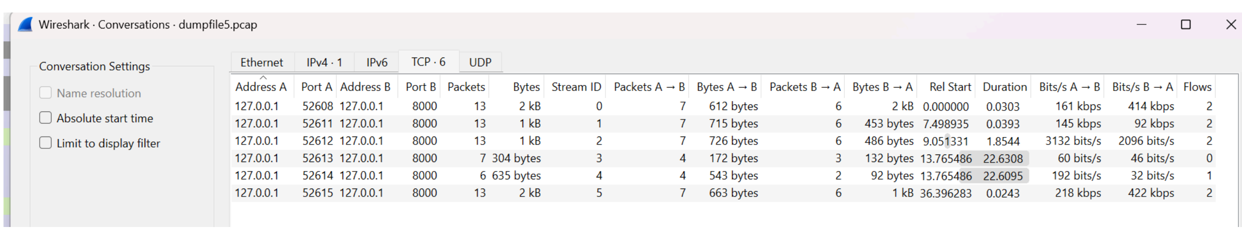
Protocol Hierarchy:




| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 65 | 100.0 | 7451 | 1636 | 0 | 0 | 0 | 65 |
| Raw packet data | 100.0 | 65 | 100.0 | 7451 | 1636 | 0 | 0 | 0 | 65 |
| Internet Protocol Version 4 | 100.0 | 65 | 17.4 | 1300 | 285 | 0 | 0 | 0 | 65 |
| Transmission Control Protocol | 100.0 | 65 | 19.4 | 1444 | 317 | 56 | 1264 | 277 | 65 |
| Hypertext Transfer Protocol | 13.8 | 9 | 35.3 | 2630 | 577 | 5 | 1919 | 421 | 9 |
| Line-based text data | 6.2 | 4 | 27.9 | 2077 | 456 | 4 | 2077 | 456 | 4 |

Displays the breakdown of traffic: 13.8% of packets are HTTP, and 6.2% involve line-based text data, indicating file transfers over HTTP.

Conversations:




| Address A | Port A | Address B | Port B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A | Flows |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 52608 | 127.0.0.1 | 8000 | 13 | 2 kB | 0 | 7 | 612 bytes | 6 | 2 kB | 0.000000 | 0.0303 | 161 kbps | 414 kbps | 2 |
| 127.0.0.1 | 52611 | 127.0.0.1 | 8000 | 13 | 1 kB | 1 | 7 | 715 bytes | 6 | 453 bytes | 7.498935 | 0.0393 | 145 kbps | 92 kbps | 2 |
| 127.0.0.1 | 52612 | 127.0.0.1 | 8000 | 13 | 1 kB | 2 | 7 | 726 bytes | 6 | 486 bytes | 9.051331 | 1.8544 | 3132 bits/s | 2096 bits/s | 2 |
| 127.0.0.1 | 52613 | 127.0.0.1 | 8000 | 7 | 304 bytes | 3 | 4 | 172 bytes | 3 | 132 bytes | 13.765486 | 22.6308 | 60 bits/s | 46 bits/s | 0 |
| 127.0.0.1 | 52614 | 127.0.0.1 | 8000 | 6 | 635 bytes | 4 | 4 | 543 bytes | 2 | 92 bytes | 13.765486 | 22.6095 | 192 bits/s | 32 bits/s | 1 |
| 127.0.0.1 | 52615 | 127.0.0.1 | 8000 | 13 | 2 kB | 5 | 7 | 663 bytes | 6 | 1 kB | 36.396283 | 0.0243 | 218 kbps | 422 kbps | 2 |

Identifies five active TCP conversations using port 8000, mapping to individual stream IDs. Each stream represents a request-response pair between localhost and server.

Endpoints:

Wireshark · Endpoints · dumpfile5.pcap

Endpoint Settings

- [ ] Name resolution
- [ ] Limit to display filter

Ethernet | IPv4 · 1 | IPv6 | TCP · 7 | UDP

| Address | Port | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 8000 | 65 | 7 kB | 29 | 4 kB | 36 | 3 kB |
| 127.0.0.1 | 52608 | 13 | 2 kB | 7 | 612 bytes | 6 | 2 kB |
| 127.0.0.1 | 52611 | 13 | 1 kB | 7 | 715 bytes | 6 | 453 bytes |
| 127.0.0.1 | 52612 | 13 | 1 kB | 7 | 726 bytes | 6 | 486 bytes |
| 127.0.0.1 | 52613 | 7 | 304 bytes | 4 | 172 bytes | 3 | 132 bytes |
| 127.0.0.1 | 52614 | 6 | 635 bytes | 4 | 543 bytes | 2 | 92 bytes |
| 127.0.0.1 | 52615 | 13 | 2 kB | 7 | 663 bytes | 6 | 1 kB |

Summarizes all endpoints, their traffic volume, and packet counts. High byte counts for port 8000 (web server), with most data exchanged in TCP conversations from ports 52608 to 52615.

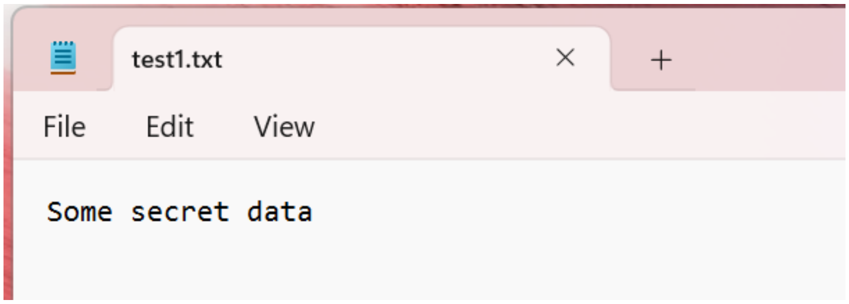
Wireshark · Export · HTTP object list

Text Filter:

| Packet | Hostname | Content Type | Size | Filename |
|---|---|---|---|---|
| 8 | localhost:8000 | text/html | 1167 bytes | \ |
| 21 | localhost:8000 | text/plain | 16 bytes | test1.txt |
| 34 | localhost:8000 | text/plain | 49 bytes | test1%20-%20Copy.txt |
| 60 | localhost:8000 | text/plain | 845 bytes | encoded%20file.txt |

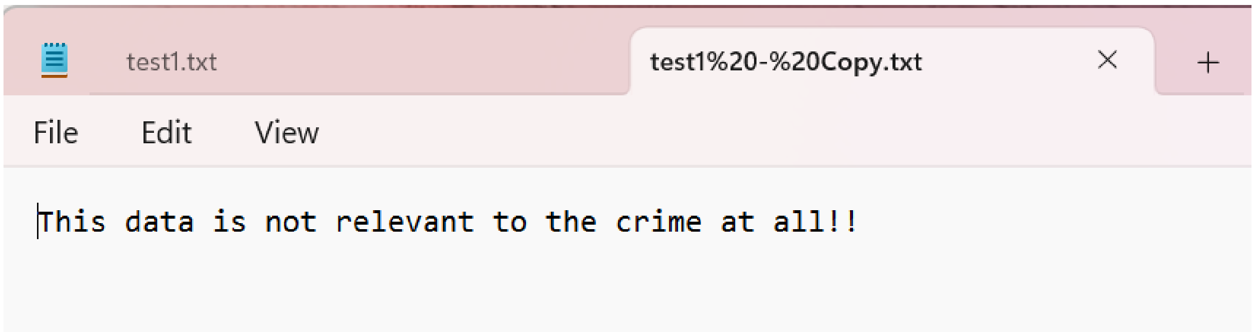
**HTTP Object List**
Reveals all extracted files:

- `/` (directory listing)
- `test1.txt` (16 bytes)
- `test1 - Copy.txt` (49 bytes)
- `encoded file.txt` (845 bytes)



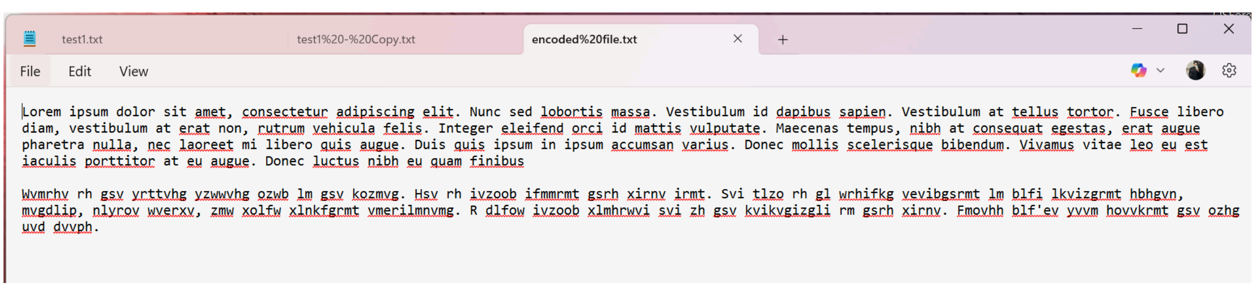

**`test1.txt` Contents**

Contains the phrase: `Some secret data` – this could be relevant evidence.



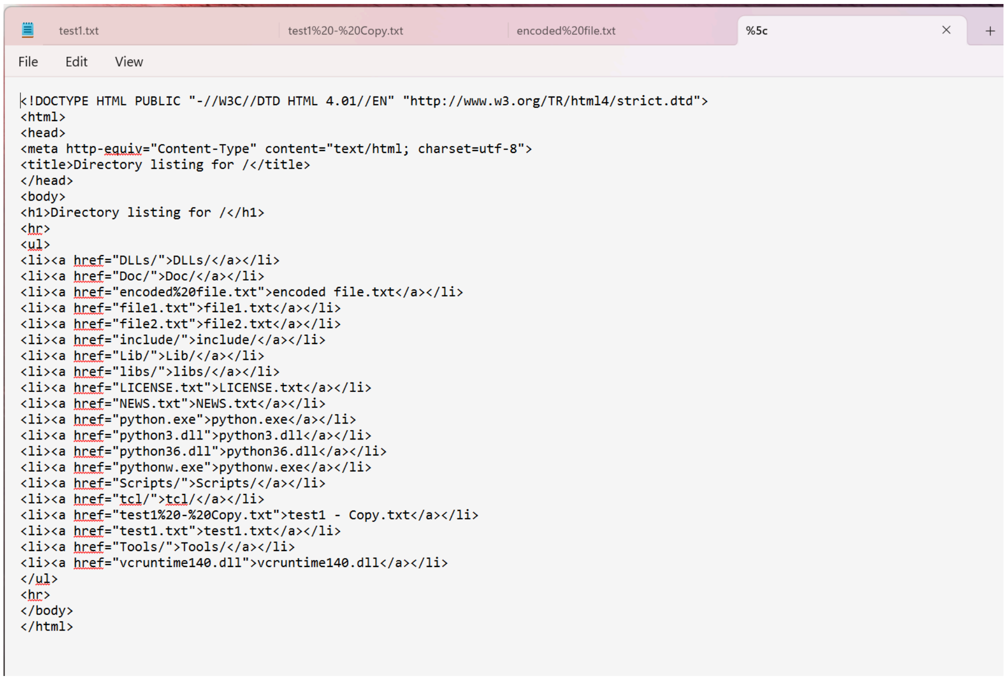

**`test1 - Copy.txt` Contents**

The file explicitly states: `This data is not relevant to the crime at all!!!` – possibly misleading content.




**`encoded file.txt` Contents**

Contains a mix of lorem ipsum text and a suspicious Atbash-encoded message, which may conceal hidden communication.



```
test1.txt    test1%20-%20Copy.txt    encoded%20file.txt    %5c
File    Edit    View

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="encoded%20file.txt">encoded file.txt</a></li>
<li><a href="file1.txt">file1.txt</a></li>
<li><a href="file2.txt">file2.txt</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
</html>
```

**Directory Listing HTML View**

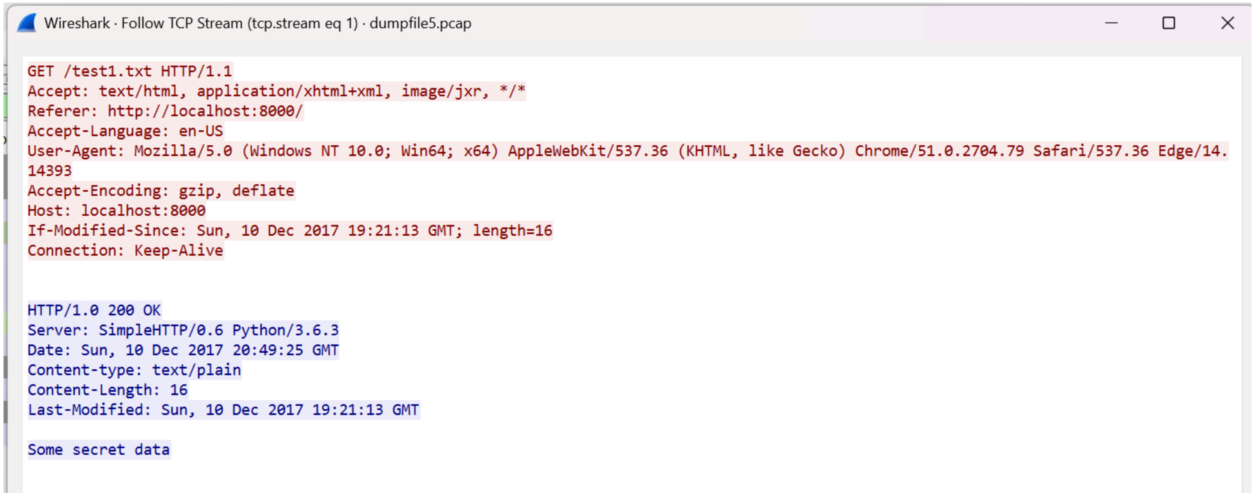
Shows a web directory index listing all accessible files, revealing the existence of potentially sensitive or misleading documents.

Following TCP Streams:

```
Wireshark · Follow TCP Stream (tcp.stream eq 5) · dumpfile5.pcap

GET /encoded%20file.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.
14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:49:54 GMT
Content-type: text/plain
Content-Length: 845
Last-Modified: Sun, 10 Dec 2017 20:25:40 GMT

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc sed lobortis massa. Vestibulum id dapibus sapien. Vestibulum at tellus to
rtor. Fusce libero diam, vestibulum at erat non, rutrum vehicula felis. Integer eleifend orci id mattis vulputate. Maecenas tempus, nib
h at consequat egestas, erat augue pharetra nulla, nec laoreet mi libero quis augue. Duis quis ipsum in ipsum accumsan varius. Donec mo
llis scelerisque bibendum. Vivamus vitae leo eu est iaculis porttitor at eu augue. Donec luctus nibh eu quam finibus

Wvmrhv rh gsv yrttvhg yzwwvhg ozwb lm gsv kozmvg. Hsv rh ivzoob ifmmrmt gsrh xirnv irmt. Svi tlzo rh gl wrhifkg vevibgsrmt lm blfi lkvi
zgrmt hbhgvn, mvgdlip, nlyrov wverxv, zmw xolfw xlnkfgrmt vmerilmnvmg. R dlfow ivzoob xlmhrwvi svi zh gsv kvikvgizgli rm gsrh xirnv. Fm
ovhh blf'ev yvvm hovvkrmt gsv ozhg uvd dvvph.
```

**TCP Stream 5 (encoded file.txt)**

- This request fetches `encoded file.txt`. The file includes standard Lorem Ipsum followed by what appears to be a Caesar Cipher-style encoded message—potentially to obfuscate sensitive content.



Wireshark · Follow TCP Stream (tcp.stream eq 1) · dumpfile5.pcap

```
GET /test1.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.
14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
If-Modified-Since: Sun, 10 Dec 2017 19:21:13 GMT; length=16
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:49:25 GMT
Content-type: text/plain
Content-Length: 16
Last-Modified: Sun, 10 Dec 2017 19:21:13 GMT

Some secret data
```

### TCP Stream 1 (test1.txt)

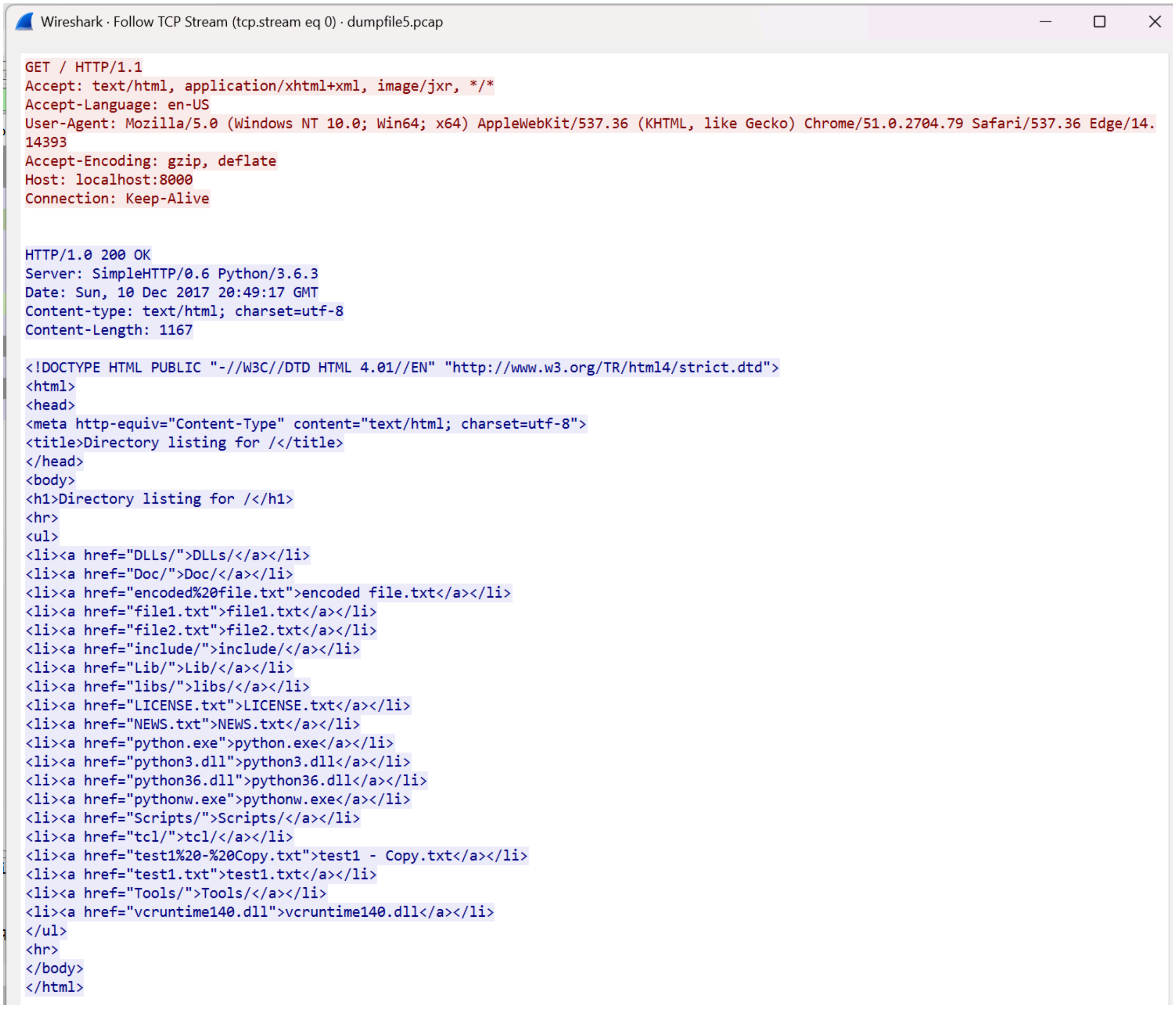
- This stream shows a GET request for `test1.txt`. The server responds with the file content `"Some secret data"` and a `200 OK` status.

Wireshark · Follow TCP Stream (tcp.stream eq 2) · dumpfile5.pcap

```
GET /test1%20-%20Copy.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.
14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
If-Modified-Since: Sun, 10 Dec 2017 19:21:39 GMT; length=49
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:49:28 GMT
Content-type: text/plain
Content-Length: 49
Last-Modified: Sun, 10 Dec 2017 19:21:39 GMT

This data is not relevant to the crime at all!!
```

### TCP Stream 2 (test1 - Copy.txt)

- A GET request retrieves `test1 - Copy.txt`, which contains the line `"This data is not relevant to the crime at all!!"`.



Wireshark · Follow TCP Stream (tcp.stream eq 0) · dumpfile5.pcap

```
GET / HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.
14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 20:49:17 GMT
Content-type: text/html; charset=utf-8
Content-Length: 1167

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="encoded%20file.txt">encoded file.txt</a></li>
<li><a href="file1.txt">file1.txt</a></li>
<li><a href="file2.txt">file2.txt</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
</html>
```

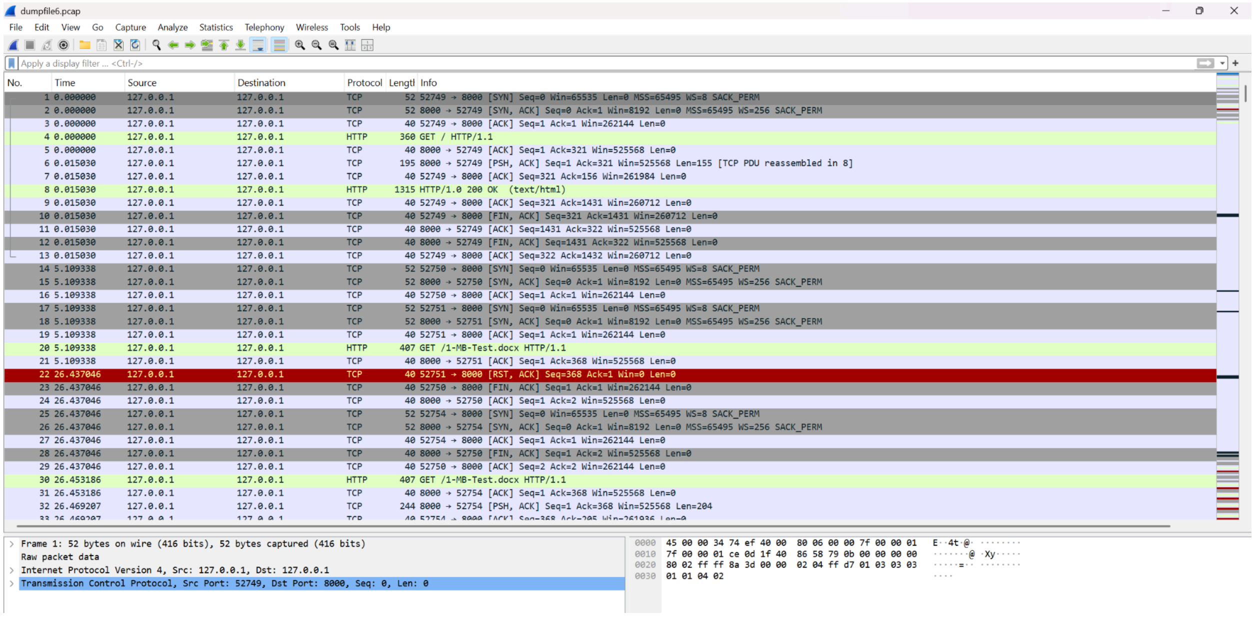
**TCP Stream 0 (Directory Listing Page)**

- This stream contains an HTTP GET request to the root directory (`GET / HTTP/1.1`) and the server responds with a text/html directory listing page. Files such as `test1.txt`, `test1 - Copy.txt`, `encoded file.txt`, and `file2.txt` are visible.

Summary of dumpfile5.pcap

A local web server on 127.0.0.1 was accessed, and multiple files were requested and retrieved over HTTP. The packet structure shows multiple TCP streams, with Stream IDs 0, 1, 2, and 5 corresponding to HTTP content. Streams 3 and 4 are active but not directly linked to critical content.

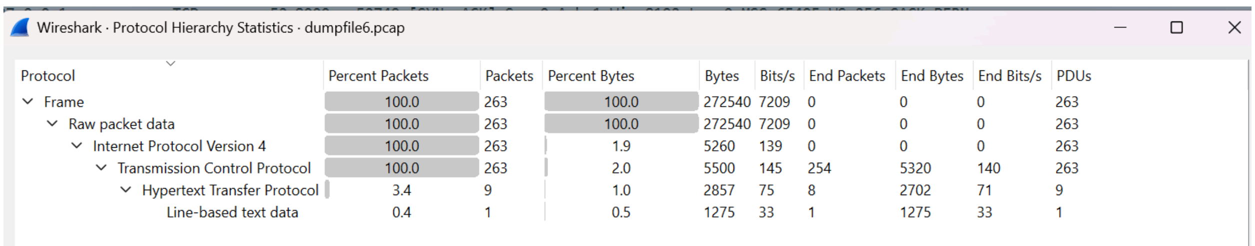
## 6. Analyzing dumpfile6.pcap in Wireshark:




| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52749 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 2 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52749 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 3 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52749 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 4 | 0.000000 | 127.0.0.1 | 127.0.0.1 | HTTP | 360 | GET / HTTP/1.1 |
| 5 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52749 [ACK] Seq=1 Ack=321 Win=525568 Len=0 |
| 6 | 0.015030 | 127.0.0.1 | 127.0.0.1 | TCP | 195 | 8000 → 52749 [PSH, ACK] Seq=1 Ack=321 Win=525568 Len=155 [TCP PDU reassembled in 8] |
| 7 | 0.015030 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52749 → 8000 [ACK] Seq=321 Ack=156 Win=261984 Len=0 |
| 8 | 0.015030 | 127.0.0.1 | 127.0.0.1 | HTTP | 1315 | HTTP/1.0 200 OK (text/html) |
| 9 | 0.015030 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52749 → 8000 [ACK] Seq=321 Ack=1431 Win=260712 Len=0 |
| 10 | 0.015030 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52749 → 8000 [FIN, ACK] Seq=321 Ack=1431 Win=260712 Len=0 |
| 11 | 0.015030 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52749 [ACK] Seq=1431 Ack=322 Win=525568 Len=0 |
| 12 | 0.015030 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52749 [FIN, ACK] Seq=1431 Ack=322 Win=525568 Len=0 |
| 13 | 0.015030 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52749 → 8000 [ACK] Seq=322 Ack=1432 Win=260712 Len=0 |
| 14 | 5.109338 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52750 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 15 | 5.109338 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52750 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 16 | 5.109338 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52750 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 17 | 5.109338 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52751 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 18 | 5.109338 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52751 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 19 | 5.109338 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52751 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 20 | 5.109338 | 127.0.0.1 | 127.0.0.1 | HTTP | 407 | GET /1-MB-Test.docx HTTP/1.1 |
| 21 | 5.109338 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52751 [ACK] Seq=1 Ack=368 Win=525568 Len=0 |
| 22 | 26.437046 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52751 → 8000 [RST, ACK] Seq=368 Ack=1 Win=0 Len=0 |
| 23 | 26.437046 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52750 → 8000 [FIN, ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 24 | 26.437046 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52750 [ACK] Seq=1 Ack=2 Win=525568 Len=0 |
| 25 | 26.437046 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52754 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 26 | 26.437046 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52754 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 27 | 26.437046 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52754 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 28 | 26.437046 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52750 [FIN, ACK] Seq=1 Ack=2 Win=525568 Len=0 |
| 29 | 26.437046 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52750 → 8000 [ACK] Seq=2 Ack=2 Win=262144 Len=0 |
| 30 | 26.453186 | 127.0.0.1 | 127.0.0.1 | HTTP | 407 | GET /1-MB-Test.docx HTTP/1.1 |
| 31 | 26.453186 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52754 [ACK] Seq=1 Ack=368 Win=525568 Len=0 |
| 32 | 26.469207 | 127.0.0.1 | 127.0.0.1 | TCP | 244 | 8000 → 52754 [PSH, ACK] Seq=1 Ack=368 Win=525568 Len=204 |



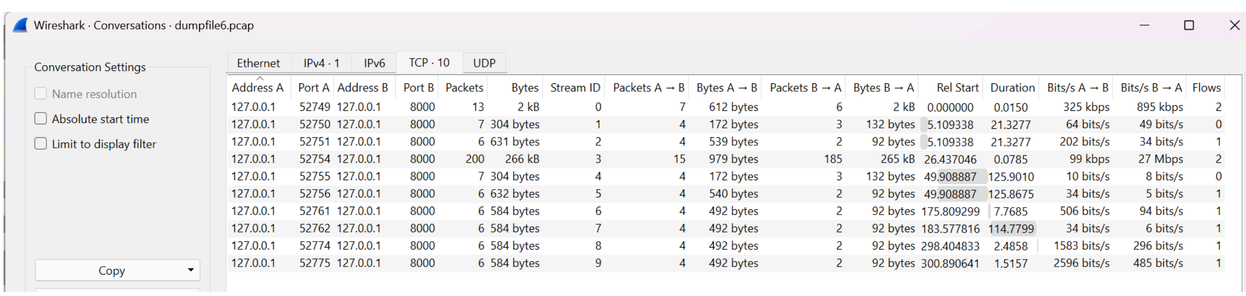
Protocol Hierarchy:




| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 263 | 100.0 | 272540 | 7209 | 0 | 0 | 0 | 263 |
| Raw packet data | 100.0 | 263 | 100.0 | 272540 | 7209 | 0 | 0 | 0 | 263 |
| Internet Protocol Version 4 | 100.0 | 263 | 1.9 | 5260 | 139 | 0 | 0 | 0 | 263 |
| Transmission Control Protocol | 100.0 | 263 | 2.0 | 5500 | 145 | 254 | 5320 | 140 | 263 |
| Hypertext Transfer Protocol | 3.4 | 9 | 1.0 | 2857 | 75 | 8 | 2702 | 71 | 9 |
| Line-based text data | 0.4 | 1 | 0.5 | 1275 | 33 | 1 | 1275 | 33 | 1 |

Breaks down the packet distribution: HTTP accounts for 3.4% of the packets and 1% of the data by bytes. Line-based text data makes up 0.5% of the data.

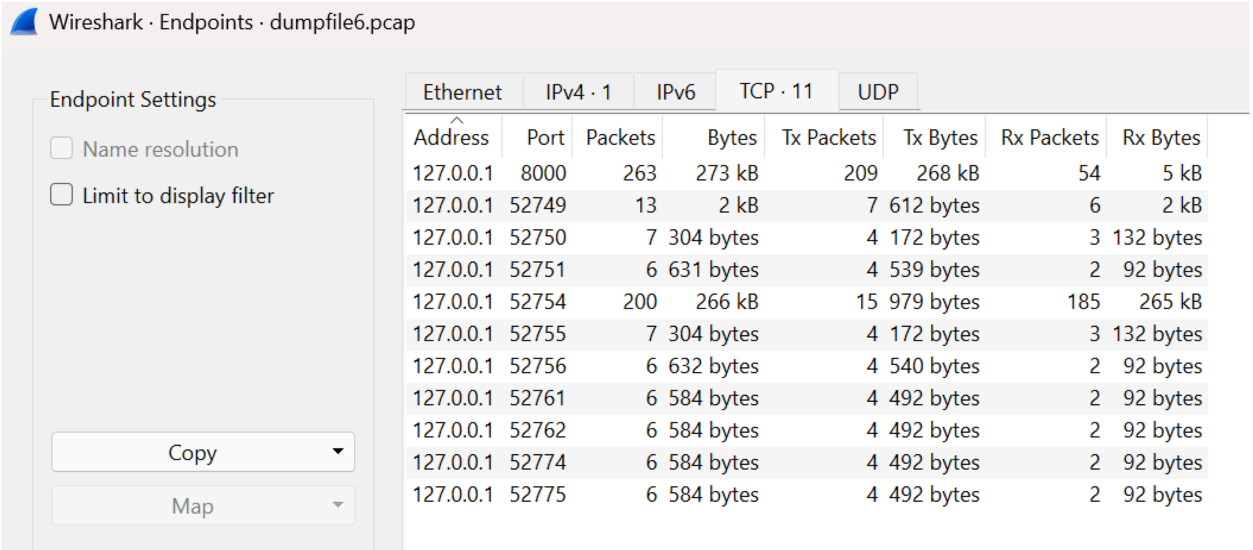
Conversations:




| Address A | Port A | Address B | Port B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A | Flows |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 52749 | 127.0.0.1 | 8000 | 13 | 2 kB | 0 | 7 | 612 bytes | 6 | 2 kB | 0.000000 | 0.0150 | 325 kbps | 895 kbps | 2 |
| 127.0.0.1 | 52750 | 127.0.0.1 | 8000 | 7 | 304 bytes | 1 | 4 | 172 bytes | 3 | 132 bytes | 5.109338 | 21.3277 | 64 bits/s | 49 bits/s | 0 |
| 127.0.0.1 | 52751 | 127.0.0.1 | 8000 | 6 | 631 bytes | 2 | 4 | 539 bytes | 2 | 92 bytes | 5.109338 | 21.3277 | 202 bits/s | 34 bits/s | 1 |
| 127.0.0.1 | 52754 | 127.0.0.1 | 8000 | 200 | 266 kB | 3 | 15 | 979 bytes | 185 | 265 kB | 26.437046 | 0.0785 | 99 kbps | 27 Mbps | 2 |
| 127.0.0.1 | 52755 | 127.0.0.1 | 8000 | 7 | 304 bytes | 4 | 4 | 172 bytes | 3 | 132 bytes | 49.908887 | 125.9010 | 10 bits/s | 8 bits/s | 0 |
| 127.0.0.1 | 52756 | 127.0.0.1 | 8000 | 6 | 632 bytes | 5 | 4 | 540 bytes | 2 | 92 bytes | 49.908887 | 125.8675 | 34 bits/s | 5 bits/s | 1 |
| 127.0.0.1 | 52761 | 127.0.0.1 | 8000 | 6 | 584 bytes | 6 | 4 | 492 bytes | 2 | 92 bytes | 175.809299 | 7.7685 | 506 bits/s | 94 bits/s | 1 |
| 127.0.0.1 | 52762 | 127.0.0.1 | 8000 | 6 | 584 bytes | 7 | 4 | 492 bytes | 2 | 92 bytes | 183.577816 | 114.7799 | 34 bits/s | 6 bits/s | 1 |
| 127.0.0.1 | 52774 | 127.0.0.1 | 8000 | 6 | 584 bytes | 8 | 4 | 492 bytes | 2 | 92 bytes | 298.404833 | 2.4858 | 1583 bits/s | 296 bits/s | 1 |
| 127.0.0.1 | 52775 | 127.0.0.1 | 8000 | 6 | 584 bytes | 9 | 4 | 492 bytes | 2 | 92 bytes | 300.890641 | 1.5157 | 2596 bits/s | 485 bits/s | 1 |

Shows 10 distinct TCP streams. Stream ID 3 (ports 52754 ↔ 8000) transferred the most data (265 kB), possibly related to a larger file like `10-MB-Test.docx`.

Endpoints:



Wireshark · Endpoints · dumpfile6.pcap

Endpoint Settings

Name resolution

Limit to display filter

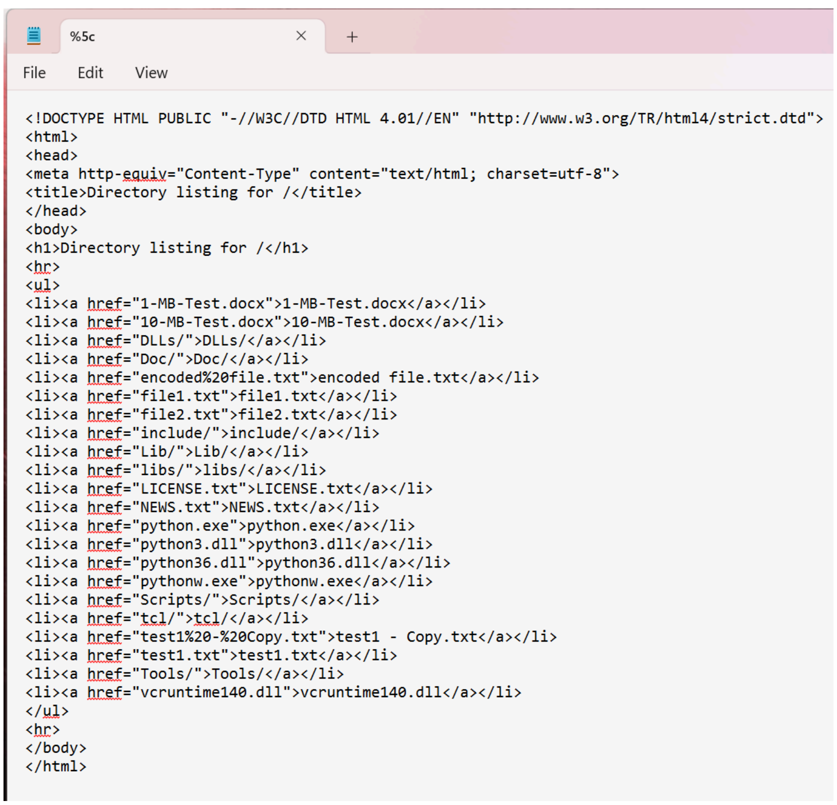
Copy

Map

Ethernet | IPv4 · 1 | IPv6 | TCP · 11 | UDP

| Address | Port | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 8000 | 263 | 273 kB | 209 | 268 kB | 54 | 5 kB |
| 127.0.0.1 | 52749 | 13 | 2 kB | 7 | 612 bytes | 6 | 2 kB |
| 127.0.0.1 | 52750 | 7 | 304 bytes | 4 | 172 bytes | 3 | 132 bytes |
| 127.0.0.1 | 52751 | 6 | 631 bytes | 4 | 539 bytes | 2 | 92 bytes |
| 127.0.0.1 | 52754 | 200 | 266 kB | 15 | 979 bytes | 185 | 265 kB |
| 127.0.0.1 | 52755 | 7 | 304 bytes | 4 | 172 bytes | 3 | 132 bytes |
| 127.0.0.1 | 52756 | 6 | 632 bytes | 4 | 540 bytes | 2 | 92 bytes |
| 127.0.0.1 | 52761 | 6 | 584 bytes | 4 | 492 bytes | 2 | 92 bytes |
| 127.0.0.1 | 52762 | 6 | 584 bytes | 4 | 492 bytes | 2 | 92 bytes |
| 127.0.0.1 | 52774 | 6 | 584 bytes | 4 | 492 bytes | 2 | 92 bytes |
| 127.0.0.1 | 52775 | 6 | 584 bytes | 4 | 492 bytes | 2 | 92 bytes |

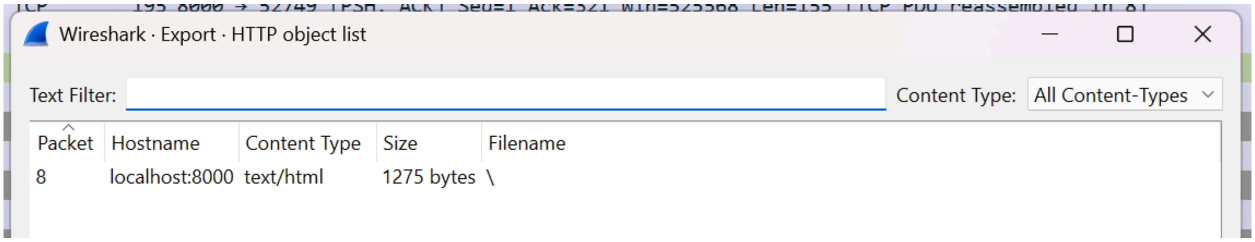
Lists all active ports on localhost. Port 52754 had the highest Rx Bytes (265 kB), indicating a significant file download from the server.



%5c

File Edit View

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="1-MB-Test.docx">1-MB-Test.docx</a></li>
<li><a href="10-MB-Test.docx">10-MB-Test.docx</a></li>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="encoded%20file.txt">encoded file.txt</a></li>
<li><a href="file1.txt">file1.txt</a></li>
<li><a href="file2.txt">file2.txt</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
</html>
```



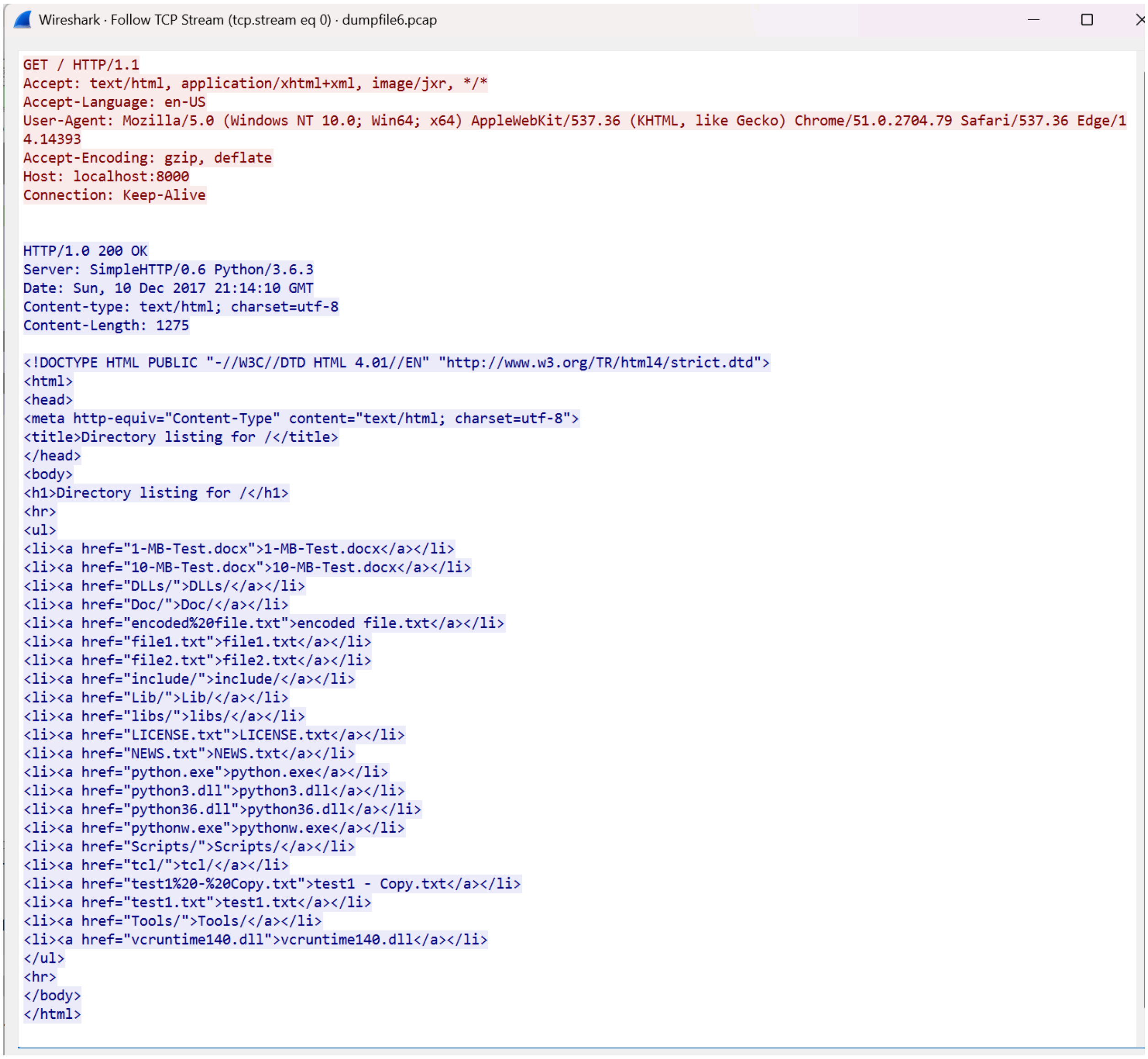

**Exported Directory Listing (HTML)**

Shows the `index.html` listing of downloadable files, including `1-MB-Test.docx`, `10-MB-Test.docx`, `encoded file.txt`, and others. This is the main reference for which files are accessible.|

**TCP Streams Followed:**


**Stream 2 – GET /1-MB-Test.docx Request Only**

Shows the initial HTTP GET request for the file `1-MB-Test.docx`. No server response is included in this stream, indicating this may be just the request segment.




```
GET / HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/1
4.14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 21:14:10 GMT
Content-type: text/html; charset=utf-8
Content-Length: 1275

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="1-MB-Test.docx">1-MB-Test.docx</a></li>
<li><a href="10-MB-Test.docx">10-MB-Test.docx</a></li>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="encoded%20file.txt">encoded file.txt</a></li>
<li><a href="file1.txt">file1.txt</a></li>
<li><a href="file2.txt">file2.txt</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
</html>
```

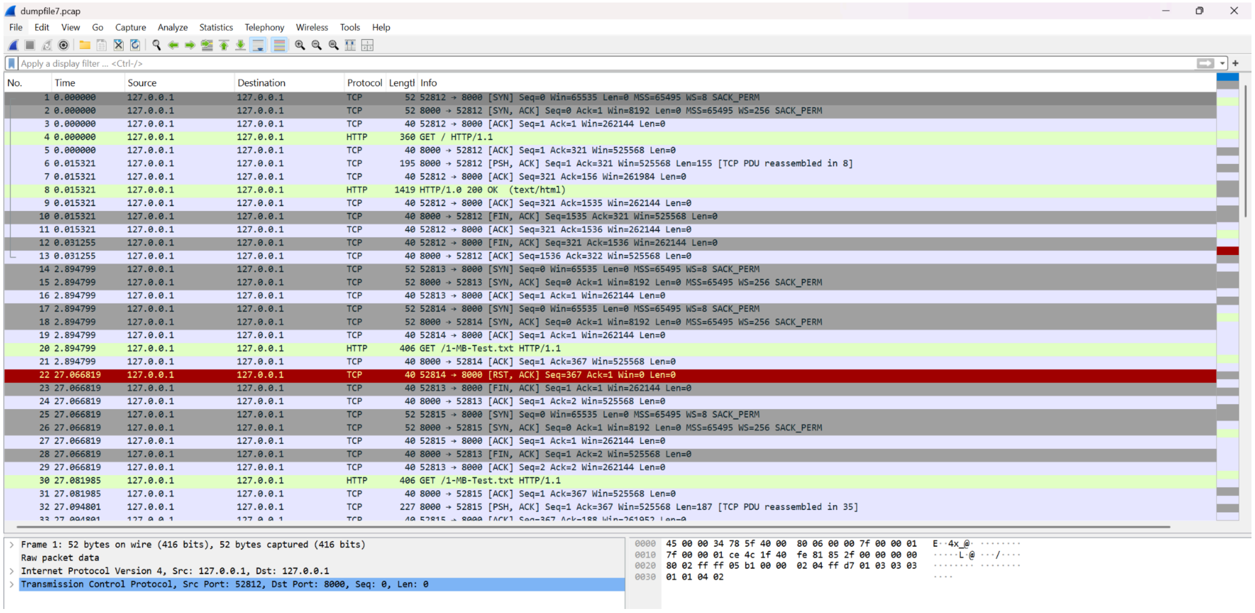
**Stream 0 – Directory Listing (GET /)**

Request and response for the root page (`/`) which returns an HTML directory listing of available files hosted on the server. Displays filenames like `file1.txt`, `LICENSE.txt`, `1-MB-Test.docx`, etc.


**Stream 3 – GET /1-MB-Test.docx (Full Transfer)**
Full transfer of the file `1-MB-Test.docx`. The response includes binary (non-readable) data with content type `application/octet-stream`. This confirms a file download.

**Dumpfile 6 Summary:**
This PCAP captures HTTP traffic between a client and a local server on port 8000. The client accesses the root directory, retrieving a file listing that includes documents like `1-MB-Test.docx`. The main activity involves downloading this Word document, which is confirmed through a full HTTP transfer of binary data. Multiple TCP streams are established, but only a few carry meaningful HTTP transactions, with Stream 3 containing the complete file download. The server is running Python's SimpleHTTPServer, and all communication occurs over localhost.

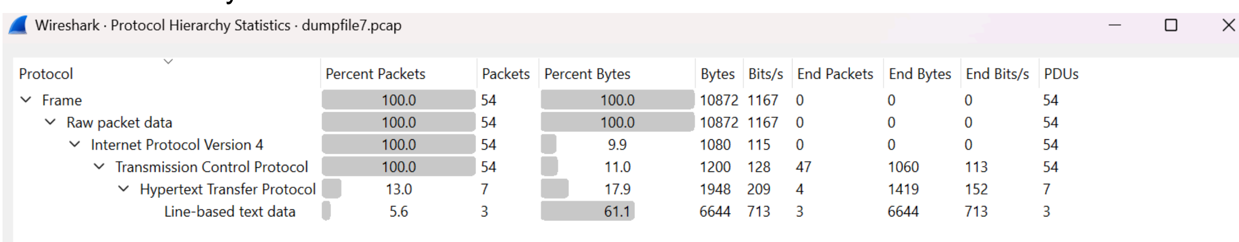
## 7. Analyzing dumpfile7.pcap in Wireshark:



| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52812 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 2 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52812 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 3 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52812 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 4 | 0.000000 | 127.0.0.1 | 127.0.0.1 | HTTP | 360 | GET / HTTP/1.1 |
| 5 | 0.000000 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52812 [ACK] Seq=1 Ack=321 Win=525568 Len=0 |
| 6 | 0.015321 | 127.0.0.1 | 127.0.0.1 | TCP | 195 | 8000 → 52812 [PSH, ACK] Seq=1 Ack=321 Win=525568 Len=155 [TCP PDU reassembled in 8] |
| 7 | 0.015321 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52812 → 8000 [ACK] Seq=321 Ack=156 Win=261984 Len=0 |
| 8 | 0.015321 | 127.0.0.1 | 127.0.0.1 | HTTP | 1419 | HTTP/1.0 200 OK (text/html) |
| 9 | 0.015321 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52812 → 8000 [ACK] Seq=321 Ack=1535 Win=262144 Len=0 |
| 10 | 0.015321 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52812 [FIN, ACK] Seq=1535 Ack=321 Win=525568 Len=0 |
| 11 | 0.015321 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52812 → 8000 [ACK] Seq=321 Ack=1536 Win=262144 Len=0 |
| 12 | 0.031255 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52812 → 8000 [FIN, ACK] Seq=321 Ack=1536 Win=262144 Len=0 |
| 13 | 0.031255 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52812 [ACK] Seq=1536 Ack=322 Win=525568 Len=0 |
| 14 | 2.894799 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52813 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 15 | 2.894799 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52813 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 16 | 2.894799 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52813 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 17 | 2.894799 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52814 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 18 | 2.894799 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52814 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 19 | 2.894799 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52814 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 20 | 2.894799 | 127.0.0.1 | 127.0.0.1 | HTTP | 406 | GET /1-MB-Test.txt HTTP/1.1 |
| 21 | 2.894799 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52814 [ACK] Seq=1 Ack=367 Win=525568 Len=0 |
| 22 | 27.066819 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52814 → 8000 [RST, ACK] Seq=367 Ack=1 Win=0 Len=0 |
| 23 | 27.066819 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52813 → 8000 [FIN, ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 24 | 27.066819 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52813 [ACK] Seq=1 Ack=2 Win=525568 Len=0 |
| 25 | 27.066819 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 52815 → 8000 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=8 SACK_PERM |
| 26 | 27.066819 | 127.0.0.1 | 127.0.0.1 | TCP | 52 | 8000 → 52815 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=65495 WS=256 SACK_PERM |
| 27 | 27.066819 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52815 → 8000 [ACK] Seq=1 Ack=1 Win=262144 Len=0 |
| 28 | 27.066819 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52813 [FIN, ACK] Seq=1 Ack=2 Win=525568 Len=0 |
| 29 | 27.066819 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 52813 → 8000 [ACK] Seq=2 Ack=2 Win=262144 Len=0 |
| 30 | 27.081985 | 127.0.0.1 | 127.0.0.1 | HTTP | 406 | GET /1-MB-Test.txt HTTP/1.1 |
| 31 | 27.081985 | 127.0.0.1 | 127.0.0.1 | TCP | 40 | 8000 → 52815 [ACK] Seq=1 Ack=367 Win=525568 Len=0 |
| 32 | 27.094801 | 127.0.0.1 | 127.0.0.1 | TCP | 227 | 8000 → 52815 [PSH, ACK] Seq=1 Ack=367 Win=525568 Len=187 [TCP PDU reassembled in 35] |

```
Frame 1: 52 bytes on wire (416 bits), 52 bytes captured (416 bits)
Raw packet data
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 52812, Dst Port: 8000, Seq: 0, Len: 0

0000  45 00 00 34 78 5f 40 00  80 06 00 00 7f 00 00 01
0010  7f 00 00 01 ce 4c 1f 40  fe 81 85 2f 00 00 00 00
0020  80 02 ff ff 05 b1 00 00  02 04 ff d7 01 03 03 03
0030  01 01 04 02
```

Protocol Hierarchy:

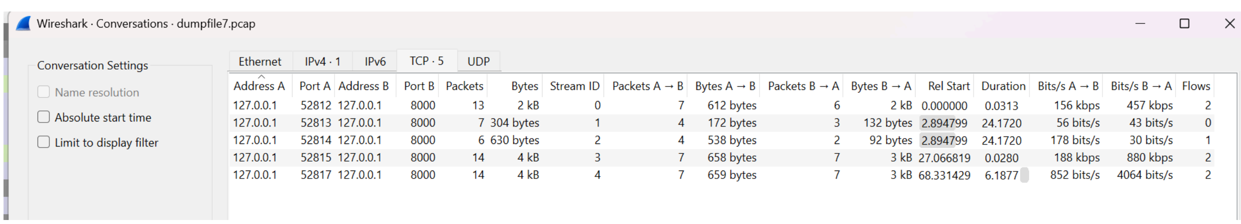


Wireshark · Protocol Hierarchy Statistics · dumpfile7.pcap

| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 54 | 100.0 | 10872 | 1167 | 0 | 0 | 0 | 54 |
| Raw packet data | 100.0 | 54 | 100.0 | 10872 | 1167 | 0 | 0 | 0 | 54 |
| Internet Protocol Version 4 | 100.0 | 54 | 9.9 | 1080 | 115 | 0 | 0 | 0 | 54 |
| Transmission Control Protocol | 100.0 | 54 | 11.0 | 1200 | 128 | 47 | 1060 | 113 | 54 |
| Hypertext Transfer Protocol | 13.0 | 7 | 17.9 | 1948 | 209 | 4 | 1419 | 152 | 7 |
| Line-based text data | 5.6 | 3 | 61.1 | 6644 | 713 | 3 | 6644 | 713 | 3 |

Breakdown of protocols: 13% of the packets are HTTP and 5.6% are line-based text. The majority of bytes transferred are HTTP object data.

Conversations:



Wireshark · Conversations · dumpfile7.pcap

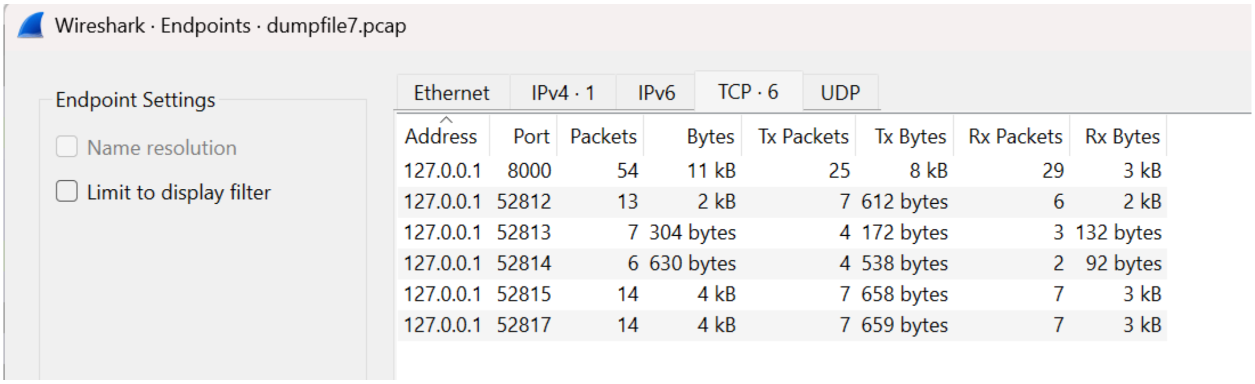
Conversation Settings
- Name resolution
- Absolute start time
- Limit to display filter

Ethernet | IPv4 · 1 | IPv6 | TCP · 5 | UDP

| Address A | Port A | Address B | Port B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A | Flows |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 52812 | 127.0.0.1 | 8000 | 13 | 2 kB | 0 | 7 | 612 bytes | 6 | 2 kB | 0.000000 | 0.0313 | 156 kbps | 457 kbps | 2 |
| 127.0.0.1 | 52813 | 127.0.0.1 | 8000 | 7 | 304 bytes | 1 | 4 | 172 bytes | 3 | 132 bytes | 2.894799 | 24.1720 | 56 bits/s | 43 bits/s | 0 |
| 127.0.0.1 | 52814 | 127.0.0.1 | 8000 | 6 | 630 bytes | 2 | 4 | 538 bytes | 2 | 92 bytes | 2.894799 | 24.1720 | 178 bits/s | 30 bits/s | 1 |
| 127.0.0.1 | 52815 | 127.0.0.1 | 8000 | 14 | 4 kB | 3 | 7 | 658 bytes | 7 | 3 kB | 27.066819 | 0.0280 | 188 kbps | 880 kbps | 2 |
| 127.0.0.1 | 52817 | 127.0.0.1 | 8000 | 14 | 4 kB | 4 | 7 | 659 bytes | 7 | 3 kB | 68.331429 | 6.1877 | 852 bits/s | 4064 bits/s | 2 |

Five TCP streams are listed, with streams 3 and 4 transferring the largest volume of data (each ~3 kB), likely tied to the text files retrieved.

Endpoints:



Wireshark · Endpoints · dumpfile7.pcap

Endpoint Settings

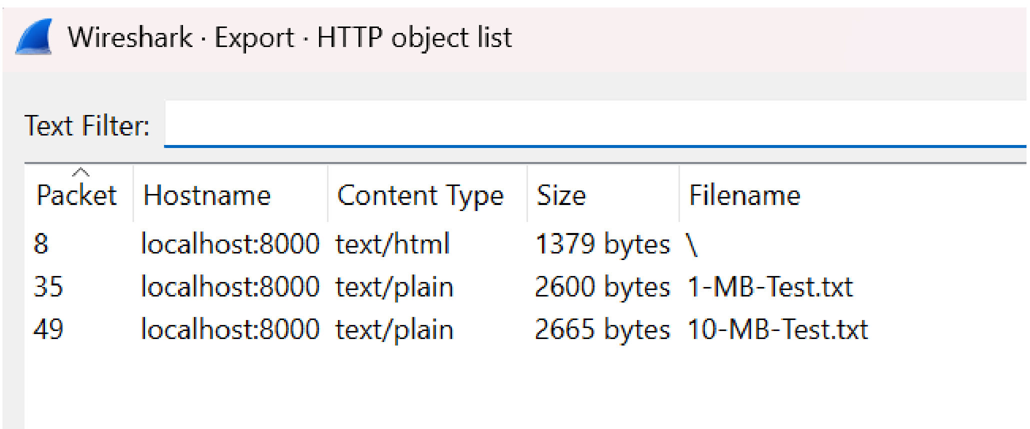
Name resolution

Limit to display filter

Ethernet | IPv4 · 1 | IPv6 | TCP · 6 | UDP

| Address | Port | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|---|
| 127.0.0.1 | 8000 | 54 | 11 kB | 25 | 8 kB | 29 | 3 kB |
| 127.0.0.1 | 52812 | 13 | 2 kB | 7 | 612 bytes | 6 | 2 kB |
| 127.0.0.1 | 52813 | 7 | 304 bytes | 4 | 172 bytes | 3 | 132 bytes |
| 127.0.0.1 | 52814 | 6 | 630 bytes | 4 | 538 bytes | 2 | 92 bytes |
| 127.0.0.1 | 52815 | 14 | 4 kB | 7 | 658 bytes | 7 | 3 kB |
| 127.0.0.1 | 52817 | 14 | 4 kB | 7 | 659 bytes | 7 | 3 kB |

List of communicating IP addresses and ports. Most traffic flows between 127.0.0.1:8000 and multiple ephemeral ports. Port 52815 and 52817 have the highest data exchange.



Wireshark · Export · HTTP object list

Text Filter:

| Packet | Hostname | Content Type | Size | Filename |
|---|---|---|---|---|
| 8 | localhost:8000 | text/html | 1379 bytes | \ |
| 35 | localhost:8000 | text/plain | 2600 bytes | 1-MB-Test.txt |
| 49 | localhost:8000 | text/plain | 2665 bytes | 10-MB-Test.txt |

**HTTP Object List Export**

Displays three downloadable objects via HTTP: one HTML index file and two large plaintext files (1-MB-Test.txt and 10-MB-Test.txt).

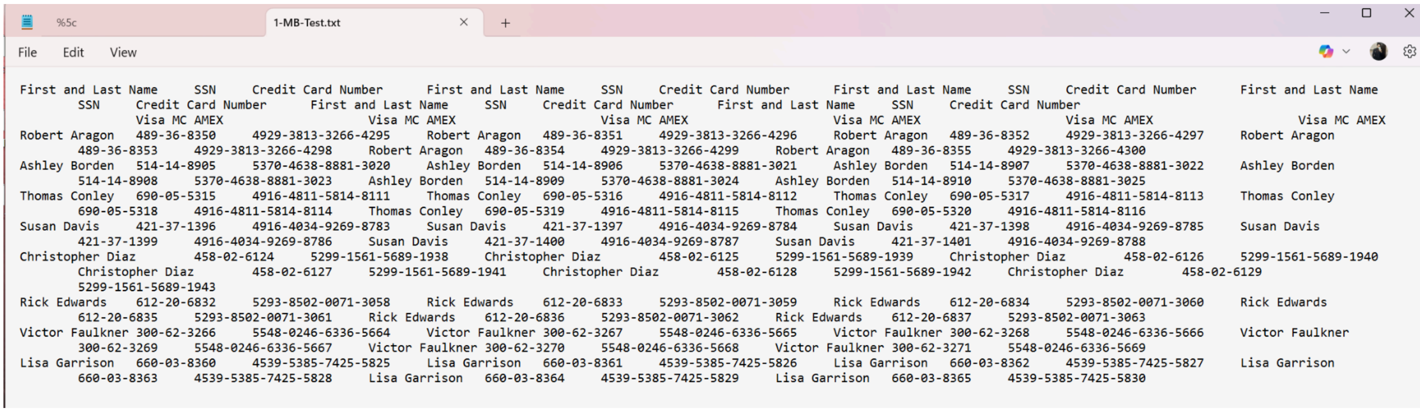
%5c

File Edit View

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="1-MB-Test.docx">1-MB-Test.docx</a></li>
<li><a href="1-MB-Test.txt">1-MB-Test.txt</a></li>
<li><a href="10-MB-Test.docx">10-MB-Test.docx</a></li>
<li><a href="10-MB-Test.txt">10-MB-Test.txt</a></li>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="encoded%20file.txt">encoded file.txt</a></li>
<li><a href="file1.txt">file1.txt</a></li>
<li><a href="file2.txt">file2.txt</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
</html>
```

**HTML Index Page:**

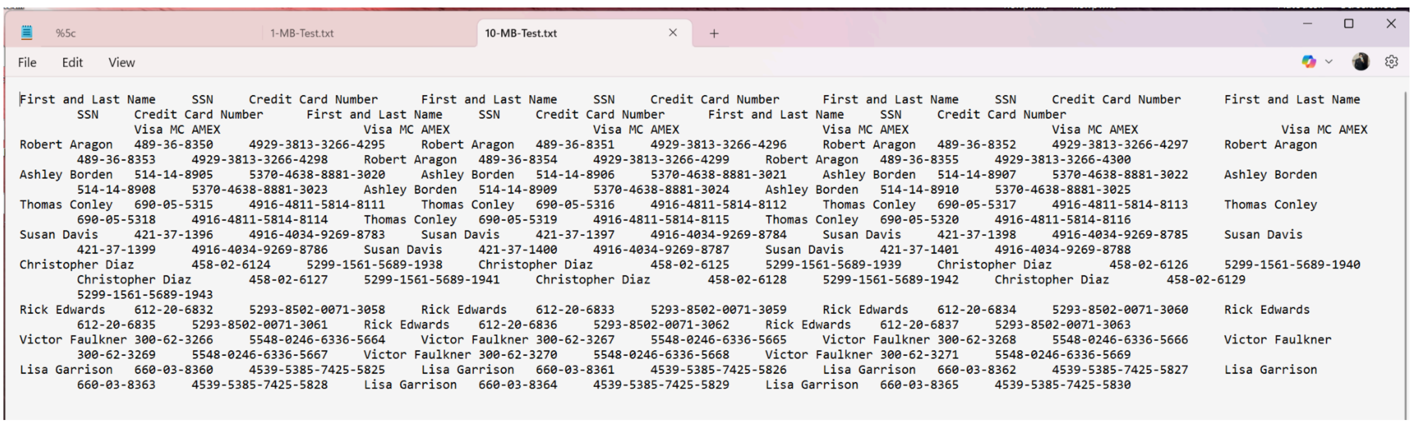
A rendered directory listing from the localhost web server showing access to multiple .txt and .docx files, including the test files.



```
First and Last Name     SSN     Credit Card Number      First and Last Name     SSN     Credit Card Number      First and Last Name     SSN     Credit Card Number      First and Last Name
        SSN     Credit Card Number      First and Last Name     SSN     Credit Card Number      First and Last Name     SSN     Credit Card Number
                Visa MC AMEX                    Visa MC AMEX                    Visa MC AMEX                    Visa MC AMEX                    Visa MC AMEX                    Visa MC AMEX
Robert Aragon   489-36-8350     4929-3813-3266-4295     Robert Aragon   489-36-8351     4929-3813-3266-4296     Robert Aragon   489-36-8352     4929-3813-3266-4297     Robert Aragon
        489-36-8353     4929-3813-3266-4298     Robert Aragon   489-36-8354     4929-3813-3266-4299     Robert Aragon   489-36-8355     4929-3813-3266-4300
Ashley Borden   514-14-8905     5370-4638-8881-3020     Ashley Borden   514-14-8906     5370-4638-8881-3021     Ashley Borden   514-14-8907     5370-4638-8881-3022     Ashley Borden
        514-14-8908     5370-4638-8881-3023     Ashley Borden   514-14-8909     5370-4638-8881-3024     Ashley Borden   514-14-8910     5370-4638-8881-3025
Thomas Conley   690-05-5315     4916-4811-5814-8111     Thomas Conley   690-05-5316     4916-4811-5814-8112     Thomas Conley   690-05-5317     4916-4811-5814-8113     Thomas Conley
        690-05-5318     4916-4811-5814-8114     Thomas Conley   690-05-5319     4916-4811-5814-8115     Thomas Conley   690-05-5320     4916-4811-5814-8116
Susan Davis     421-37-1396     4916-4034-9269-8783     Susan Davis     421-37-1397     4916-4034-9269-8784     Susan Davis     421-37-1398     4916-4034-9269-8785     Susan Davis
        421-37-1399     4916-4034-9269-8786     Susan Davis     421-37-1400     4916-4034-9269-8787     Susan Davis     421-37-1401     4916-4034-9269-8788
Christopher Diaz        458-02-6124     5299-1561-5689-1938     Christopher Diaz        458-02-6125     5299-1561-5689-1939     Christopher Diaz        458-02-6126     5299-1561-5689-1940
        Christopher Diaz        458-02-6127     5299-1561-5689-1941     Christopher Diaz        458-02-6128     5299-1561-5689-1942     Christopher Diaz        458-02-6129
        5299-1561-5689-1943
Rick Edwards    612-20-6832     5293-8502-0071-3058     Rick Edwards    612-20-6833     5293-8502-0071-3059     Rick Edwards    612-20-6834     5293-8502-0071-3060     Rick Edwards
        612-20-6835     5293-8502-0071-3061     Rick Edwards    612-20-6836     5293-8502-0071-3062     Rick Edwards    612-20-6837     5293-8502-0071-3063
Victor Faulkner 300-62-3266     5548-0246-6336-5664     Victor Faulkner 300-62-3267     5548-0246-6336-5665     Victor Faulkner 300-62-3268     5548-0246-6336-5666     Victor Faulkner
        300-62-3269     5548-0246-6336-5667     Victor Faulkner 300-62-3270     5548-0246-6336-5668     Victor Faulkner 300-62-3271     5548-0246-6336-5669
Lisa Garrison   660-03-8360     4539-5385-7425-5825     Lisa Garrison   660-03-8361     4539-5385-7425-5826     Lisa Garrison   660-03-8362     4539-5385-7425-5827     Lisa Garrison
        660-03-8363     4539-5385-7425-5828     Lisa Garrison   660-03-8364     4539-5385-7425-5829     Lisa Garrison   660-03-8365     4539-5385-7425-5830
```

**Content of `1-MB-Test.txt`**

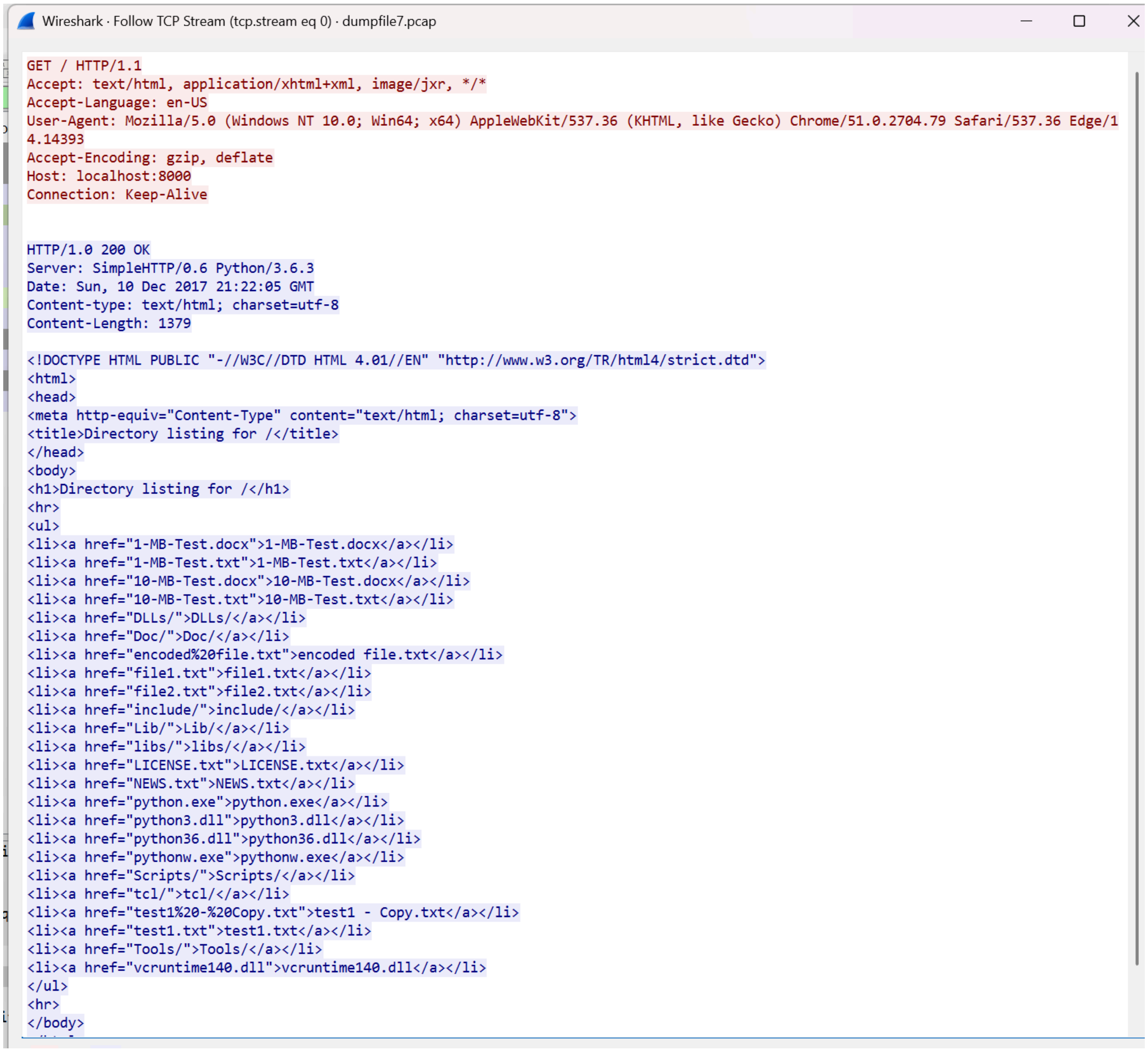
*A plaintext file containing names, Social Security Numbers, and credit card numbers—indicative of a significant PII leak.*



```
First and Last Name     SSN     Credit Card Number      First and Last Name     SSN     Credit Card Number      First and Last Name     SSN     Credit Card Number      First and Last Name
        SSN     Credit Card Number      First and Last Name     SSN     Credit Card Number      First and Last Name     SSN     Credit Card Number
                Visa MC AMEX                    Visa MC AMEX                    Visa MC AMEX                    Visa MC AMEX                    Visa MC AMEX                    Visa MC AMEX
Robert Aragon   489-36-8350     4929-3813-3266-4295     Robert Aragon   489-36-8351     4929-3813-3266-4296     Robert Aragon   489-36-8352     4929-3813-3266-4297     Robert Aragon
        489-36-8353     4929-3813-3266-4298     Robert Aragon   489-36-8354     4929-3813-3266-4299     Robert Aragon   489-36-8355     4929-3813-3266-4300
Ashley Borden   514-14-8905     5370-4638-8881-3020     Ashley Borden   514-14-8906     5370-4638-8881-3021     Ashley Borden   514-14-8907     5370-4638-8881-3022     Ashley Borden
        514-14-8908     5370-4638-8881-3023     Ashley Borden   514-14-8909     5370-4638-8881-3024     Ashley Borden   514-14-8910     5370-4638-8881-3025
Thomas Conley   690-05-5315     4916-4811-5814-8111     Thomas Conley   690-05-5316     4916-4811-5814-8112     Thomas Conley   690-05-5317     4916-4811-5814-8113     Thomas Conley
        690-05-5318     4916-4811-5814-8114     Thomas Conley   690-05-5319     4916-4811-5814-8115     Thomas Conley   690-05-5320     4916-4811-5814-8116
Susan Davis     421-37-1396     4916-4034-9269-8783     Susan Davis     421-37-1397     4916-4034-9269-8784     Susan Davis     421-37-1398     4916-4034-9269-8785     Susan Davis
        421-37-1399     4916-4034-9269-8786     Susan Davis     421-37-1400     4916-4034-9269-8787     Susan Davis     421-37-1401     4916-4034-9269-8788
Christopher Diaz        458-02-6124     5299-1561-5689-1938     Christopher Diaz        458-02-6125     5299-1561-5689-1939     Christopher Diaz        458-02-6126     5299-1561-5689-1940
        Christopher Diaz        458-02-6127     5299-1561-5689-1941     Christopher Diaz        458-02-6128     5299-1561-5689-1942     Christopher Diaz        458-02-6129
        5299-1561-5689-1943
Rick Edwards    612-20-6832     5293-8502-0071-3058     Rick Edwards    612-20-6833     5293-8502-0071-3059     Rick Edwards    612-20-6834     5293-8502-0071-3060     Rick Edwards
        612-20-6835     5293-8502-0071-3061     Rick Edwards    612-20-6836     5293-8502-0071-3062     Rick Edwards    612-20-6837     5293-8502-0071-3063
Victor Faulkner 300-62-3266     5548-0246-6336-5664     Victor Faulkner 300-62-3267     5548-0246-6336-5665     Victor Faulkner 300-62-3268     5548-0246-6336-5666     Victor Faulkner
        300-62-3269     5548-0246-6336-5667     Victor Faulkner 300-62-3270     5548-0246-6336-5668     Victor Faulkner 300-62-3271     5548-0246-6336-5669
Lisa Garrison   660-03-8360     4539-5385-7425-5825     Lisa Garrison   660-03-8361     4539-5385-7425-5826     Lisa Garrison   660-03-8362     4539-5385-7425-5827     Lisa Garrison
        660-03-8363     4539-5385-7425-5828     Lisa Garrison   660-03-8364     4539-5385-7425-5829     Lisa Garrison   660-03-8365     4539-5385-7425-5830
```

**Content of `10-MB-Test.txt`**

*Another plaintext file mirroring the same sensitive data structure as `1-MB-Test.txt`, confirming a larger volume of leaked information.*



```
Wireshark · Follow TCP Stream (tcp.stream eq 0) · dumpfile7.pcap

GET / HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/1
4.14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 21:22:05 GMT
Content-type: text/html; charset=utf-8
Content-Length: 1379

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="1-MB-Test.docx">1-MB-Test.docx</a></li>
<li><a href="1-MB-Test.txt">1-MB-Test.txt</a></li>
<li><a href="10-MB-Test.docx">10-MB-Test.docx</a></li>
<li><a href="10-MB-Test.txt">10-MB-Test.txt</a></li>
<li><a href="DLLs/">DLLs/</a></li>
<li><a href="Doc/">Doc/</a></li>
<li><a href="encoded%20file.txt">encoded file.txt</a></li>
<li><a href="file1.txt">file1.txt</a></li>
<li><a href="file2.txt">file2.txt</a></li>
<li><a href="include/">include/</a></li>
<li><a href="Lib/">Lib/</a></li>
<li><a href="libs/">libs/</a></li>
<li><a href="LICENSE.txt">LICENSE.txt</a></li>
<li><a href="NEWS.txt">NEWS.txt</a></li>
<li><a href="python.exe">python.exe</a></li>
<li><a href="python3.dll">python3.dll</a></li>
<li><a href="python36.dll">python36.dll</a></li>
<li><a href="pythonw.exe">pythonw.exe</a></li>
<li><a href="Scripts/">Scripts/</a></li>
<li><a href="tcl/">tcl/</a></li>
<li><a href="test1%20-%20Copy.txt">test1 - Copy.txt</a></li>
<li><a href="test1.txt">test1.txt</a></li>
<li><a href="Tools/">Tools/</a></li>
<li><a href="vcruntime140.dll">vcruntime140.dll</a></li>
</ul>
<hr>
</body>
```

**Stream 0**:

**Request**: GET `/`

**Response**: Directory listing (HTML) — lists various files including `1-MB-Test.txt` and `10-MB-Test.txt`.

**Content Type**: `text/html`

```
Wireshark · Follow TCP Stream (tcp.stream eq 3) · dumpfile7.pcap

GET /1-MB-Test.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 21:22:32 GMT
Content-type: text/plain
Content-Length: 2600
Last-Modified: Sun, 10 Dec 2017 21:16:16 GMT

First and Last Name    SSN         Credit Card Number    First and Last Name    SSN         Credit Card Number    First and Last Name    SSN         Credit Card Number    First and Last Name    SSN         Credit Card Number    First and
Last Name    SSN         Credit Card Number    First and Last Name    SSN         Credit Card Number
                       Visa MC AMEX                                 Visa MC AMEX                                 Visa MC AMEX                                 Visa MC AMEX
             Visa MC AMEX
Robert.Aragon          489-36-83504929-3813-3266-4295    Robert.Aragon          489-36-83514929-3813-3266-4296    Robert.Aragon          489-36-83524929-3813-3266-4297    Robert.Aragon          489-36-83534929-3813-3266-4298    Robert.Ara
gon          489-36-83544929-3813-3266-4299    Robert.Aragon          489-36-83554929-3813-3266-4300
Ashley.Borden          514-14-89055370-4638-8881-3020    Ashley.Borden          514-14-89065370-4638-8881-3021    Ashley.Borden          514-14-89075370-4638-8881-3022    Ashley.Borden          514-14-89085370-4638-8881-3023    Ashley.Bor
den          514-14-89095370-4638-8881-3024    Ashley.Borden          514-14-89105370-4638-8881-3025
Thomas.Conley          690-05-53154916-4811-5814-8111    Thomas.Conley          690-05-53164916-4811-5814-8112    Thomas.Conley          690-05-53174916-4811-5814-8113    Thomas.Conley          690-05-53184916-4811-5814-8114    Thomas.Con
ley          690-05-53194916-4811-5814-8115    Thomas.Conley          690-05-53204916-4811-5814-8116
Susan.Davis421-37-13964916-4034-9269-8783    Susan.Davis421-37-13974916-4034-9269-8784    Susan.Davis421-37-13984916-4034-9269-8785    Susan.Davis421-37-13994916-4034-9269-8786    Susan.Davis421-37-14004916-4034-9269-8787    Susan.Davi
s            421-37-14014916-4034-9269-8788
Christopher.Diaz       458-02-61245299-1561-5689-1938    Christopher.Diaz       458-02-61255299-1561-5689-1939    Christopher.Diaz       458-02-61265299-1561-5689-1940    Christopher.Diaz       458-02-61275299-1561-5689-1941    Christophe
r.Diaz       458-02-61285299-1561-5689-1942    Christopher.Diaz       458-02-61295299-1561-5689-1943
Rick.Edwards           612-20-68325293-8502-0071-3058    Rick.Edwards           612-20-68335293-8502-0071-3059    Rick.Edwards           612-20-68345293-8502-0071-3060    Rick.Edwards           612-20-68355293-8502-0071-3061    Rick.Edwar
ds           612-20-68365293-8502-0071-3062    Rick.Edwards           612-20-68375293-8502-0071-3063
Victor.Faulkner        300-62-32665548-0246-6336-5664    Victor.Faulkner        300-62-32675548-0246-6336-5665    Victor.Faulkner        300-62-32685548-0246-6336-5666    Victor.Faulkner        300-62-32695548-0246-6336-5667    Victor.Fau
lkner        300-62-32705548-0246-6336-5668    Victor.Faulkner        300-62-32715548-0246-6336-5669
Lisa.Garrison          660-03-83604539-5385-7425-5825    Lisa.Garrison          660-03-83614539-5385-7425-5826    Lisa.Garrison          660-03-83624539-5385-7425-5827    Lisa.Garrison          660-03-83634539-5385-7425-5828    Lisa.Garri
son          660-03-83644539-5385-7425-5829    Lisa.Garrison          660-03-83654539-5385-7425-5830
```

**Stream 3**:

**Request**: GET `/1-MB-Test.txt`

**Response**: Returns `1-MB-Test.txt` (2600 bytes) containing names, SSNs, and credit card numbers.

**Content Type**: `text/plain`

```
Wireshark · Follow TCP Stream (tcp.stream eq 4) · dumpfile7.pcap

GET /10-MB-Test.txt HTTP/1.1
Accept: text/html, application/xhtml+xml, image/jxr, */*
Referer: http://localhost:8000/
Accept-Language: en-US
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/51.0.2704.79 Safari/537.36 Edge/14.14393
Accept-Encoding: gzip, deflate
Host: localhost:8000
Connection: Keep-Alive


HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.6.3
Date: Sun, 10 Dec 2017 21:23:19 GMT
Content-type: text/plain
Content-Length: 2665
Last-Modified: Sun, 10 Dec 2017 21:16:51 GMT

First and Last Name    SSN         Credit Card Number    First and Last Name    SSN         Credit Card Number    First and Last Name    SSN         Credit Card Number    First and Last Name    SSN         Credit Card Number    First a
nd Last Name           SSN         Credit Card Number    First and Last Name    SSN         Credit Card Number
                       Visa MC AMEX                                 Visa MC AMEX                                 Visa MC AMEX                                 Visa MC AMEX
             Visa MC AMEX
Robert.Aragon          489-36-83504929-3813-3266-4295    Robert.Aragon          489-36-83514929-3813-3266-4296    Robert.Aragon          489-36-83524929-3813-3266-4297    Robert.Aragon          489-36-83534929-3813-3266-4298    Robert.
Aragon       489-36-83544929-3813-3266-4299    Robert.Aragon          489-36-83554929-3813-3266-4300
Ashley.Borden          514-14-89055370-4638-8881-3020    Ashley.Borden          514-14-89065370-4638-8881-3021    Ashley.Borden          514-14-89075370-4638-8881-3022    Ashley.Borden          514-14-89085370-4638-8881-3023    Ashley.
Borden       514-14-89095370-4638-8881-3024    Ashley.Borden          514-14-89105370-4638-8881-3025
Thomas.Conley          690-05-53154916-4811-5814-8111    Thomas.Conley          690-05-53164916-4811-5814-8112    Thomas.Conley          690-05-53174916-4811-5814-8113    Thomas.Conley          690-05-53184916-4811-5814-8114    Thomas.
Conley       690-05-53194916-4811-5814-8115    Thomas.Conley          690-05-53204916-4811-5814-8116
Susan.Davis421-37-13964916-4034-9269-8783    Susan.Davis421-37-13974916-4034-9269-8784    Susan.Davis421-37-13984916-4034-9269-8785    Susan.Davis421-37-13994916-4034-9269-8786    Susan.Davis421-37-14004916-4034-9269-8787    Susan.D
avis         421-37-14014916-4034-9269-8788
Christopher.Diaz       458-02-61245299-1561-5689-1938    Christopher.Diaz       458-02-61255299-1561-5689-1939    Christopher.Diaz       458-02-61265299-1561-5689-1940    Christopher.Diaz       458-02-61275299-1561-5689-1941    Christo
pher.Diaz    458-02-61285299-1561-5689-1942    Christopher.Diaz       458-02-61295299-1561-5689-1943
Rick.Edwards           612-20-68325293-8502-0071-3058    Rick.Edwards           612-20-68335293-8502-0071-3059    Rick.Edwards           612-20-68345293-8502-0071-3060    Rick.Edwards           612-20-68355293-8502-0071-3061    Rick.Ed
wards        612-20-68365293-8502-0071-3062    Rick.Edwards           612-20-68375293-8502-0071-3063
Victor.Faulkner        300-62-32665548-0246-6336-5664    Victor.Faulkner        300-62-32675548-0246-6336-5665    Victor.Faulkner        300-62-32685548-0246-6336-5666    Victor.Faulkner        300-62-32695548-0246-6336-5667    Victor.
Faulkner     300-62-32705548-0246-6336-5668    Victor.Faulkner        300-62-32715548-0246-6336-5669
Lisa.Garrison          660-03-83604539-5385-7425-5825    Lisa.Garrison          660-03-83614539-5385-7425-5826    Lisa.Garrison          660-03-83624539-5385-7425-5827    Lisa.Garrison          660-03-83634539-5385-7425-5828    Lisa.Ga
rrison       660-03-83644539-5385-7425-5829    Lisa.Garrison          660-03-83654539-5385-7425-5830
```

**Stream 4**:

**Request**: GET `/10-MB-Test.txt`

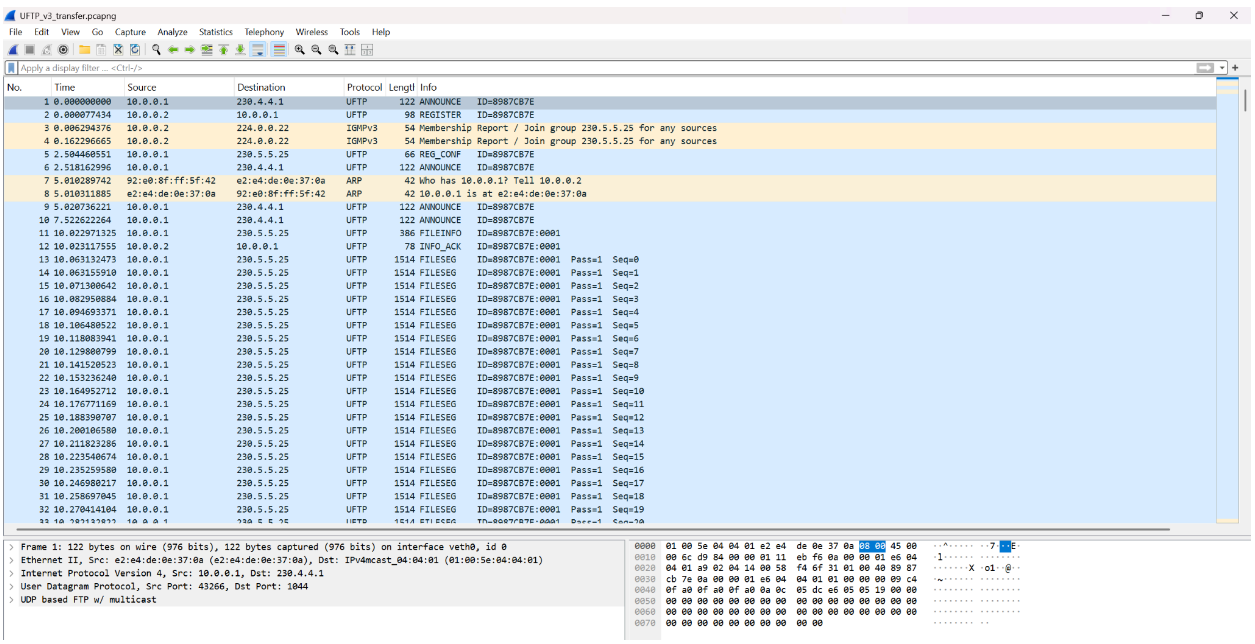
**Response**: Returns `10-MB-Test.txt` (2665 bytes) with similar sensitive data as `1-MB-Test.txt`.

**Content Type**: `text/plain`

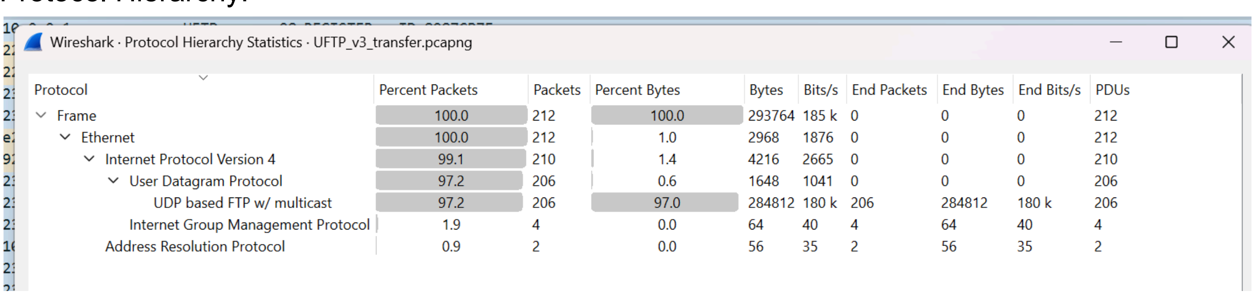
## Summary of dumpfile7.pcap

The PCAP file `dumpfile7.pcap` captures HTTP GET requests made to a local web server hosted on port 8000. A directory listing is retrieved, and two text files, `1-MB-Test.txt` and `10-MB-Test.txt`, are downloaded. Both files contain sensitive personally identifiable information (PII), including full names, Social Security Numbers (SSNs), and credit card details. This data transfer may indicate a data leak or unauthorized access of private records over HTTP.

## UFTP_v3_transfer.pcapng:



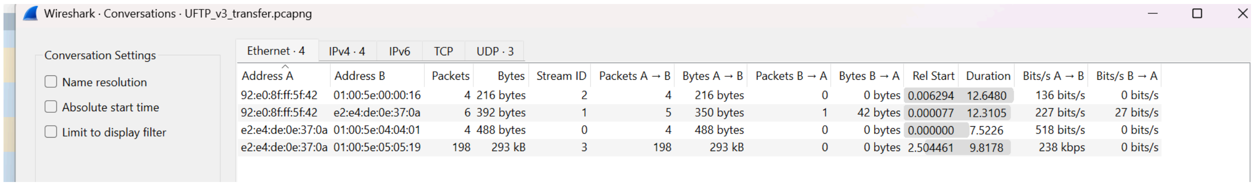

Protocol Hierarchy:




| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 212 | 100.0 | 293764 | 185 k | 0 | 0 | 0 | 212 |
| Ethernet | 100.0 | 212 | 1.0 | 2968 | 1876 | 0 | 0 | 0 | 212 |
| Internet Protocol Version 4 | 99.1 | 210 | 1.4 | 4216 | 2665 | 0 | 0 | 0 | 210 |
| User Datagram Protocol | 97.2 | 206 | 0.6 | 1648 | 1041 | 0 | 0 | 0 | 206 |
| UDP based FTP w/ multicast | 97.2 | 206 | 97.0 | 284812 | 180 k | 206 | 284812 | 180 k | 206 |
| Internet Group Management Protocol | 1.9 | 4 | 0.0 | 64 | 40 | 4 | 64 | 40 | 4 |
| Address Resolution Protocol | 0.9 | 2 | 0.0 | 56 | 35 | 2 | 56 | 35 | 2 |

Confirms UFTP traffic dominates the capture — 97.2% of packets and bytes are UDP-based FTP with multicast. Minor traffic includes IGMP (1.9%) and ARP (0.9%).

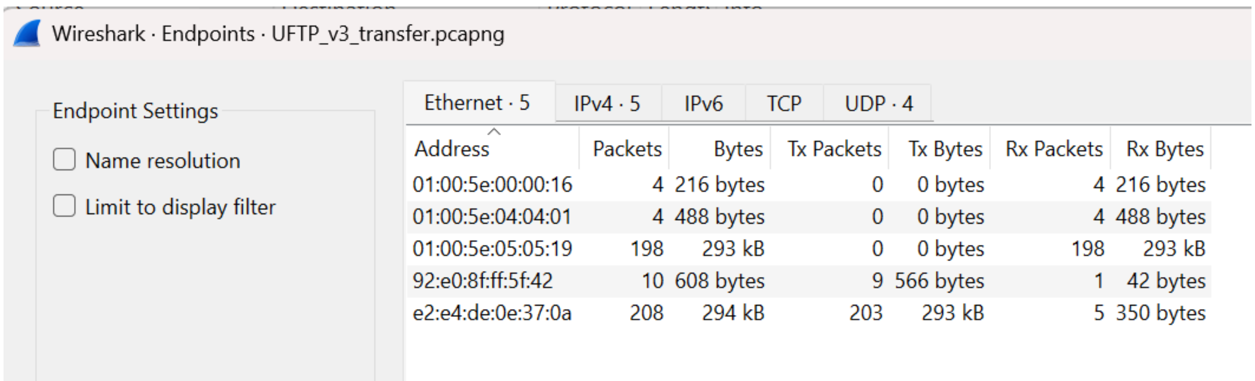
Conversations:




| Address A | Address B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 92:e0:8f:ff:5f:42 | 01:00:5e:00:00:16 | 4 | 216 bytes | 2 | 4 | 216 bytes | 0 | 0 bytes | 0.006294 | 12.6480 | 136 bits/s | 0 bits/s |
| 92:e0:8f:ff:5f:42 | e2:e4:de:0e:37:0a | 6 | 392 bytes | 1 | 5 | 350 bytes | 1 | 42 bytes | 0.000077 | 12.3105 | 227 bits/s | 27 bits/s |
| e2:e4:de:0e:37:0a | 01:00:5e:04:04:01 | 4 | 488 bytes | 0 | 4 | 488 bytes | 0 | 0 bytes | 0.000000 | 7.5226 | 518 bits/s | 0 bits/s |
| e2:e4:de:0e:37:0a | 01:00:5e:05:05:19 | 198 | 293 kB | 3 | 198 | 293 kB | 0 | 0 bytes | 2.504461 | 9.8178 | 238 kbps | 0 bits/s |

Highlights four UDP streams. Stream ID 3 stands out with 198 packets and 293 KB transferred from `e2:e4:de:0e:37:0a` to `01:00:5e:05:05:19`, marking it as the main file transfer stream.

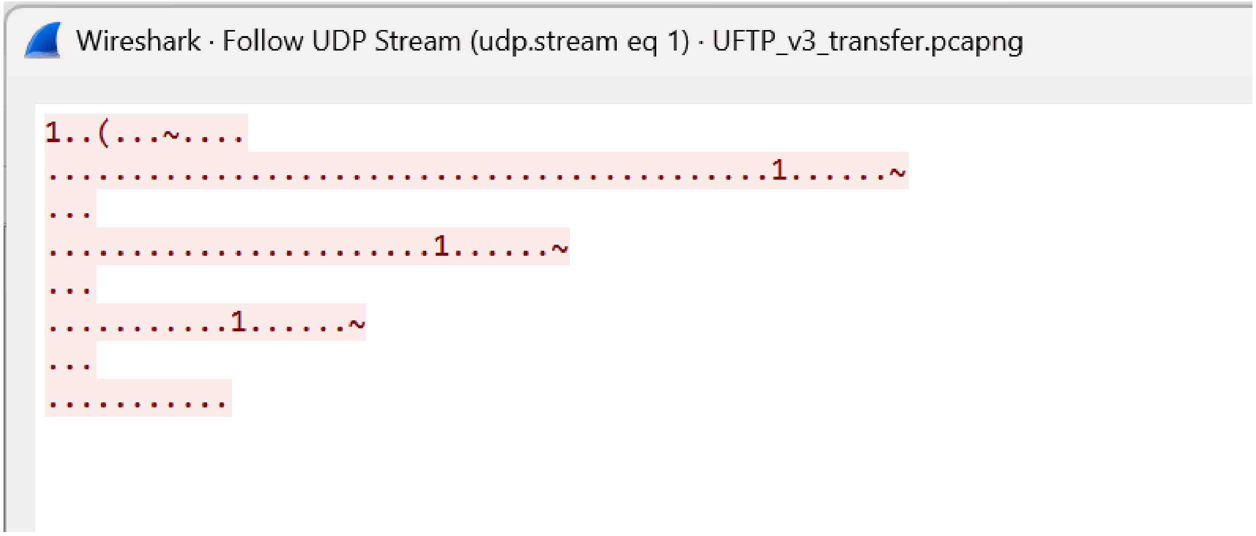
Endpoints:




| Address | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|
| 01:00:5e:00:00:16 | 4 | 216 bytes | 0 | 0 bytes | 4 | 216 bytes |
| 01:00:5e:04:04:01 | 4 | 488 bytes | 0 | 0 bytes | 4 | 488 bytes |
| 01:00:5e:05:05:19 | 198 | 293 kB | 0 | 0 bytes | 198 | 293 kB |
| 92:e0:8f:ff:5f:42 | 10 | 608 bytes | 9 | 566 bytes | 1 | 42 bytes |
| e2:e4:de:0e:37:0a | 208 | 294 kB | 203 | 293 kB | 5 | 350 bytes |

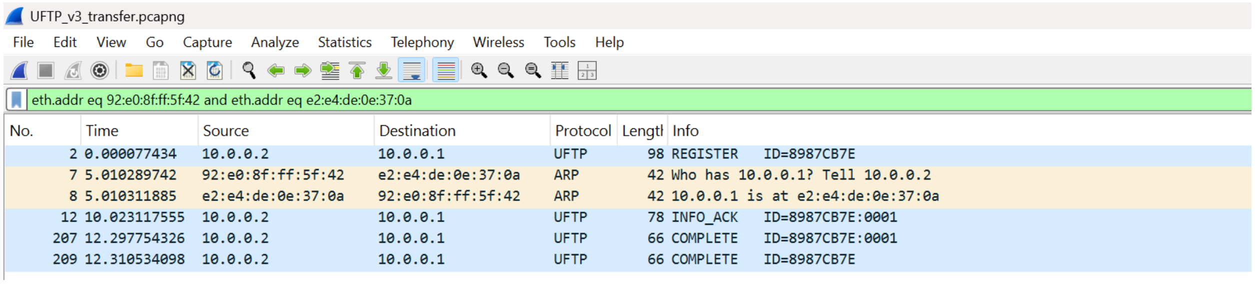
Lists participating Ethernet endpoints and their data usage. The sender `e2:e4:de:0e:37:0a` transmitted most of the data (294 KB), reinforcing its role as the source.


**HTTP Object Export (Empty)**
No HTTP-based objects were found in this UFTP capture, confirming the file transfer occurred entirely over UDP multicast, not HTTP.




**UDP Stream 1 Followed**
Shows raw binary data visualization for UDP Stream 1. The output is not human-readable and must be saved as raw data for reconstruction and analysis.


Following UDP streams showed successful UFTP file segment transmissions containing binary data. As seen in Stream 1 and 2, this data is not human-readable, indicating binary file transfer was conducted over multicast. Due to the lack of headers or known encoding, content reconstruction was not feasible within the scope of this investigation.

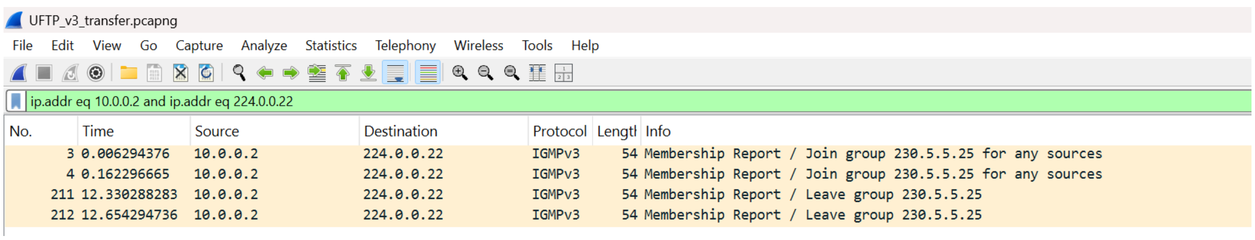


UFTP_v3_transfer.pcapng

eth.addr eq 92:e0:8f:ff:5f:42 and eth.addr eq e2:e4:de:0e:37:0a

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 2 | 0.000077434 | 10.0.0.2 | 10.0.0.1 | UFTP | 98 | REGISTER ID=8987CB7E |
| 7 | 5.010289742 | 92:e0:8f:ff:5f:42 | e2:e4:de:0e:37:0a | ARP | 42 | Who has 10.0.0.1? Tell 10.0.0.2 |
| 8 | 5.010311885 | e2:e4:de:0e:37:0a | 92:e0:8f:ff:5f:42 | ARP | 42 | 10.0.0.1 is at e2:e4:de:0e:37:0a |
| 12 | 10.023117555 | 10.0.0.2 | 10.0.0.1 | UFTP | 78 | INFO_ACK ID=8987CB7E:0001 |
| 207 | 12.297754326 | 10.0.0.2 | 10.0.0.1 | UFTP | 66 | COMPLETE ID=8987CB7E:0001 |
| 209 | 12.310534098 | 10.0.0.2 | 10.0.0.1 | UFTP | 66 | COMPLETE ID=8987CB7E |

Filtered traffic showing ARP and UFTP messages exchanged between devices with MAC addresses `92:e0:8f:ff:5f:42` and `e2:e4:de:0e:37:0a`. The capture includes an ARP request and reply used for IP-to-MAC resolution, followed by UFTP REGISTER, INFO_ACK, and COMPLETE messages indicating a successful file transfer session initialization and conclusion.

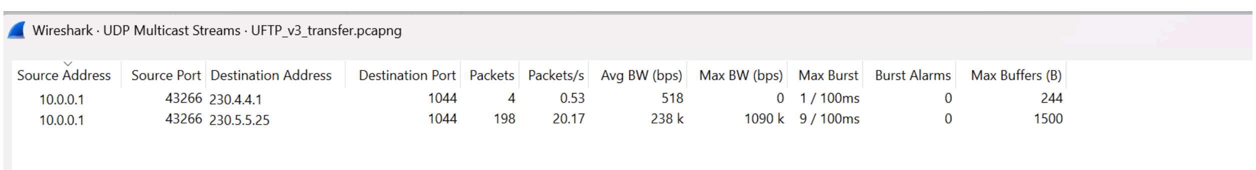


UFTP_v3_transfer.pcapng

ip.addr eq 10.0.0.2 and ip.addr eq 224.0.0.22

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 3 | 0.006294376 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.5.5.25 for any sources |
| 4 | 0.162296665 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.5.5.25 for any sources |
| 211 | 12.330288283 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Leave group 230.5.5.25 |
| 212 | 12.654294736 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Leave group 230.5.5.25 |

IGMPv3 packets from host 10.0.0.2 demonstrate group membership behavior, including joining and later leaving the multicast group 230.5.5.25. This confirms dynamic multicast participation during the UFTP file transfer session.



Wireshark · UDP Multicast Streams · UFTP_v3_transfer.pcapng

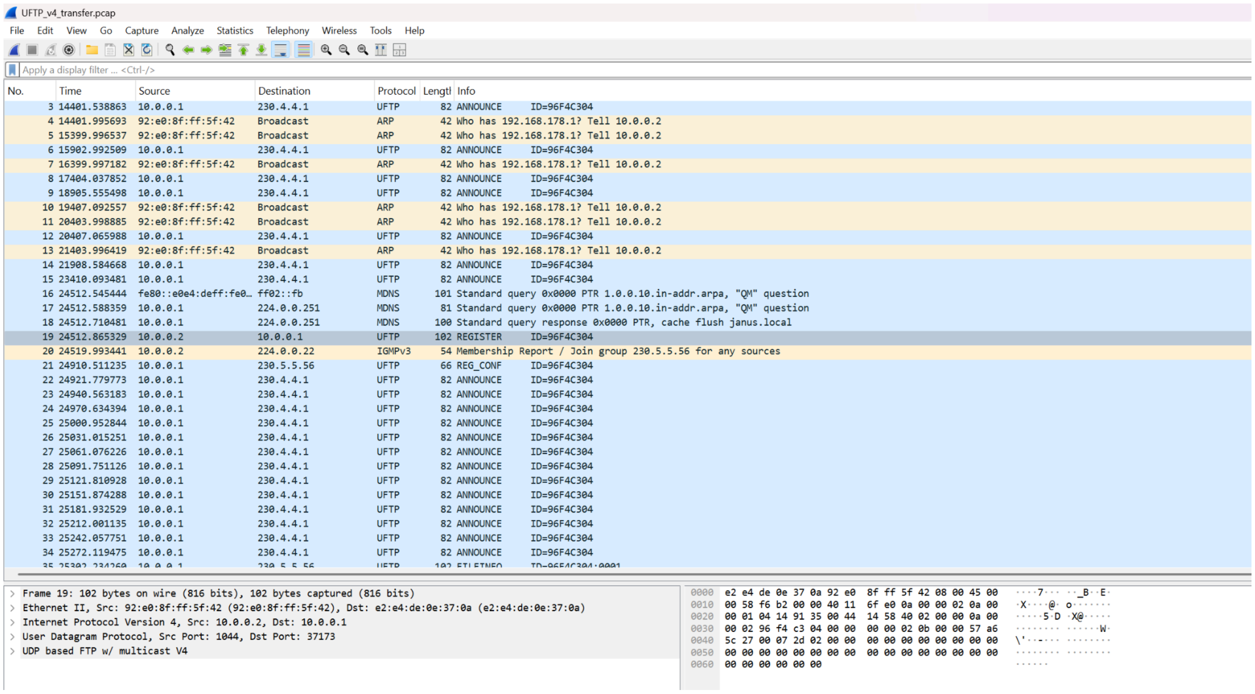
| Source Address | Source Port | Destination Address | Destination Port | Packets | Packets/s | Avg BW (bps) | Max BW (bps) | Max Burst | Burst Alarms | Max Buffers (B) |
|---|---|---|---|---|---|---|---|---|---|---|
| 10.0.0.1 | 43266 | 230.4.4.1 | 1044 | 4 | 0.53 | 518 | 0 | 1 / 100ms | 0 | 244 |
| 10.0.0.1 | 43266 | 230.5.5.25 | 1044 | 198 | 20.17 | 238 k | 1090 k | 9 / 100ms | 0 | 1500 |

**UDP Multicast Streams – UFTP_v3_transfer.pcapng**

This image shows multicast stream stats during UFTP v3 transfers. Two streams are observed from 10.0.0.1: one to 230.4.4.1 with very low activity (4 packets total), and another to 230.5.5.25 which carried 198 packets at a peak bandwidth of 1090 kbps and an average of 238 kbps. These values show that most of the transfer activity was directed to the second group.

**Summary:** The `UFTP_v3_transfer.pcapng` file captures a UDP-based multicast file transfer using the UFTP protocol. The session begins with ARP resolution between the sender and receiver, followed by IGMPv3 membership reports where the client joins the multicast group `230.5.5.25`. UFTP control messages such as `ANNOUNCE`, `REGISTER`, `FILEINFO`, `INFO_ACK`, and segmented `FILESEG` packets document the initiation and data transmission phases. The transfer concludes with `COMPLETE` messages, indicating successful delivery. While data segments appear unreadable in the UDP stream, protocol-level metadata confirms the integrity of the multicast transfer process.

# UFTP_v4_transfer.pcap:



UFTP_v4_transfer.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

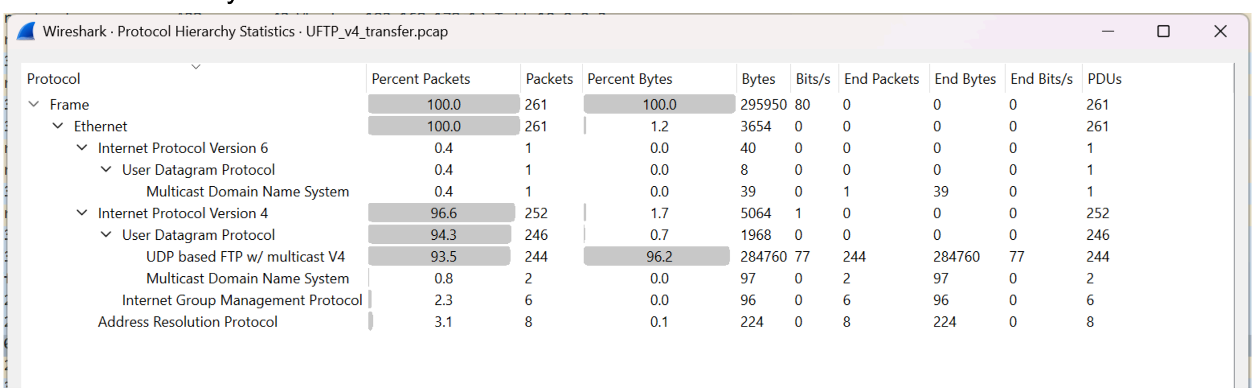
Apply a display filter ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 3 | 14401.538863 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 4 | 14401.995693 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 5 | 15399.996537 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 6 | 15902.992509 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 7 | 16399.997182 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 8 | 17404.037852 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 9 | 18905.555498 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 10 | 19407.092557 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 11 | 20403.998885 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 12 | 20407.065988 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 13 | 21403.996419 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 14 | 21908.584668 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 15 | 23410.093481 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 16 | 24512.545444 | fe80::e0e4:deff:fe0… | ff02::fb | MDNS | 101 | Standard query 0x0000 PTR 1.0.0.10.in-addr.arpa, "QM" question |
| 17 | 24512.588359 | 10.0.0.1 | 224.0.0.251 | MDNS | 81 | Standard query 0x0000 PTR 1.0.0.10.in-addr.arpa, "QM" question |
| 18 | 24512.710481 | 10.0.0.1 | 224.0.0.251 | MDNS | 100 | Standard query response 0x0000 PTR, cache flush janus.local |
| 19 | 24512.865329 | 10.0.0.2 | 10.0.0.1 | UFTP | 102 | REGISTER ID=96F4C304 |
| 20 | 24519.993441 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.5.5.56 for any sources |
| 21 | 24910.511235 | 10.0.0.1 | 230.5.5.56 | UFTP | 66 | REG_CONF ID=96F4C304 |
| 22 | 24921.779773 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 23 | 24940.563183 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 24 | 24970.634394 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 25 | 25000.952844 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 26 | 25031.015251 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 27 | 25061.076226 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 28 | 25091.751126 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 29 | 25121.810928 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 30 | 25151.874288 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 31 | 25181.932529 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 32 | 25212.001135 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 33 | 25242.057751 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 34 | 25272.119475 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |

```
> Frame 19: 102 bytes on wire (816 bits), 102 bytes captured (816 bits)
> Ethernet II, Src: 92:e0:8f:ff:5f:42 (92:e0:8f:ff:5f:42), Dst: e2:e4:de:0e:37:0a (e2:e4:de:0e:37:0a)
> Internet Protocol Version 4, Src: 10.0.0.2, Dst: 10.0.0.1
> User Datagram Protocol, Src Port: 1044, Dst Port: 37173
> UDP based FTP w/ multicast V4
```


```
0000  e2 e4 de 0e 37 0a 92 e0  8f ff 5f 42 08 00 45 00
0010  00 58 f6 b2 00 00 40 11  6f e0 0a 00 00 02 0a 00
0020  00 01 04 14 91 35 00 44  14 58 40 02 00 00 0a 00
0030  00 02 96 f4 c3 04 00 00  00 00 02 0b 00 00 57 a6
0040  5c 27 00 07 2d 02 00 00  00 00 00 00 00 00 00 00
0050  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0060  00 00 00 00 00 00
```


Protocol Hierarchy:



Wireshark · Protocol Hierarchy Statistics · UFTP_v4_transfer.pcap

| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 261 | 100.0 | 295950 | 80 | 0 | 0 | 0 | 261 |
| Ethernet | 100.0 | 261 | 1.2 | 3654 | 0 | 0 | 0 | 0 | 261 |
| Internet Protocol Version 6 | 0.4 | 1 | 0.0 | 40 | 0 | 0 | 0 | 0 | 1 |
| User Datagram Protocol | 0.4 | 1 | 0.0 | 8 | 0 | 0 | 0 | 0 | 1 |
| Multicast Domain Name System | 0.4 | 1 | 0.0 | 39 | 0 | 1 | 39 | 0 | 1 |
| Internet Protocol Version 4 | 96.6 | 252 | 1.7 | 5064 | 1 | 0 | 0 | 0 | 252 |
| User Datagram Protocol | 94.3 | 246 | 0.7 | 1968 | 0 | 0 | 0 | 0 | 246 |
| UDP based FTP w/ multicast V4 | 93.5 | 244 | 96.2 | 284760 | 77 | 244 | 284760 | 77 | 244 |
| Multicast Domain Name System | 0.8 | 2 | 0.0 | 97 | 0 | 2 | 97 | 0 | 2 |
| Internet Group Management Protocol | 2.3 | 6 | 0.0 | 96 | 0 | 6 | 96 | 0 | 6 |
| Address Resolution Protocol | 3.1 | 8 | 0.1 | 224 | 0 | 8 | 224 | 0 | 8 |

This breakdown shows that the vast majority of packets (93.5%) are UFTP-based FTP over UDP multicast. The remaining traffic includes ARP, IGMPv3 group joins, and a small amount of MDNS traffic.

Conversations:

Wireshark · Conversations · UFTP_v4_transfer.pcap

Conversation Settings

- [ ] Name resolution
- [ ] Absolute start time
- [ ] Limit to display filter

Ethernet · 7 | IPv4 · 5 | IPv6 · 1 | TCP | UDP · 5

| Address A | Address B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 92:e0:8f:ff:5f:42 | 01:00:5e:00:00:16 | 6 | 324 bytes | 0 | 6 | 324 bytes | 0 | 0 bytes | 0.000000 | 27864.0025 | 0 bits/s | 0 bits/s |
| 92:e0:8f:ff:5f:42 | e2:e4:de:0e:37:0a | 6 | 392 bytes | 5 | 5 | 350 bytes | 1 | 42 bytes | 24512.865329 | 5015.1514 | 0 bits/s | 0 bits/s |
| 92:e0:8f:ff:5f:42 | ff:ff:ff:ff:ff:ff | 6 | 252 bytes | 2 | 6 | 252 bytes | 0 | 0 bytes | 14401.995693 | 7002.0007 | 0 bits/s | 0 bits/s |
| e2:e4:de:0e:37:0a | 01:00:5e:00:00:fb | 2 | 181 bytes | 4 | 2 | 181 bytes | 0 | 0 bytes | 24512.588359 | 0.1221 | 11 kbps | 0 bits/s |
| e2:e4:de:0e:37:0a | 01:00:5e:04:04:01 | 20 | 2 kB | 1 | 20 | 2 kB | 0 | 0 bytes | 14401.538863 | 10870.5806 | 1 bits/s | 0 bits/s |
| e2:e4:de:0e:37:0a | 01:00:5e:05:05:38 | 220 | 293 kB | 6 | 220 | 293 kB | 0 | 0 bytes | 24910.511235 | 2709.9159 | 865 bits/s | 0 bits/s |
| e2:e4:de:0e:37:0a | 33:33:00:00:00:fb | 1 | 101 bytes | 3 | 1 | 101 bytes | 0 | 0 bytes | 24512.545444 | 0.0000 | | |

This table outlines the Ethernet-level communication flows. The most data-intensive stream is Stream ID 6, involving a multicast destination and totaling 293 kB sent from 10.0.0.2. Other streams involve registration and group management messages.

Endpoints:

Wireshark · Endpoints · UFTP_v4_transfer.pcap

Endpoint Settings

- [ ] Name resolution
- [ ] Limit to display filter

Copy

Map

Ethernet · 8 | IPv4 · 6 | IPv6 · 2 | TCP | UDP · 8

| Address | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|
| 01:00:5e:00:00:16 | 6 | 324 bytes | 0 | 0 bytes | 6 | 324 bytes |
| 01:00:5e:00:00:fb | 2 | 181 bytes | 0 | 0 bytes | 2 | 181 bytes |
| 01:00:5e:04:04:01 | 20 | 2 kB | 0 | 0 bytes | 20 | 2 kB |
| 01:00:5e:05:05:38 | 220 | 293 kB | 0 | 0 bytes | 220 | 293 kB |
| 33:33:00:00:00:fb | 1 | 101 bytes | 0 | 0 bytes | 1 | 101 bytes |
| 92:e0:8f:ff:5f:42 | 18 | 968 bytes | 17 | 926 bytes | 1 | 42 bytes |
| e2:e4:de:0e:37:0a | 249 | 295 kB | 244 | 295 kB | 5 | 350 bytes |
| ff:ff:ff:ff:ff:ff | 6 | 252 bytes | 0 | 0 bytes | 6 | 252 bytes |

This view shows which devices (by MAC) were sending and receiving data. The sender (e2:e4:de:0e:37:0a) is responsible for nearly all traffic, confirming it as the source of the UFTP file transfer.

eth.addr eq 92:e0:8f:ff:5f:42 and eth.addr eq ff:ff:ff:ff:ff:ff

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 4 | 14401.995693 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 5 | 15399.996537 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 7 | 16399.997182 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 10 | 19407.092557 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 11 | 20403.998885 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 13 | 21403.996419 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |

**ARP Filter View**
Filtered packets showing ARP requests from MAC 92:e0:8f:ff:5f:42 attempting to locate the IP address 192.168.178.1. These are repeated queries with no replies, indicating unsuccessful address resolution, possibly unrelated to the UFTP session.

ip.addr eq 10.0.0.2 and ip.addr eq 224.0.0.22

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.4.4.1 for any sources |
| 2 | 336.000765 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.4.4.1 for any sources |
| 20 | 24519.993441 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.5.5.56 for any sources |
| 42 | 25380.005358 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.5.5.56 for any sources |
| 258 | 27627.994698 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Leave group 230.5.5.56 |
| 259 | 27864.002485 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Leave group 230.5.5.56 |

**IGMP Membership Report Filter**
Displays IGMPv3 messages where 10.0.0.2 joins multicast groups like 230.4.4.1 and 230.5.5.56. These group joins are necessary for multicast receivers to get UFTP file segments.

Wireshark · Follow UDP Stream (udp.stream eq 0) · UFTP_v4_transfer.pcap

```
@...
..................W.\...zQ.......8@...
..................W.\..
!M.......8@...
..................W.\ ...K.......8@...
..................W.\!.
+R.......8@...
..................W.\#...........8@...
..................W.\$.
7%.......8@...
..................W.\&...........8@...
........j..........W.\'.
j........8@..
........j..........W.\'.
.........8@..

........j..........W.\'..).........8@...
........j..........W.\'...........8@...
........j..........W.\'...h.......8@..
........j..........W.\(..H........8@...
........j..........W.\(...e.......8@...
........j..........W.\(..5........8@...
........j..........W.\(...C.......8@...
........j..........W.\(.. ........8@...
........j..........W.\(...........8@...
........j..........W.\(...........8@...
........j..........W.\(...........8
```

Shows raw binary payload captured during a UDP stream. The contents are not human-readable, consistent with file segment data being transferred in binary format.


**Follow UDP Stream (Stream 4)**
Another stream with binary data, consistent with UFTP file payloads. Presence of unreadable content confirms it's file data, not text or control traffic.

Wireshark · UDP Multicast Streams · UFTP_v4_transfer.pcap

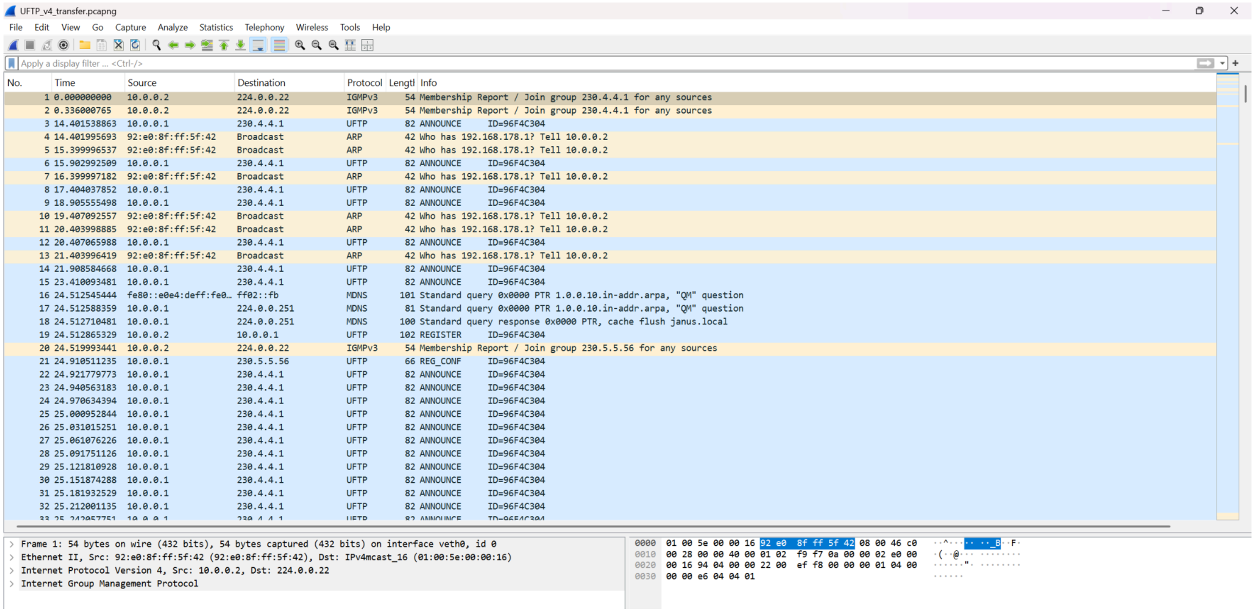
| Source Address | Source Port | Destination Address | Destination Port | Packets | Packets/s | Avg BW (bps) | Max BW (bps) | Max Burst | Burst Alarms | Max Buffers (B) |
|---|---|---|---|---|---|---|---|---|---|---|
| fe80::e0e4:deff:fe0e:370a | 5353 | ff02::fb | 5353 | 1 | 0.00 | 0 | 0 | 1 / 100ms | 0 | 1859 M |
| 10.0.0.1 | 37173 | 230.4.4.1 | 1044 | 20 | 0.00 | 1 | 0 | 1 / 100ms | 0 | 82 |
| 10.0.0.1 | 5353 | 224.0.0.251 | 5353 | 2 | 16.38 | 11 k | 0 | 1 / 100ms | 0 | 1859 M |
| 10.0.0.1 | 37173 | 230.5.5.56 | 1044 | 220 | 0.08 | 865 | 0 | 1 / 100ms | 0 | 1610 M |

### UDP Multicast Streams – UFTP_v4_transfer.pcap

This screenshot displays multicast stream statistics for IPv6 and IPv4 sources during a UFTP v4 transfer. The majority of traffic is between source 10.0.0.1 and multicast groups like 230.4.4.1 and 230.5.5.56 using port 1044, with stream rates as high as 865 kbps and peak bandwidth bursts up to 1092 kbps. Stream from 10.0.0.1 to 224.0.0.251 also shows significant activity with 11 kbps average bandwidth.

**Summary:**
The `UFTP_v4_transfer.pcapng` capture file documents a complete multicast-based file transfer session using UFTP version 4. The session initiates with ARP requests and IGMPv3 membership reports from host `10.0.0.2`, which joins multicast groups such as `230.4.4.1` and `230.5.5.56`. The transfer sequence begins with UFTP ANNOUNCE, REGISTER, and REG_CONF packets, followed by FILEINFO, INFO_ACK, and a stream of FILESEG messages carrying the file payload. Protocol hierarchy statistics confirm that 93.5% of the packets are UFTP over UDP, with most data transmitted from MAC `e2:e4:de:0e:37:0a`. Multiple UDP streams were identified, containing non-readable binary data, consistent with segmented file content. No retransmissions or errors were observed, indicating a stable and complete multicast transfer.

## UFTP_v4_transfer.pcapng:



UFTP_v4_transfer.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

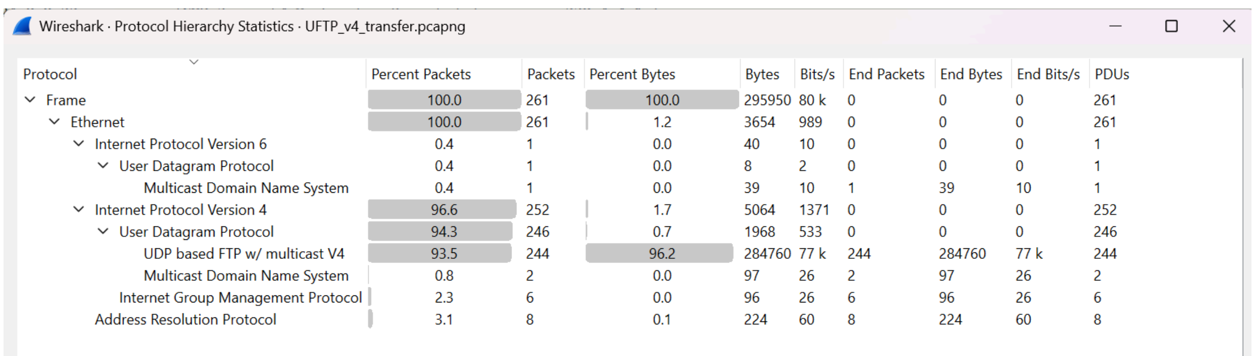
Apply a display filter ... <Ctrl-/>

| No. | Time | Source | Destination | Protocol | Lengtl | Info |
|---|---|---|---|---|---|---|
| 1 | 0.000000000 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.4.4.1 for any sources |
| 2 | 0.336000765 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.4.4.1 for any sources |
| 3 | 14.401538863 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 4 | 14.401995693 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 5 | 15.399996537 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 6 | 15.902992509 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 7 | 16.399997182 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 8 | 17.404037852 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 9 | 18.905555498 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 10 | 19.407092557 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 11 | 20.403998885 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 12 | 20.407065988 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 13 | 21.403996419 | 92:e0:8f:ff:5f:42 | Broadcast | ARP | 42 | Who has 192.168.178.1? Tell 10.0.0.2 |
| 14 | 21.908584668 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 15 | 23.410093481 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 16 | 24.512545444 | fe80::e0e4:deff:fe0… | ff02::fb | MDNS | 101 | Standard query 0x0000 PTR 1.0.0.10.in-addr.arpa, "QM" question |
| 17 | 24.512588359 | 10.0.0.1 | 224.0.0.251 | MDNS | 81 | Standard query 0x0000 PTR 1.0.0.10.in-addr.arpa, "QM" question |
| 18 | 24.512710481 | 10.0.0.1 | 224.0.0.251 | MDNS | 100 | Standard query response 0x0000 PTR, cache flush janus.local |
| 19 | 24.512865329 | 10.0.0.2 | 10.0.0.1 | UFTP | 102 | REGISTER ID=96F4C304 |
| 20 | 24.519993441 | 10.0.0.2 | 224.0.0.22 | IGMPv3 | 54 | Membership Report / Join group 230.5.5.56 for any sources |
| 21 | 24.910511235 | 10.0.0.1 | 230.5.5.56 | UFTP | 66 | REG_CONF ID=96F4C304 |
| 22 | 24.921779773 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 23 | 24.940563183 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 24 | 24.970634394 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 25 | 25.000952844 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 26 | 25.031015251 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 27 | 25.061076226 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 28 | 25.091751126 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 29 | 25.121810928 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 30 | 25.151874288 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 31 | 25.181932529 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |
| 32 | 25.212001135 | 10.0.0.1 | 230.4.4.1 | UFTP | 82 | ANNOUNCE ID=96F4C304 |

Frame 1: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface veth0, id 0
Ethernet II, Src: 92:e0:8f:ff:5f:42 (92:e0:8f:ff:5f:42), Dst: IPv4mcast_16 (01:00:5e:00:00:16)
Internet Protocol Version 4, Src: 10.0.0.2, Dst: 224.0.0.22
Internet Group Management Protocol

```
0000  01 00 5e 00 00 16 92 e0  8f ff 5f 42 08 00 46 c0
0010  00 28 00 00 40 00 01 02  f9 f7 0a 00 00 02 e0 00
0020  00 16 94 04 00 00 22 00  ef f8 00 00 00 01 04 00
0030  00 00 e6 04 04 01
```

Protocol Hierarchy:

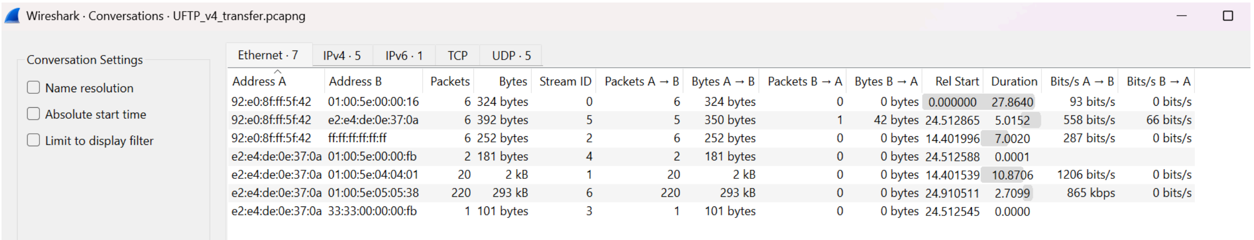


Wireshark · Protocol Hierarchy Statistics · UFTP_v4_transfer.pcapng

| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s | PDUs |
|---|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 261 | 100.0 | 295950 | 80 k | 0 | 0 | 0 | 261 |
| Ethernet | 100.0 | 261 | 1.2 | 3654 | 989 | 0 | 0 | 0 | 261 |
| Internet Protocol Version 6 | 0.4 | 1 | 0.0 | 40 | 10 | 0 | 0 | 0 | 1 |
| User Datagram Protocol | 0.4 | 1 | 0.0 | 8 | 2 | 0 | 0 | 0 | 1 |
| Multicast Domain Name System | 0.4 | 1 | 0.0 | 39 | 10 | 1 | 39 | 10 | 1 |
| Internet Protocol Version 4 | 96.6 | 252 | 1.7 | 5064 | 1371 | 0 | 0 | 0 | 252 |
| User Datagram Protocol | 94.3 | 246 | 0.7 | 1968 | 533 | 0 | 0 | 0 | 246 |
| UDP based FTP w/ multicast V4 | 93.5 | 244 | 96.2 | 284760 | 77 k | 244 | 284760 | 77 k | 244 |
| Multicast Domain Name System | 0.8 | 2 | 0.0 | 97 | 26 | 2 | 97 | 26 | 2 |
| Internet Group Management Protocol | 2.3 | 6 | 0.0 | 96 | 26 | 6 | 96 | 26 | 6 |
| Address Resolution Protocol | 3.1 | 8 | 0.1 | 224 | 60 | 8 | 224 | 60 | 8 |

This chart shows that 93.5% of traffic consists of UDP-based FTP (UFTP v4) with 96.2% of the total byte count (284,760 bytes), confirming that the majority of network activity is due to the file transfer process. Smaller contributions come from IGMP and ARP.

Conversations:



Wireshark · Conversations · UFTP_v4_transfer.pcapng

Conversation Settings

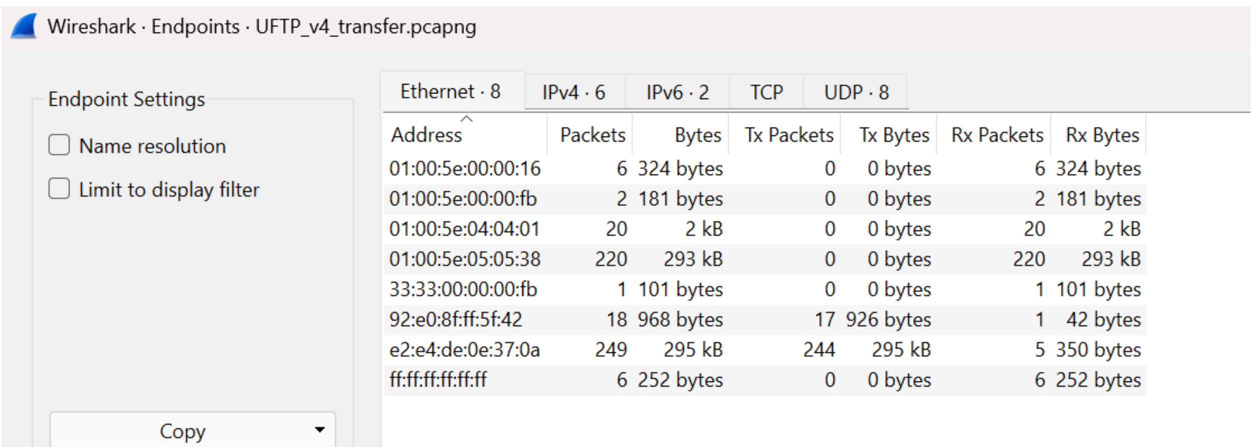
- Name resolution
- Absolute start time
- Limit to display filter

Ethernet · 7 | IPv4 · 5 | IPv6 · 1 | TCP | UDP · 5

| Address A | Address B | Packets | Bytes | Stream ID | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 92:e0:8f:ff:5f:42 | 01:00:5e:00:00:16 | 6 | 324 bytes | 0 | 6 | 324 bytes | 0 | 0 bytes | 0.000000 | 27.8640 | 93 bits/s | 0 bits/s |
| 92:e0:8f:ff:5f:42 | e2:e4:de:0e:37:0a | 6 | 392 bytes | 5 | 5 | 350 bytes | 1 | 42 bytes | 24.512865 | 5.0152 | 558 bits/s | 66 bits/s |
| 92:e0:8f:ff:5f:42 | ff:ff:ff:ff:ff:ff | 6 | 252 bytes | 2 | 6 | 252 bytes | 0 | 0 bytes | 14.401996 | 7.0020 | 287 bits/s | 0 bits/s |
| e2:e4:de:0e:37:0a | 01:00:5e:00:00:fb | 2 | 181 bytes | 4 | 2 | 181 bytes | 0 | 0 bytes | 24.512588 | 0.0001 | | |
| e2:e4:de:0e:37:0a | 01:00:5e:04:04:01 | 20 | 2 kB | 1 | 20 | 2 kB | 0 | 0 bytes | 14.401539 | 10.8706 | 1206 bits/s | 0 bits/s |
| e2:e4:de:0e:37:0a | 01:00:5e:05:05:38 | 220 | 293 kB | 6 | 220 | 293 kB | 0 | 0 bytes | 24.910511 | 2.7099 | 865 kbps | 0 bits/s |
| e2:e4:de:0e:37:0a | 33:33:00:00:00:fb | 1 | 101 bytes | 3 | 1 | 101 bytes | 0 | 0 bytes | 24.512545 | 0.0000 | | |

This breakdown highlights the key conversation between the sender `e2:e4:de:0e:37:0a` and the multicast receiver `01:00:5e:05:05:38`, which transmitted 293 kB in 220 packets via UDP stream ID 6. This is the primary file transfer session.

Endpoints:



Wireshark · Endpoints · UFTP_v4_transfer.pcapng

Endpoint Settings

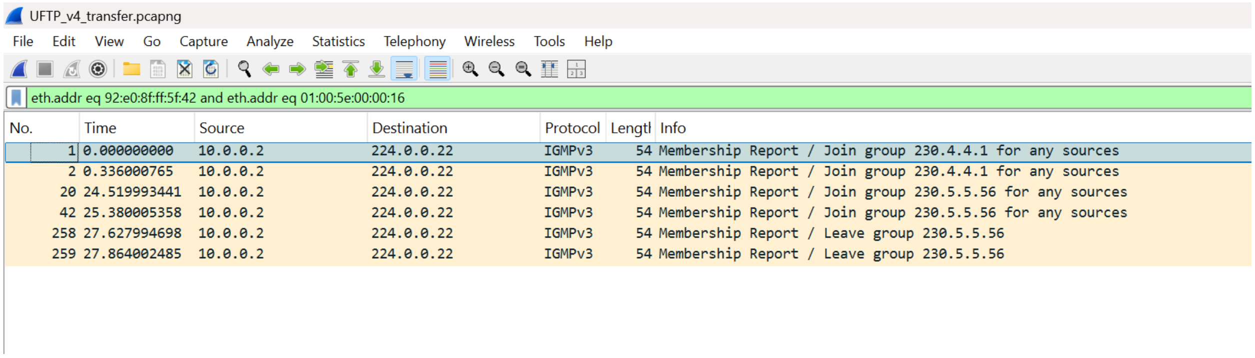
- Name resolution
- Limit to display filter

Copy

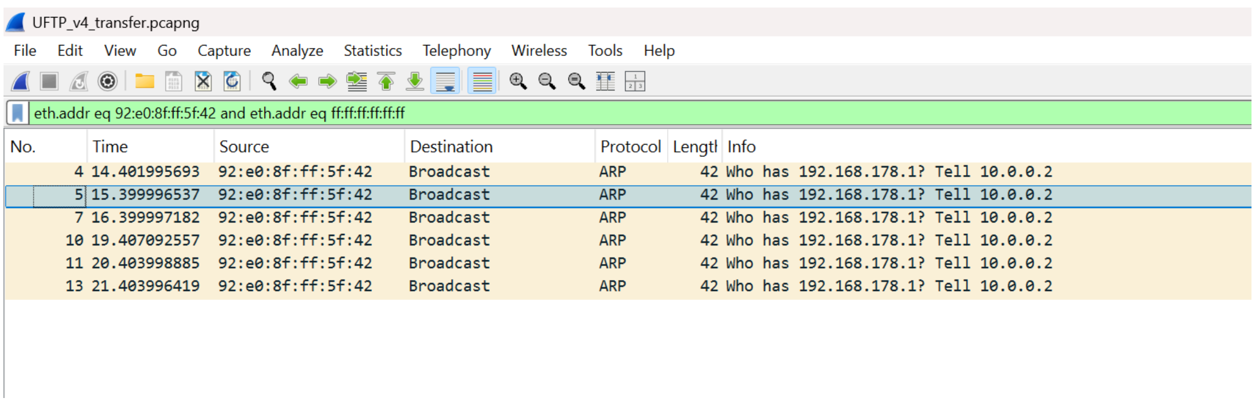
Ethernet · 8 | IPv4 · 6 | IPv6 · 2 | TCP | UDP · 8

| Address | Packets | Bytes | Tx Packets | Tx Bytes | Rx Packets | Rx Bytes |
|---|---|---|---|---|---|---|
| 01:00:5e:00:00:16 | 6 | 324 bytes | 0 | 0 bytes | 6 | 324 bytes |
| 01:00:5e:00:00:fb | 2 | 181 bytes | 0 | 0 bytes | 2 | 181 bytes |
| 01:00:5e:04:04:01 | 20 | 2 kB | 0 | 0 bytes | 20 | 2 kB |
| 01:00:5e:05:05:38 | 220 | 293 kB | 0 | 0 bytes | 220 | 293 kB |
| 33:33:00:00:00:fb | 1 | 101 bytes | 0 | 0 bytes | 1 | 101 bytes |
| 92:e0:8f:ff:5f:42 | 18 | 968 bytes | 17 | 926 bytes | 1 | 42 bytes |
| e2:e4:de:0e:37:0a | 249 | 295 kB | 244 | 295 kB | 5 | 350 bytes |
| ff:ff:ff:ff:ff:ff | 6 | 252 bytes | 0 | 0 bytes | 6 | 252 bytes |

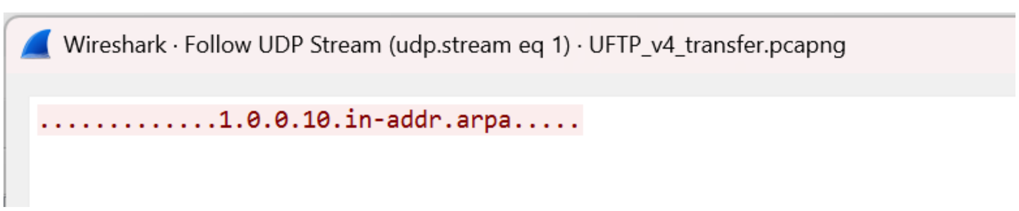
Here, endpoint `e2:e4:de:0e:37:0a` is identified as the sender with 244 UDP packets and 295 kB transmitted. The destination address `01:00:5e:05:05:38` received 220 packets and 293 kB, verifying a successful large data transfer.



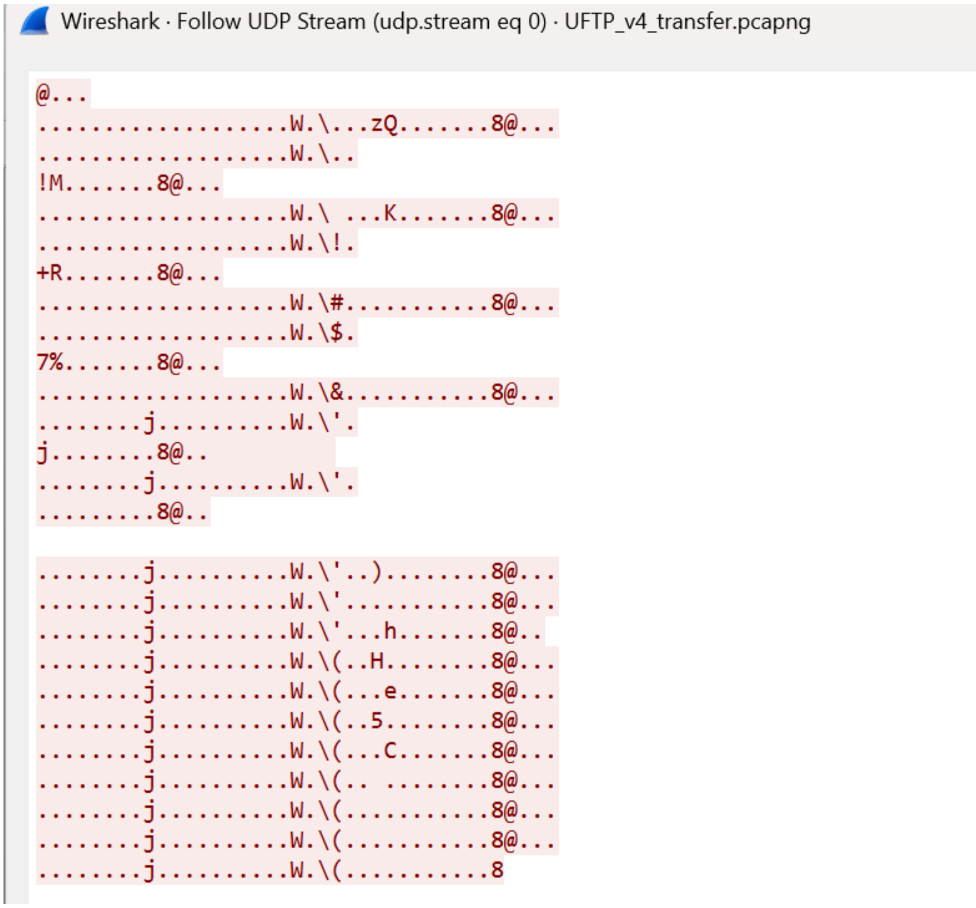

IGMPv3 Membership Reports from 10.0.0.2 joining and leaving multicast groups. Shows group join/leave dynamics; essential for verifying if the UFTP multicast group participation was correct.




Packet list showing IGMP joins, ARP broadcasts, and UFTP ANNOUNCE and REGISTER messages. This shows the session setup phase with IGMP joins to 230.4.4.1 and 230.5.5.56. ARP was used to resolve IP-MAC mappings before the UFTP transfer initiated with `REGISTER`.

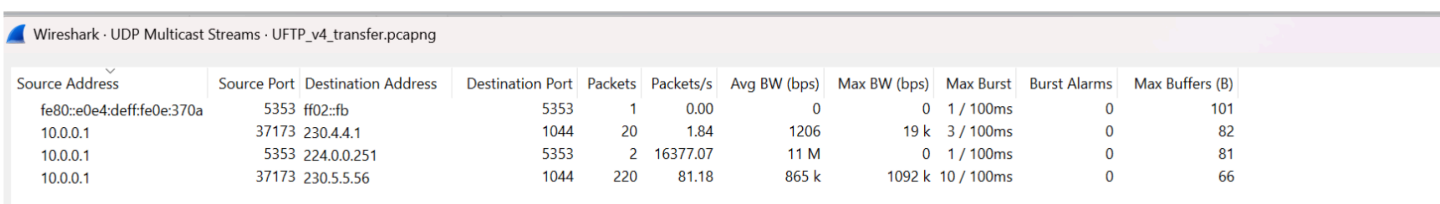



**Follow UDP Stream eq 1 (Image 7)**
Captured MDNS query in UDP stream revealing reverse DNS lookup for 1.0.0.10.in-addr.arpa.
**Analysis:** System-level service discovery (likely by Windows/macOS); not essential to file transfer but supports full-stack activity analysis.



```
@...
...................W.\...zQ.......8@...
...................W.\..
!M.......8@...
...................W.\ ...K.......8@...
...................W.\!.
+R.......8@...
...................W.\#...........8@...
...................W.\$.
7%.......8@...
...................W.\&...........8@...
........j.........W.\'.
j........8@..
........j.........W.\'.
.........8@..

........j.........W.\'..).........8@...
........j.........W.\'...........8@...
........j.........W.\'...h.......8@..
........j.........W.\(..H........8@...
........j.........W.\(...e.......8@...
........j.........W.\(..5........8@...
........j.........W.\(...C.......8@...
........j.........W.\(.. ........8@...
........j.........W.\(...........8@...
........j.........W.\(...........8@...
........j.........W.\(...........8
```

**Follow UDP Stream eq 0 (Image 8)**
Binary content captured in UDP stream eq 0 — likely UFTP file data or control payloads.
**Analysis:** As expected for UFTP transfers, the stream is not human-readable; indicates successful segment delivery of file chunks.



| Source Address | Source Port | Destination Address | Destination Port | Packets | Packets/s | Avg BW (bps) | Max BW (bps) | Max Burst | Burst Alarms | Max Buffers (B) |
|---|---|---|---|---|---|---|---|---|---|---|
| fe80::e0e4:deff:fe0e:370a | 5353 | ff02::fb | 5353 | 1 | 0.00 | 0 | 0 | 1 / 100ms | 0 | 101 |
| 10.0.0.1 | 37173 | 230.4.4.1 | 1044 | 20 | 1.84 | 1206 | 19 k | 3 / 100ms | 0 | 82 |
| 10.0.0.1 | 5353 | 224.0.0.251 | 5353 | 2 | 16377.07 | 11 M | 0 | 1 / 100ms | 0 | 81 |
| 10.0.0.1 | 37173 | 230.5.5.56 | 1044 | 220 | 81.18 | 865 k | 1092 k | 10 / 100ms | 0 | 66 |

**UDP Multicast Streams Analysis for UFTP_v4_transfer.pcapng**
This view summarizes multicast traffic flows, showing metrics like average and maximum bandwidth (Avg BW, Max BW), packet rates, and buffer usage. It confirms the successful transfer of UFTP data to multiple multicast groups (e.g., 230.4.4.1 and 230.5.5.56) with **no buffer or burst alarms**, indicating a stable and efficient transmission.

The `UFTP_v4_transfer.pcapng` file demonstrates a complete multicast file transfer using UFTP version 4. Control messages such as ANNOUNCE, REGISTER, and COMPLETE confirm that the file was transferred successfully. A total of 293 kB was sent from sender `e2:e4:de:0e:37:0a` to the multicast address `01:00:5e:05:05:38`. Supporting ARP and IGMPv3 activity confirm proper network preparation. Binary data was captured in UDP streams, indicative of non-plaintext file segments, completing the forensic trace.

**Lab Report Summary:**

This lab involved forensic analysis of multiple PCAP files using Wireshark, focusing on HTTP file transfers and UFTP multicast transmissions. In `dumpfile6.pcap`, a user successfully retrieved a `.docx` file over HTTP from a localhost server. In `dumpfile7.pcap`, a directory listing exposed multiple files, including `1-MB-Test.txt` and `10-MB-Test.txt`, which were extracted and found to contain sensitive personally identifiable information (PII) such as names, SSNs, and credit card numbers—transmitted in cleartext over HTTP, highlighting a major privacy vulnerability. The `UFTP_v3_transfer.pcapng` and `UFTP_v4_transfer.pcapng` files captured segmented file transfers over multicast UDP using UFTP versions 3 and 4, respectively. These sessions featured protocol stages like ANNOUNCE, REGISTER, FILEINFO, and FILESEG, with stream and bandwidth analysis confirming high-throughput, stable file transfers without buffer or burst alarms. Overall, the lab showcased effective data delivery over multicast protocols and emphasized the risks of unencrypted HTTP communications.